unix-api

POSIX process management

essential operations:

process information: `getpid`

process creation: `fork`

running programs: `exec*`

waiting for processes to finish: `waitpid` (or `wait`)

process destruction, ‘signaling’: `exit`, `kill`

fork

`pid_t fork()` — copy the current process

returns twice:

- in *parent* (original process): pid of new *child* process
- in *child* (new process): `0`

everything (but pid) duplicated in parent, child:

- memory
- file descriptors (later)
- registers

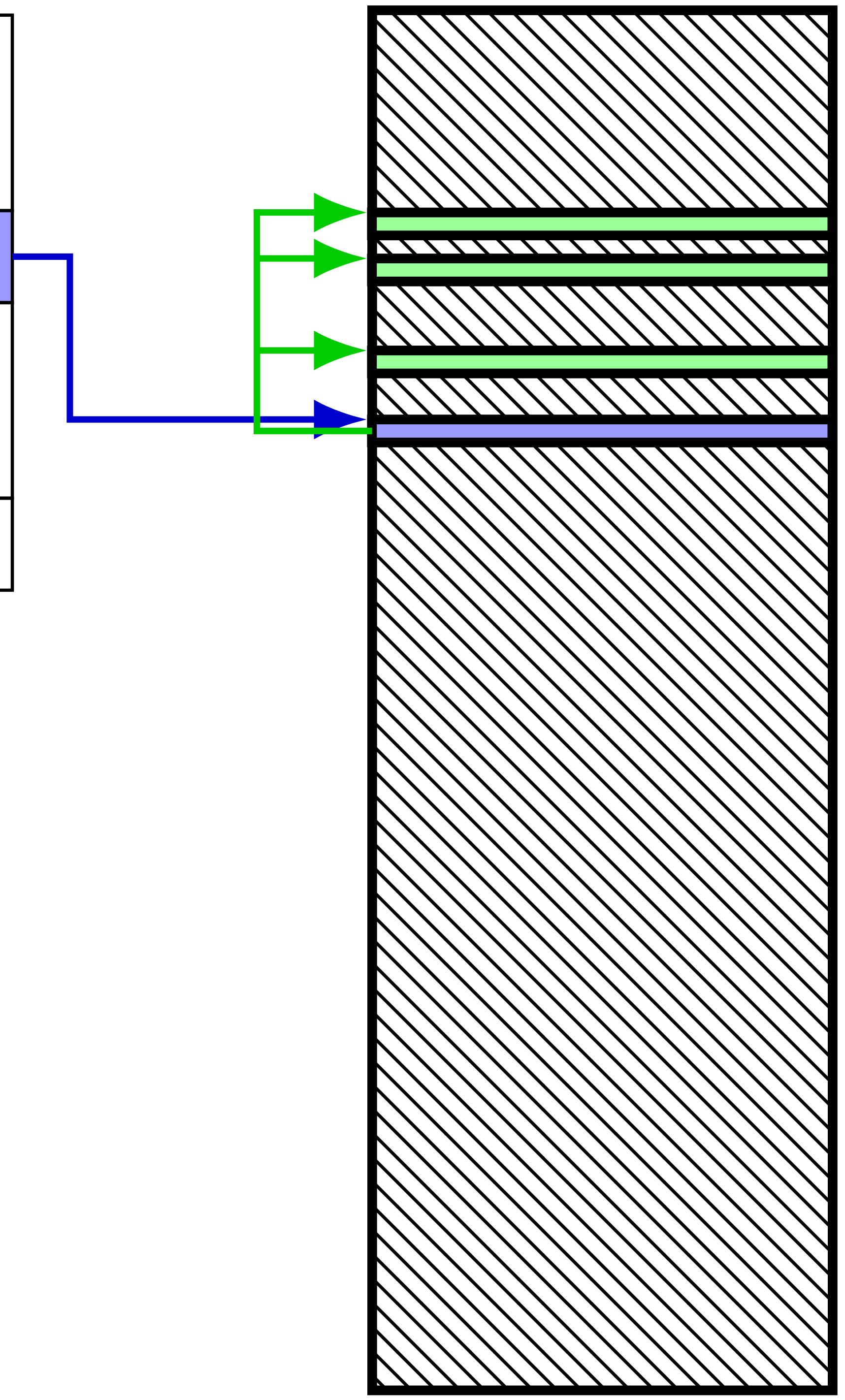
fork and process info (w/o copy-on-write)

fork and process info (w/o copy-on-write)

parent process info

user regs	`rax (return val.)=42,` `rcx=133, …`
memory mapping	
open files	`fd 0: …` `fd 1: …`
…	…

memory



fork and process info (w/o copy-on-write)

parent process info

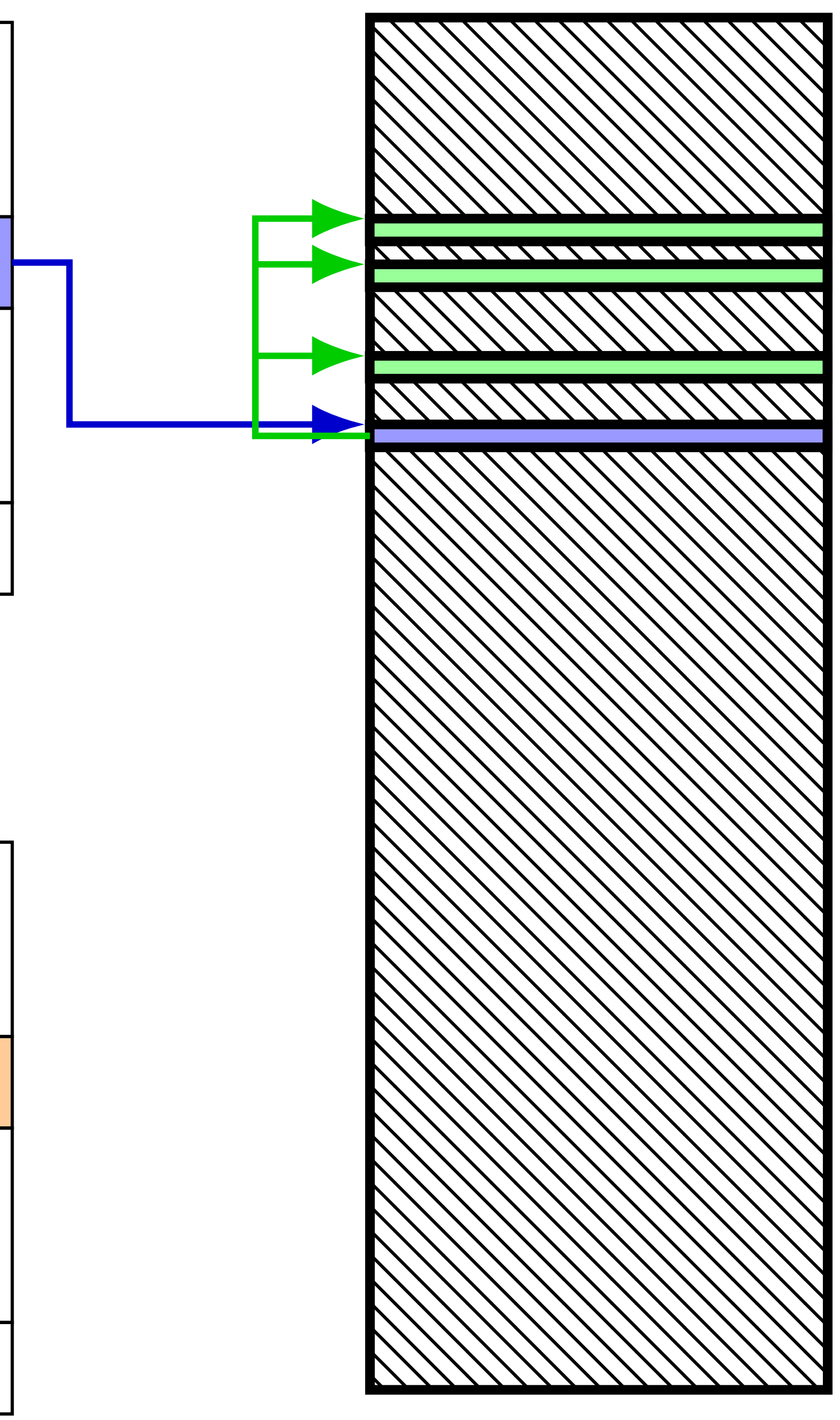
user regs	`rax (return val.)=42,` `rcx=133, …`
memory mapping	
open files	`fd 0: …` `fd 1: …`
…	…

copy

child process info

user regs	`rax (return val.)=42,` `rcx=133, …`
memory mapping	
open files	`fd 0: …` `fd 1: …`
…	…

memory



fork and process info (w/o copy-on-write)

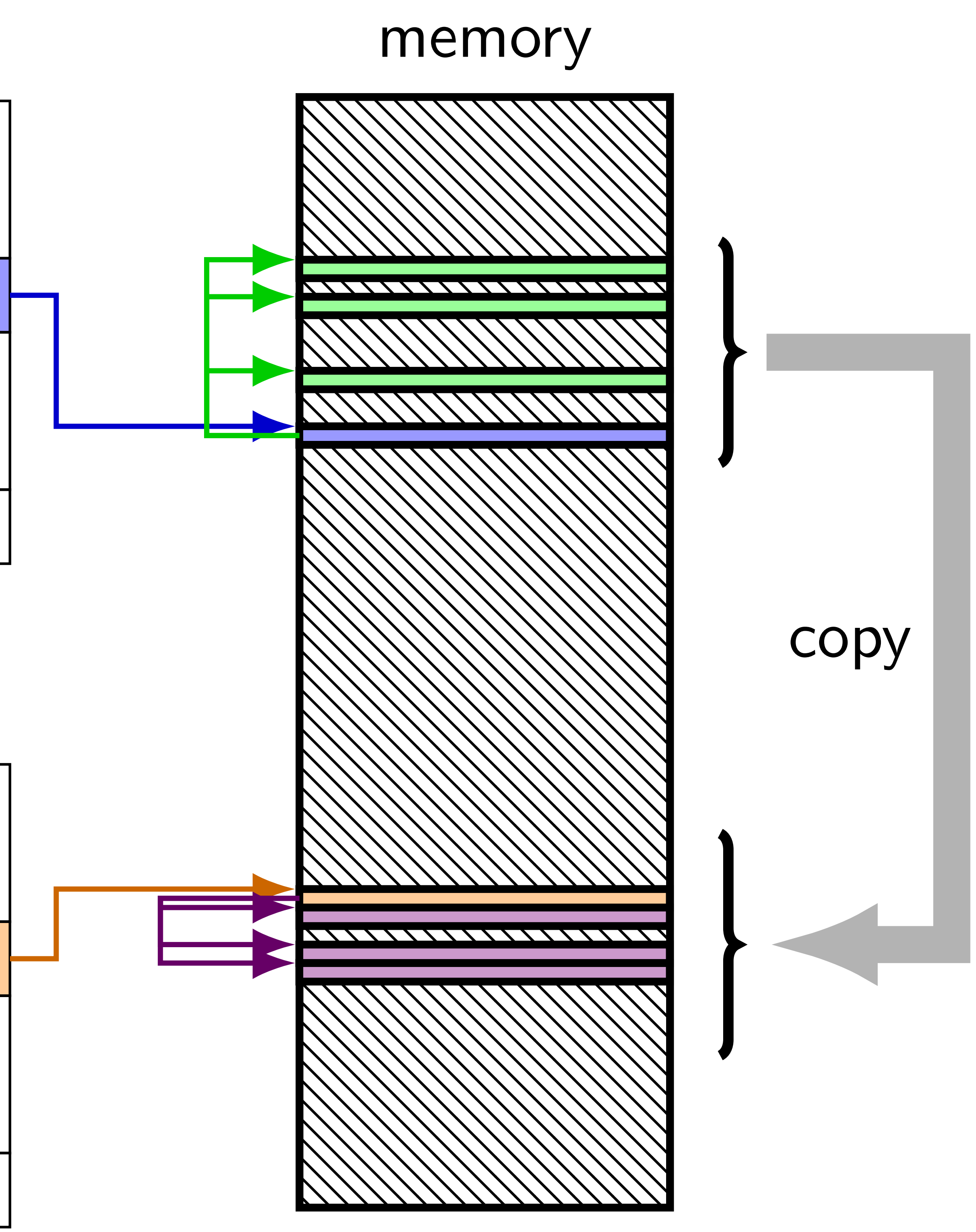
parent process info

user regs	`rax (return val.)=42,` `rcx=133, …`
memory mapping	
open files	`fd 0: …` `fd 1: …`
…	…

copy

child process info

user regs	`rax (return val.)=42,` `rcx=133, …`
memory mapping	
open files	`fd 0: …` `fd 1: …`
…	…



fork and process info (w/o copy-on-write)

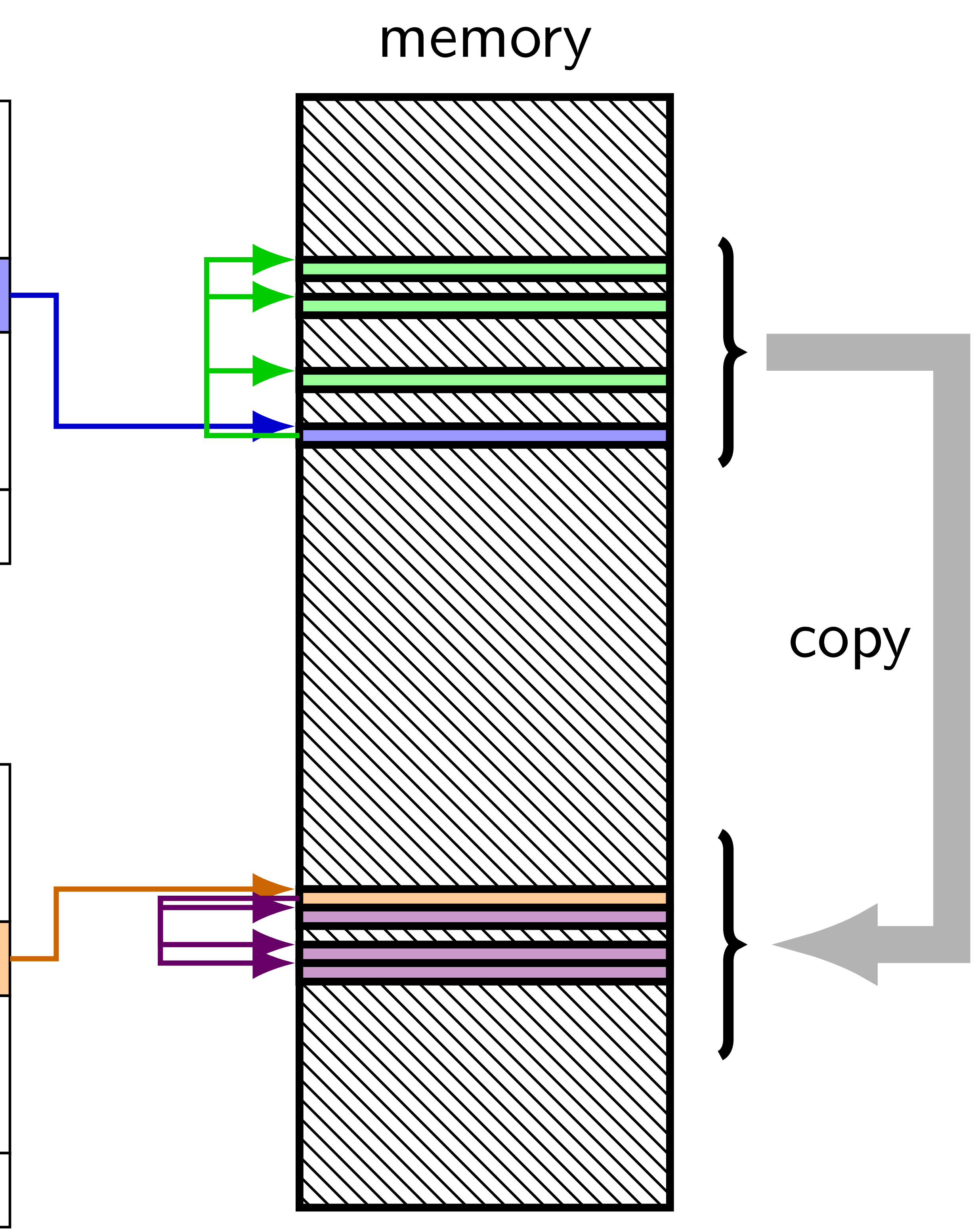
parent process info

user regs	rax (return val.)=42, rcx=133, …
memory mapping	
open files	fd 0: … fd 1: …
…	…

copy

child process info

user regs	rax (return val.)=42, rcx=133, …
memory mapping	
open files	fd 0: … fd 1: …
…	…



fork and process info (w/o copy-on-write)

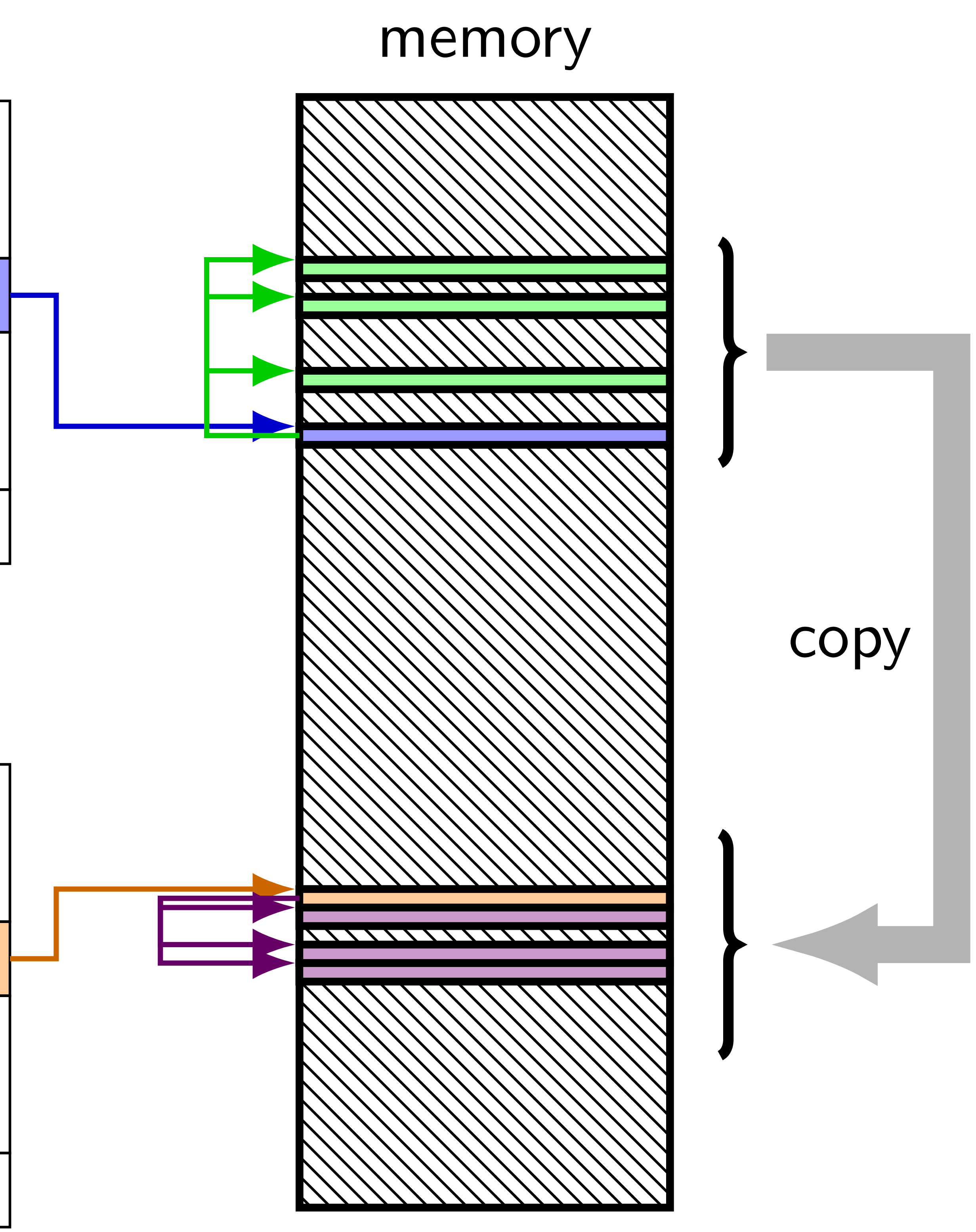
parent process info

user regs	rax (return val.)=~~42~~*child pid*, rcx=133, …
memory mapping	
open files	fd 0: … fd 1: …
…	…

copy

child process info

user regs	rax (return val.)=~~42~~*0*, rcx=133, …
memory mapping	
open files	fd 0: … fd 1: …
…	…



fork example

```c
// not shown: #include various headers
int main(int argc, char *argv[]) {
    pid_t pid = getpid();
    printf("Parent pid: %d\n", (int) pid);
    pid_t child_pid = fork();
    if (child_pid > 0) {
        /* Parent Process */
        pid_t my_pid = getpid();
        printf("[%d] parent of [%d]\n", (int) my_pid, (int) child_pid);
    } else if (child_pid == 0) {
        /* Child Process */
        pid_t my_pid = getpid();
        printf("[%d] child\n", (int) my_pid);
    } else {
        perror("Fork failed");
    }
    return 0;
}
```

fork example

```c
// not shown: #include various headers
int main(int argc, char *argv[]) {
    pid_t pid = getpid();
    printf("Parent pid: %d\n", (int) pid);
    pid_t child_pid = fork();
    if (child_pid > 0) {
        /* Parent Process */
        pid_t my_pid = getpid();
        printf("[%d] parent of [%d]\n", (int) my_pid, (int) child_pid);
    } else if (child_pid == 0) {
        /* Child Process */
        pid_t my_pid = getpid();
        printf("[%d] child\n", (int) my_pid);
    } else {
        perror("Fork failed");
    }
    return 0;
}
```

`getpid` — returns current process ID

fork example

```c
// not shown: #include various headers
int main(int argc, char *argv[]) {
    pid_t pid = getpid();
    printf("Parent pid: %d\n", (int) pid);
    pid_t child_pid = fork();
    if (child_pid > 0) {
        /* Parent Process */
        pid_t my_pid = getpid();
        printf("[%d] parent of [%d]\n", (int) my_pid, (int) child_pid);
    } else if (child_pid == 0) {
        /* Child Process */
        pid_t my_pid = getpid();
        printf("[%d] child\n", (int) my_pid);
    } else {
        perror("Fork failed");
    }
    return 0;
}
```

casts on printf because `pid_t` might not be `int`

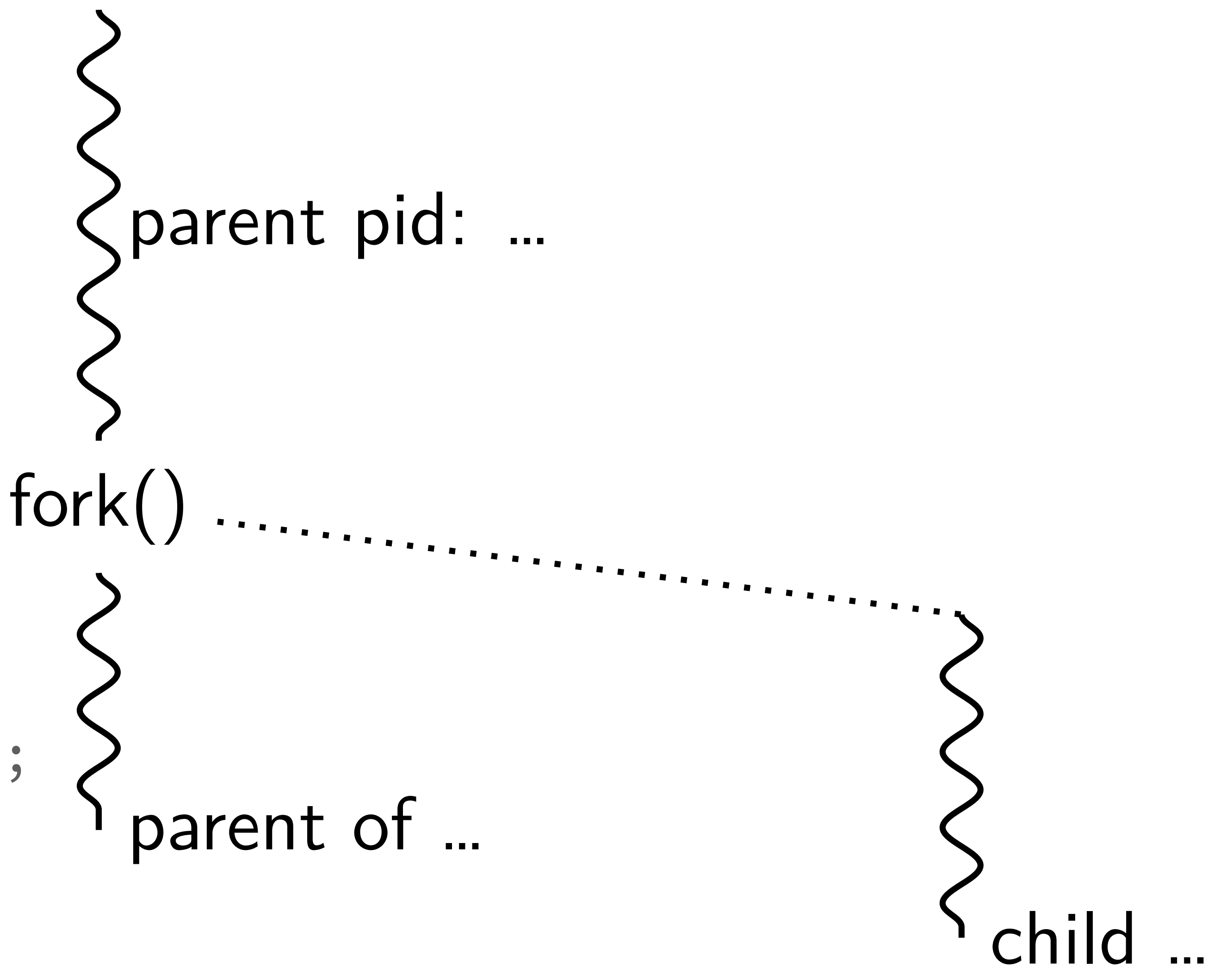
fork example

```
// not shown: #include various headers
int main(int argc, char *argv[]) {
    pid_t pid = getpid();
    printf("Parent pid: %d\n", (int) pid);
    pid_t child_pid = fork();
    if (child_pid > 0) {
        /* Parent Process */
        pid_t my_pid = getpid();
        printf("[%d] parent of [%d]\n", (int) my_pid, (int) child_pid);
    } else if (child_pid == 0) {
        /* Child Process */
        pid_t my_pid = getpid();
        printf("[%d] child\n", (int) my_pid);
    } else {
        perror("Fork failed");
    }
    return 0;
}
```

`perror("Fork failed")` prints out `Fork failed: *error message*`
example *error message*: “Resource temporarily unavailable”

fork example

```c
// not shown: #include various headers
int main(int argc, char *argv[]) {
    pid_t pid = getpid();
    printf("Parent pid: %d\n", (int) pid);
    pid_t child_pid = fork();
    if (child_pid > 0) {
        /* Parent Process */
        pid_t my_pid = getpid();
        printf("[%d] parent of [%d]\n", (int) my_pid, (int) child_pid);
    } else if (child_pid == 0) {
        /* Child Process */
        pid_t my_pid = getpid();
        printf("[%d] child\n", (int) my_pid);
    } else {
        perror("Fork failed");
    }
    return 0;
}
```



Example output:

```
Parent pid: 100
[100] parent of [432]
[432] child
```

a fork question

```
int main() {
    pid_t pid = fork();
    if (pid == 0) {
        printf("In child\n");
    } else {
        printf("Child %d\n", pid);
    }
    printf("Done!\n");
}
```

Exercise: Suppose the pid of the parent process is `99` and child is `100`. Give *two* possible outputs. (Assume no crashes, etc.)

a fork question

```c
int main() {
    pid_t pid = fork();
    if (pid == 0) {
        printf("In child\n");
    } else {
        printf("Child %d\n", pid);
    }
    printf("Done!\n");
}
```

Exercise: Suppose the pid of the parent process is `99` and child is `100`. Give *two* possible outputs. (Assume no crashes, etc.)

a fork question (2)

```
int x = 0;
int main() {
    pid_t pid = fork();
    int y = 0;
    if (pid == 0) {
        x += 1;
        y += 2;
    } else {
        x += 3;
        y += 4;
    }
    printf("%d %d\n", x, y);
}
```

Which (possibly multiple) are possible outputs? (assume trailing newline)

A. 1 2 (newline) 3 4   B. 1 2 (newline) 4 4   C. 1 2 (newline) 4 6
D. 3 4 (newline) 1 2   E. 3 4 (newline) 4 6   F. 4 6 (newline) 4 6

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exec*

exec* — *replace* current program with new program

- $\star$: multiple variants
- same pid, new process image

```
int execv(const char *path, const char **argv)
```

- path: new program to run
- argv: array of arguments, terminated by null pointer

also other variants that take argv in different form and/or environment variables

- environment variables = list of key-value pairs

execv example

```
...
child_pid = fork();
if (child_pid == 0) {
 /* child process */
 char *args[] = {"ls", "-l", NULL};
 execv("/bin/ls", args);
 /* execv doesn't return when it works.
    So, if we got here, it failed. */
 perror("execv");
 exit(1);
} else if (child_pid > 0) {
 /* parent process */
 ...
}
```

execv example

```
...
child_pid = fork();
if (child_pid == 0) {
  /* child process */
  char *args[] = {"ls", "-l", NULL};
  execv("/bin/ls", args);
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     So, if we got here, it failed. */
  perror("execv");
  exit(1);
} else if (child_pid > 0) {
  /* parent process */
  ...
}
```

execv example

```
...
child_pid = fork();
if (child_pid == 0) {
 /* child process */
 char *args[] = {"ls", "-l", NULL};
 execv("/bin/ls", args);
 /* execv doesn't return when it works.
    So, if we got here, it failed. */
 perror("execv");
 exit(1);
} else if (child_pid > 0) {
 /* parent process */
 ...
}
```

used to compute argv, argc
when program's `main` is run

convention: first argument is program name

execv example

```
...
child_pid = fork();
if (child_pid == 0) {
 /* child process */
 char *args[] = {"ls", "-l", NULL};
 execv("/bin/ls", args);
 /* execv doesn't return when it works.
    So, if we got here, it failed. */
 perror("execv");
 exit(1);
} else if (child_pid > 0) {
 /* parent process */
 ...
}
```

path of executable to run
need not match first argument
(but probably should match it)

on Unix /bin is a directory
containing many common programs,
including `ls` (‘list directory’)

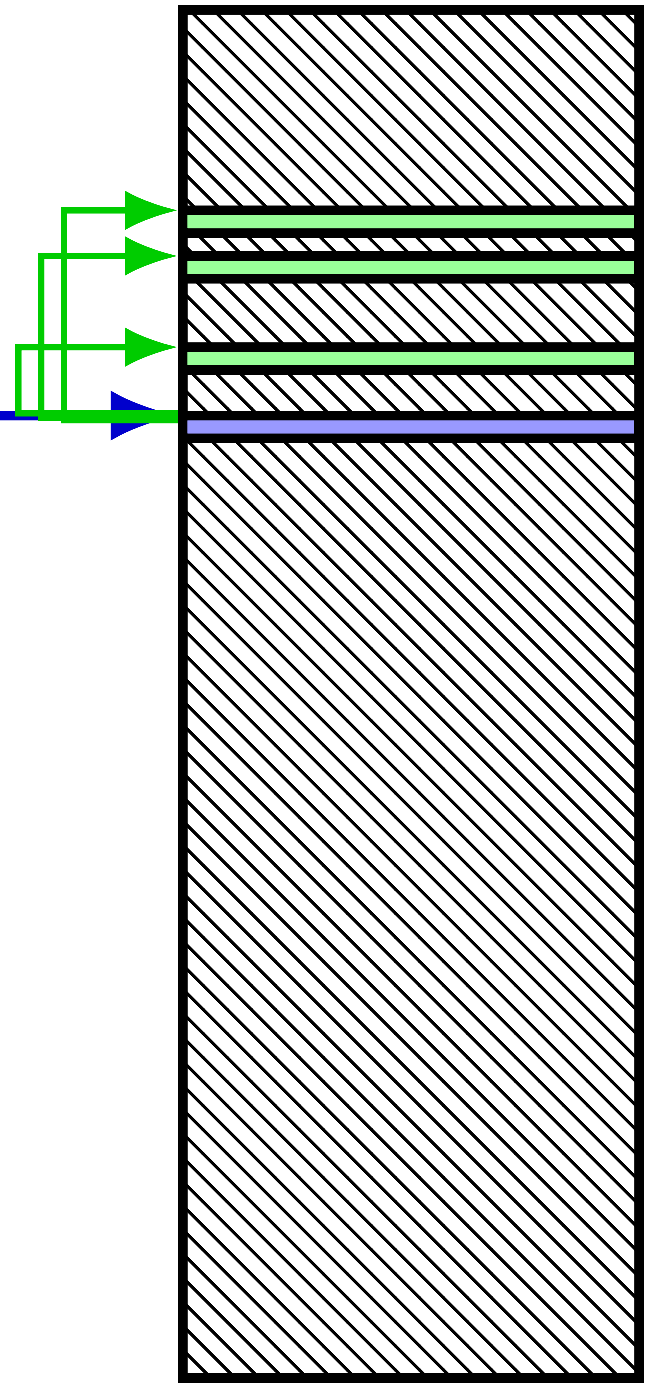
exec in the kernel

exec in the kernel

the process control block

user regs	eax=42, ecx=133, …
memory mapping	
open files	fd 0: (terminal …) fd 1: …
…	…

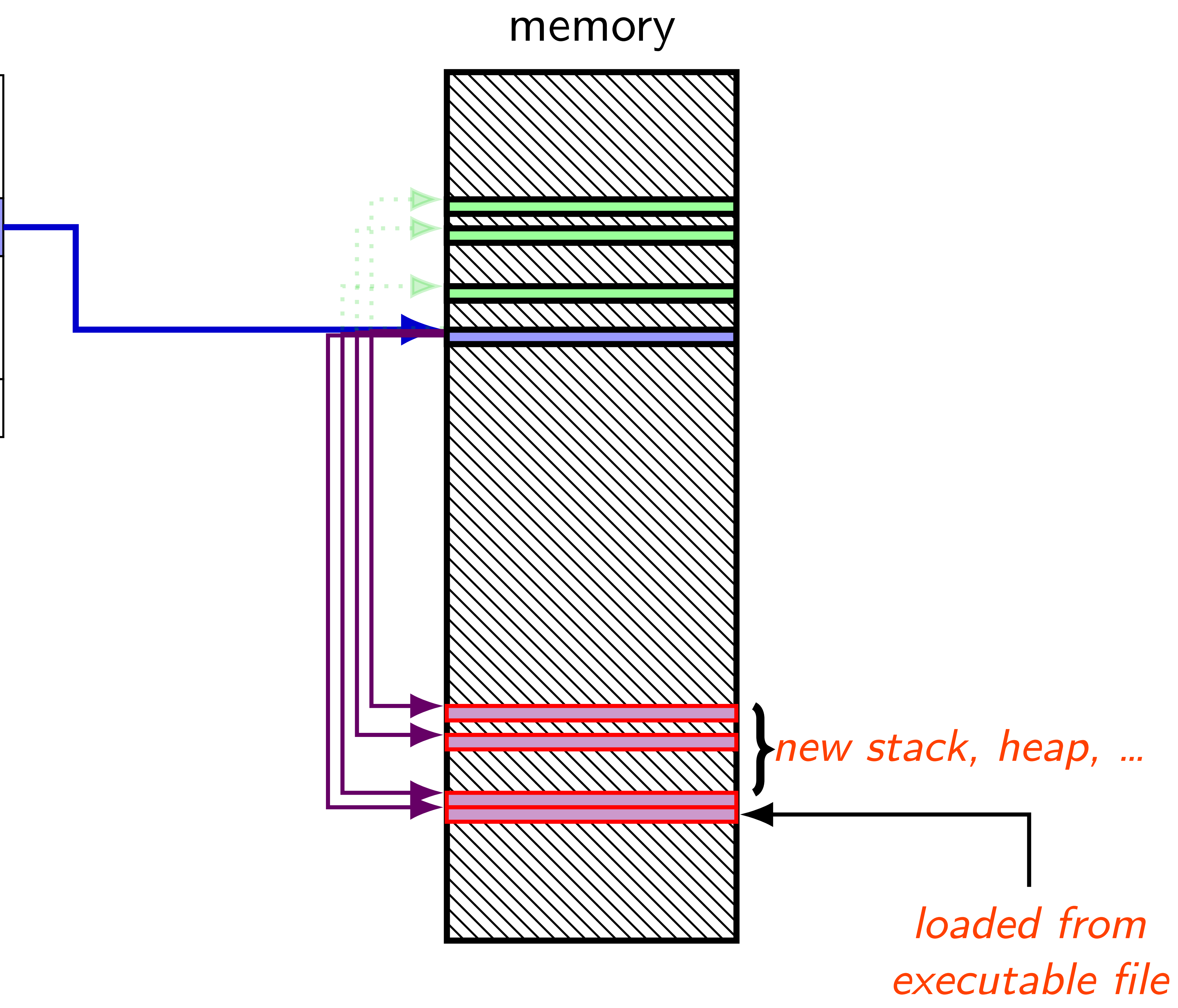
memory



exec in the kernel

the process control block

user regs	eax=~~42~~*init. val.*, ecx=~~133~~*init. val.*, …
memory mapping	
open files	fd 0: (terminal …) fd 1: …
…	…



exec in the kernel

the process control block

user regs	eax=~~42~~*init. val.*, ecx=~~133~~*init. val.*, ...
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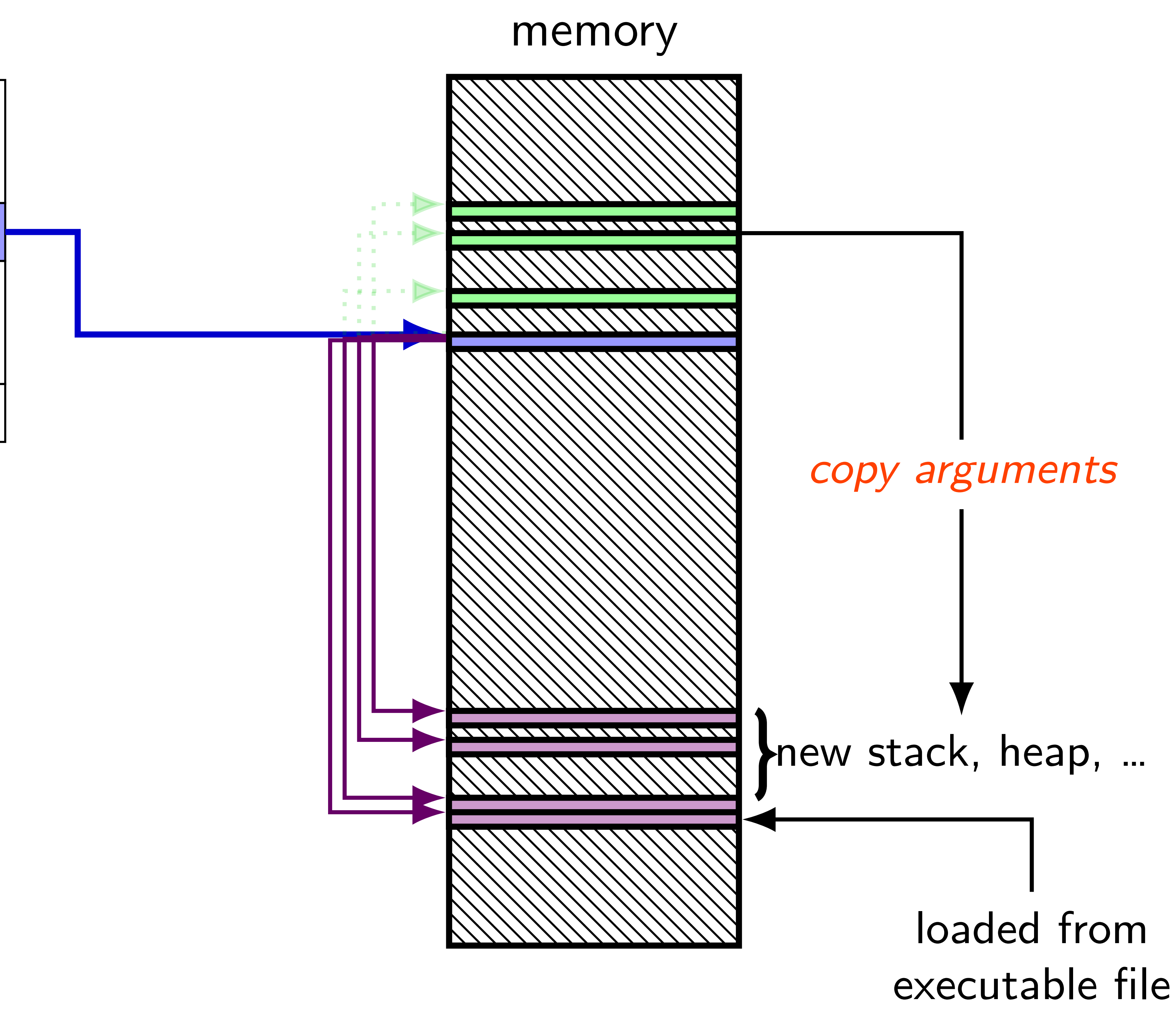
open files	fd 0: (terminal ...) fd 1: ...
...	...

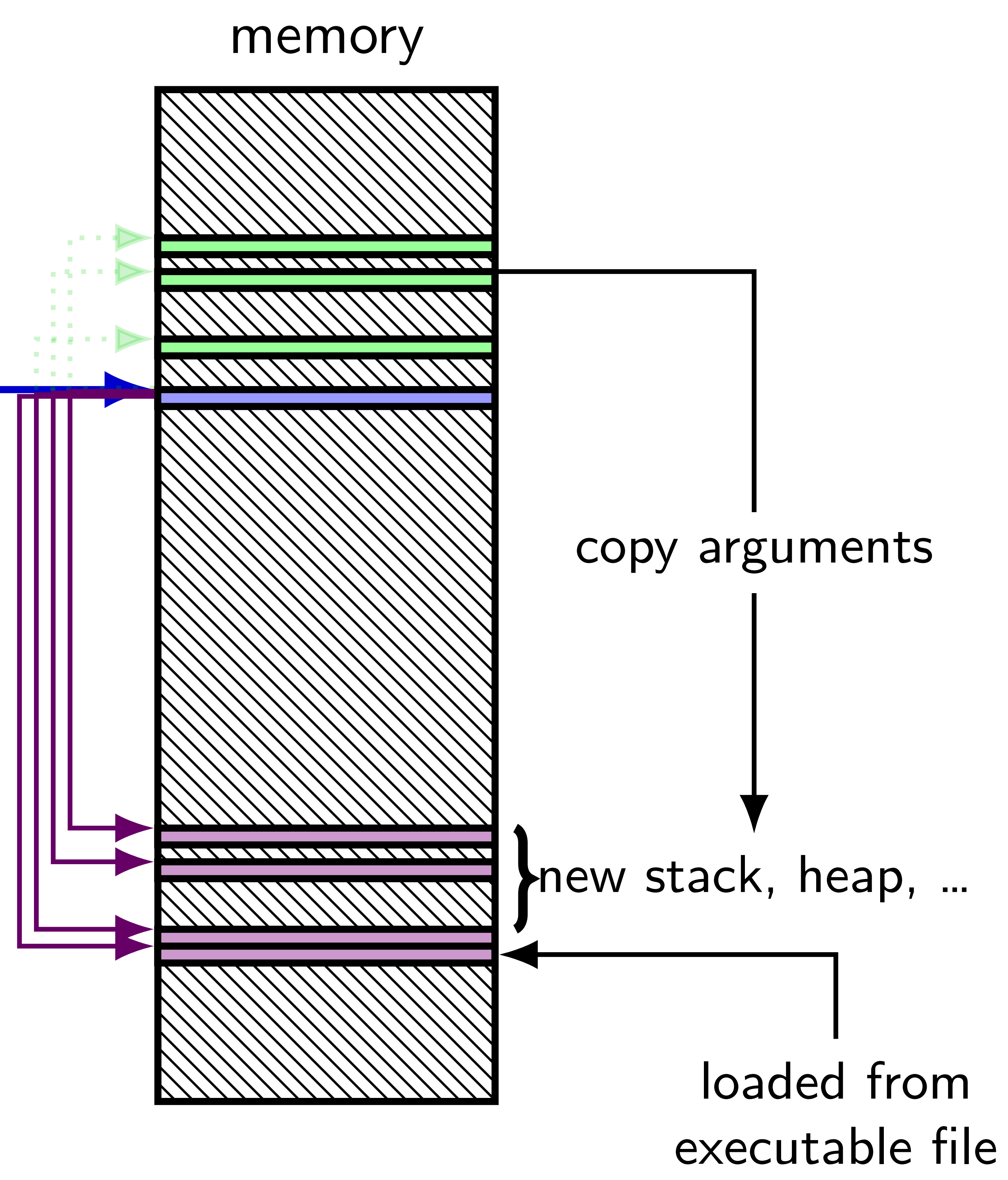


exec in the kernel

the process control block

user regs	eax=~~42~~*init. val.*, ecx=~~133~~*init. val.*, …
memory mapping	
open files	`fd 0: (terminal …)` `fd 1: …`
…	…

files + some other things not changed!
(more on this later)



exec in the kernel

the process control block

user regs	eax=~~42~~*init. val.*, ecx=~~133~~*init. val.*, …
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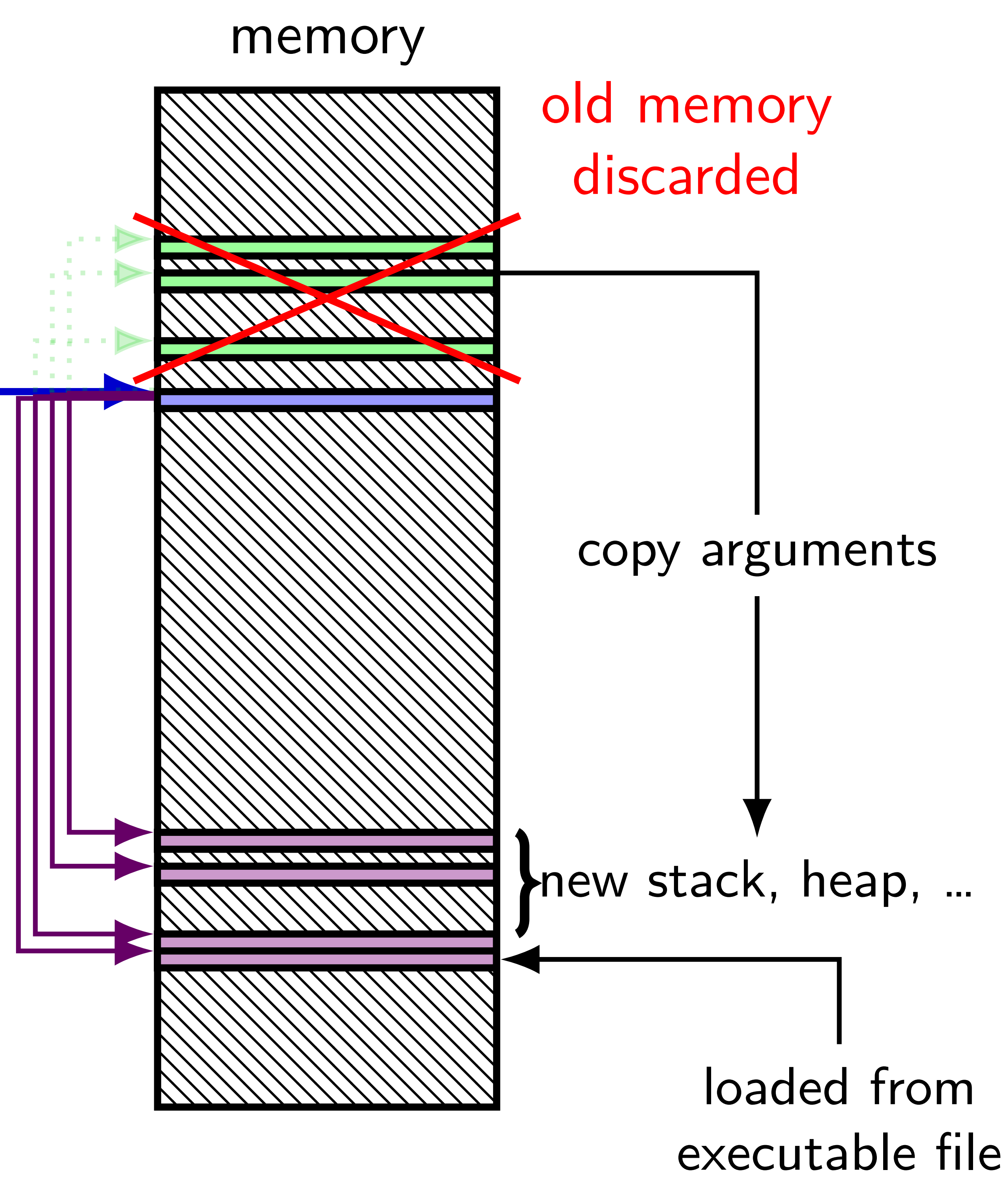
open files	fd 0: (terminal …) fd 1: …
…	…



files + some other things not changed!
(more on this later)

why fork/exec?

could just have a function to spawn a new program

- Windows `CreateProcess()`; POSIX's (rarely used) `posix_spawn`

some other OSs do this (e.g. Windows)

needs to include API to set new program's state

- e.g. without fork: either:
- need function to set new program's current directory, *or*
- need to change your directory, then start program, then change back
- e.g. with fork: just change your current directory before exec

but allows OS to avoid 'copy everything' code

- probably makes OS implementation easier

posix_spawn example

```
pid_t new_pid;
const char argv[] = { "ls", "-l", NULL };
int error_code = posix_spawn(
    &new_pid,
    "/bin/ls",
    NULL /* null = copy current process's open files;
            if not null, do something else */,
    NULL /* null = no special settings for new process */,
    argv,
    NULL /* null = copy current "environment variables",
            if not null, do something else */
);
if (error_code == 0) {
    /* handle error */
}
```

some opinions (via HotOS '19)

A fork() in the road

Andrew Baumann
Microsoft Research

Jonathan Appavoo
Boston University

Orran Krieger
Boston University

Timothy Roscoe
ETH Zurich

ABSTRACT

The received wisdom suggests that Unix's unusual combination of `fork()` and `exec()` for process creation was an inspired design. In this paper, we argue that fork was a clever hack for machines and programs of the 1970s that has long outlived its usefulness and is now a liability. We catalog the ways in which fork is a terrible abstraction for the modern programmer to use, describe how it compromises OS implementations, and propose alternatives.

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wait/waitpid

`pid_t waitpid(pid_t pid, int *status, int options)`

wait for a child process (with pid=`pid`) to finish

sets `*status` to its “status information”

`pid=-1` → wait for *any* child process instead

- `pid=0` almost the same

options? see manual page (command `man waitpid`)

- `0` — no options

waitpid example

```c
#include <sys/wait.h>
...

child_pid = fork();
if (child_pid > 0) {
    /* Parent process */
    int status;
    waitpid(child_pid, &status, 0);
} else if (child_pid == 0) {
    /* Child process */
    ...
    /* Eventually exit */
    ...
```

exit statuses

```
int main() {
    return 0;  /* or exit(0); */
}
```

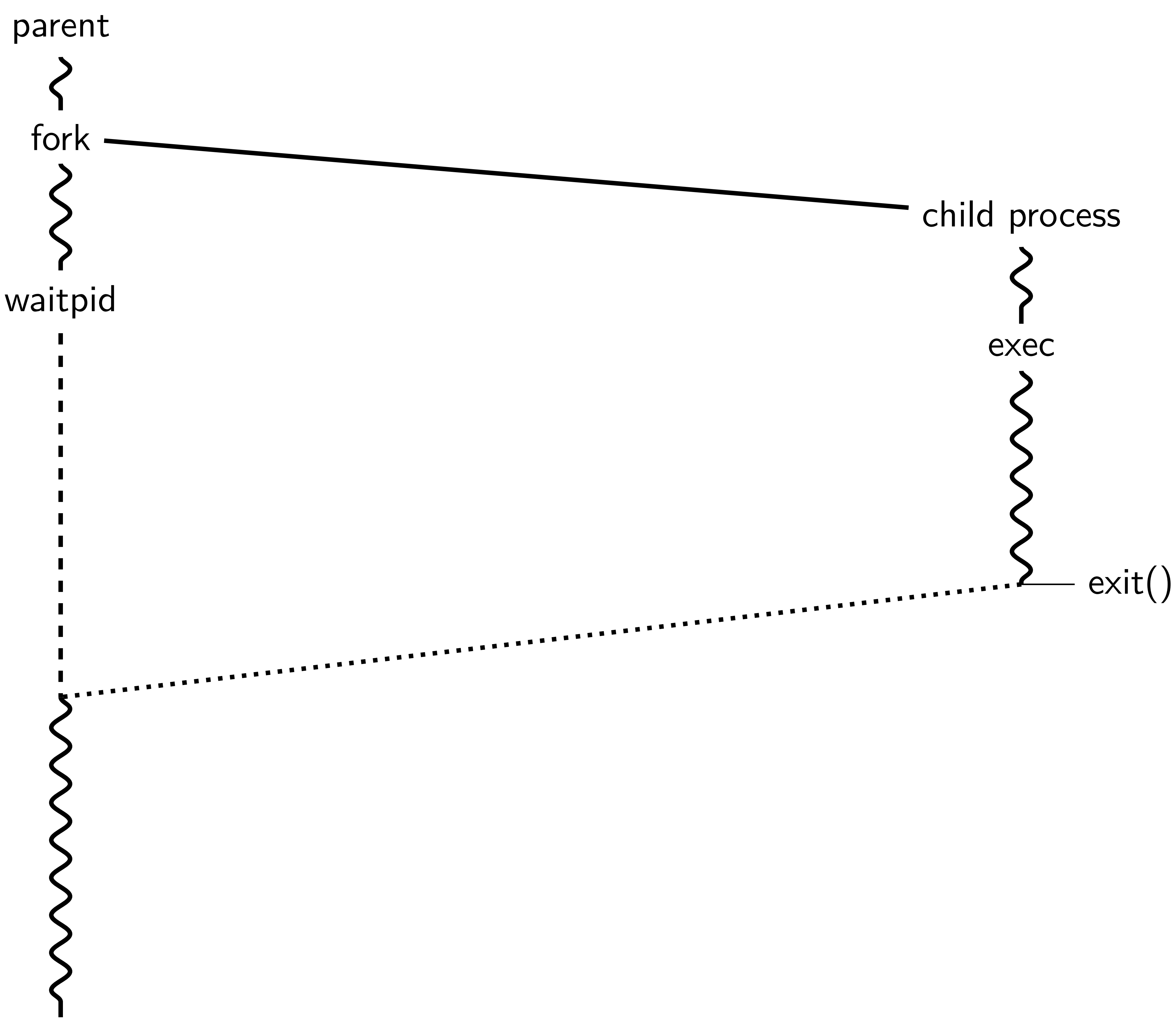
the status

```
#include <sys/wait.h>
...
waitpid(child_pid, &status, 0);
if (WIFEXITED(status)) {
  printf("main returned or exit called with %d\n", WEXITSTATUS(status));
} else if (WIFSIGNALED(status)) {
  printf("killed by signal %d\n", WTERMSIG(status));
} else {
    ...
}
```

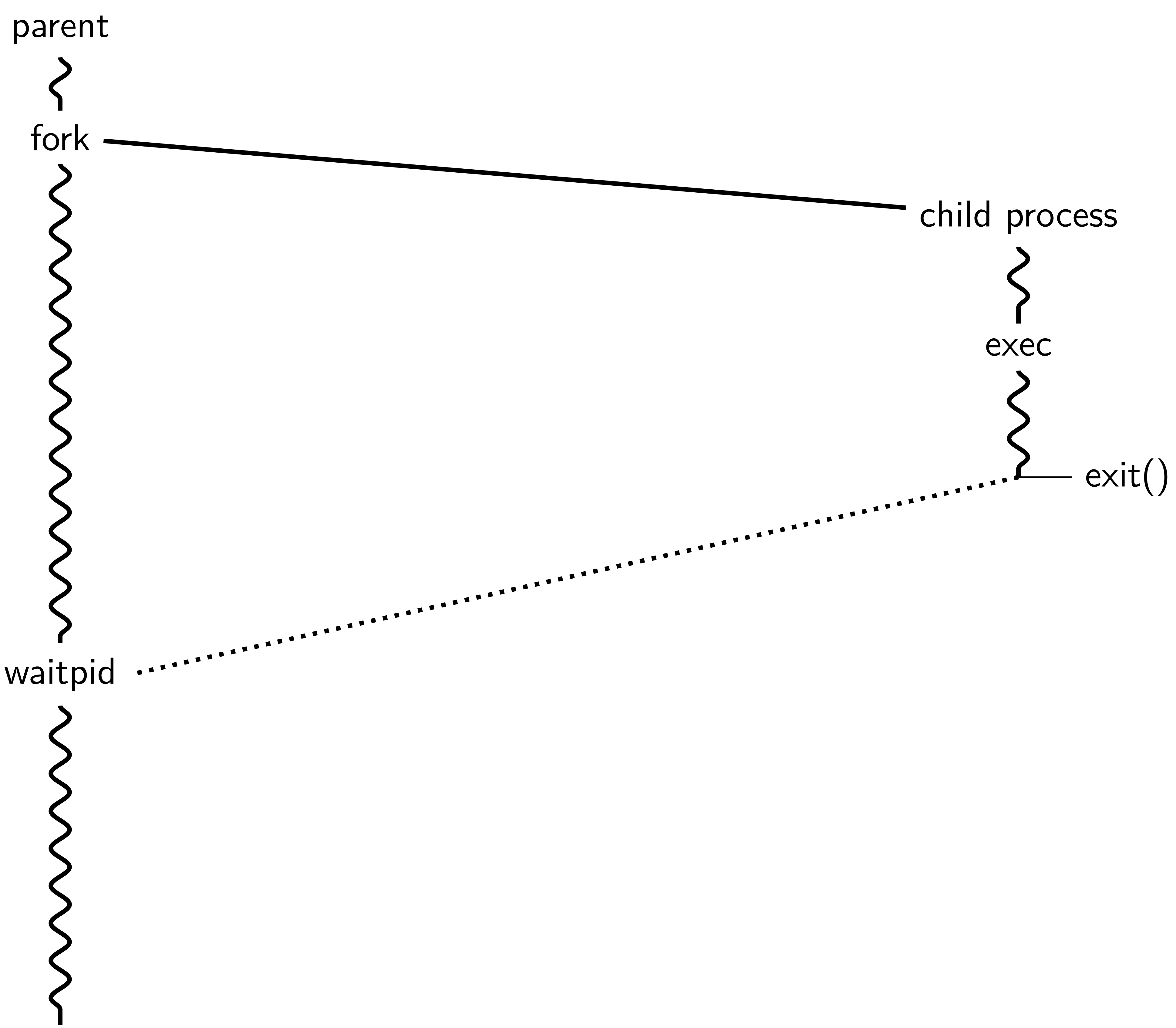
"status code" encodes *both return value and if exit was abnormal*

`W*` macros to decode it

typical pattern



typical pattern (alt)



typical pattern (detail)

```
pid = fork();
if (pid == 0) {
    exec…(…);
    …
} else if (pid > 0) {
    waitpid(pid,…);
    …
}
…
```

```
pid = fork();
if (pid == 0) {
    exec…(…);
    …
} else if (pid > 0) {
    waitpid(pid,…);
    …
}
…
```

```
pid = fork();
if (pid == 0) {
    exec…(…);
    …
} else if (pid > 0) {
    waitpid(pid,…);
    …
}
…
```

```
main() {
    …
}
```

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exercise (1)

```
int main() {
    pid_t pids[2]; const char *args[] = {"echo", "ARG", NULL};
    const char *extra[] = {"L1", "L2"};
    for (int i = 0; i < 2; ++i) {
        pids[i] = fork();
        if (pids[i] == 0) {
            args[1] = extra[i];
            execv("/bin/echo", args);
        }
    }
    for (int i = 0; i < 2; ++i) {
        waitpid(pids[i], NULL, 0);
    }
}
```

Assuming fork and execv do not fail, which are possible outputs?

A. L1 (newline) L2
B. L1 (newline) L2 (newline) L2
C. L2 (newline) L1
D. A and B
E. A and C
F. all of the above
G. something else

exercise (2)

```
int main() {
    pid_t pids[2]; const char *args[] = {"echo", "0", NULL};
    for (int i = 0; i < 2; ++i) {
        pids[i] = fork();
        if (pids[i] == 0) { execv("/bin/echo", args); }
    }
    printf("1\n"); fflush(stdout);
    for (int i = 0; i < 2; ++i) {
        waitpid(pids[i], NULL, 0);
    }
    printf("2\n"); fflush(stdout);
}
```

Assuming fork and execv do not fail, which are possible outputs?

A. 0 (newline) 0 (newline) 1 (newline) 2
B. 0 (newline) 1 (newline) 0 (newline) 2
C. 1 (newline) 0 (newline) 0 (newline) 2
D. 1 (newline) 0 (newline) 2 (newline) 0
E. A, B, and C
F. C and D
G. all of the above
H. something else

warning re: forking in loops

```
while (true) {
    fork();
}
```

what’s going to happen?

- lots of CPU being used
- lots of memory being allocated
- control-C might only terminate one of the processes

probably very bad and hard to recover from

some recommendations re: timing

in timing assignment, you'll need to run `fork()` multiple times to measure it

I recommend:

- always `waitpid` *successfully* before `fork()`ing a second time
- test `fork()`ing exactly once before doing a loop

some POSIX command-line features

searching for programs

`ls -l` → `/bin/ls -l`

`make` → `/usr/bin/make`

running in background

```
./someprogram &
```

redirection

```
./someprogram >output.txt
./someprogram <input.txt
```

pipelines:

```
./someprogram | ./somefilter
```

file descriptors

```
struct process_info {  /* <-- in the kernel somewhere */
    ...
    struct open_file_description *files[SIZE];
    ...
};
...
process->files[file_descriptor]
```

Unix: every process has an array (or similar) of *open file descriptions*

“open file”: terminal, socket, regular file, pipe

file descriptor = index into array

- usually what’s used with system calls
- stdio.h FILE*s usually have file descriptor + buffer

special file descriptors

file descriptor 0 = standard input

file descriptor 1 = standard output

file descriptor 2 = standard error

constants in `unistd.h`

- `STDIN_FILENO`, `STDOUT_FILENO`, `STDERR_FILENO`

but you can't choose which number `open` assigns...?

- more on this later

getting file descriptors

```
int read_fd  = open("dir/file1", O_RDONLY);
int write_fd = open("/other/file2", O_WRONLY | ...);
int rdwr_fd  = open("file3", O_RDWR);
```

used internally by `fopen()`, etc.

also for files without normal filenames…:

```
int fd        = shm_open("/shared_memory", O_RDWR, 0666); // shared memory
int socket_fd = socket(AF_INET, SOCK_STREAM, 0);          // TCP socket
int term_fd   = posix_openpt(O_RDWR); // pseudo-terminal
int pipe_fds[2]; pipe(pipefds);       // "pipes" (implements ./command | ./other_cmd)
...
```

open: flags

```
int open(const char *path, int flags);
int open(const char *path, int flags, int mode);
```

flags: bitwise or of:

- `O_RDWR`, `O_RDONLY`, or `O_WRONLY`
  - read/write, read-only, write-only
- `O_APPEND`
  - append to end of file
- `O_TRUNC`
  - truncate (set length to 0) file if it already exists
- `O_CREAT`
  - create a new file if one doesn't exist
  - (default: file must already exist)
- … and more

`man 2 open`

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- ... and more

`man 2 open`

open: mode

```
int open(const char *path, int flags);
int open(const char *path, int flags, int mode);
```

mode: permissions of newly created file

- like numbers provided to `chmod` command
- filtered by a “umask”

simple advice: always use `0666`

- = readable/writeable by everyone, except where umask prohibits
- (typical umask: prohibit other/group writing)

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- = readable/writeable by everyone, except where umask prohibits
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close

```
int close(int fd);
```

close the file descriptor, deallocating that array index

- does not affect other file descriptors that refer to same “open file description”
- (e.g. in `fork()`ed child or created via (later) `dup2`)

if last file descriptor for open file description, resources deallocated

returns `0 on success

returns `-1` on error

- e.g. ran out of disk space while finishing saving file

shell redirection

`./my_program ... < input.txt`:

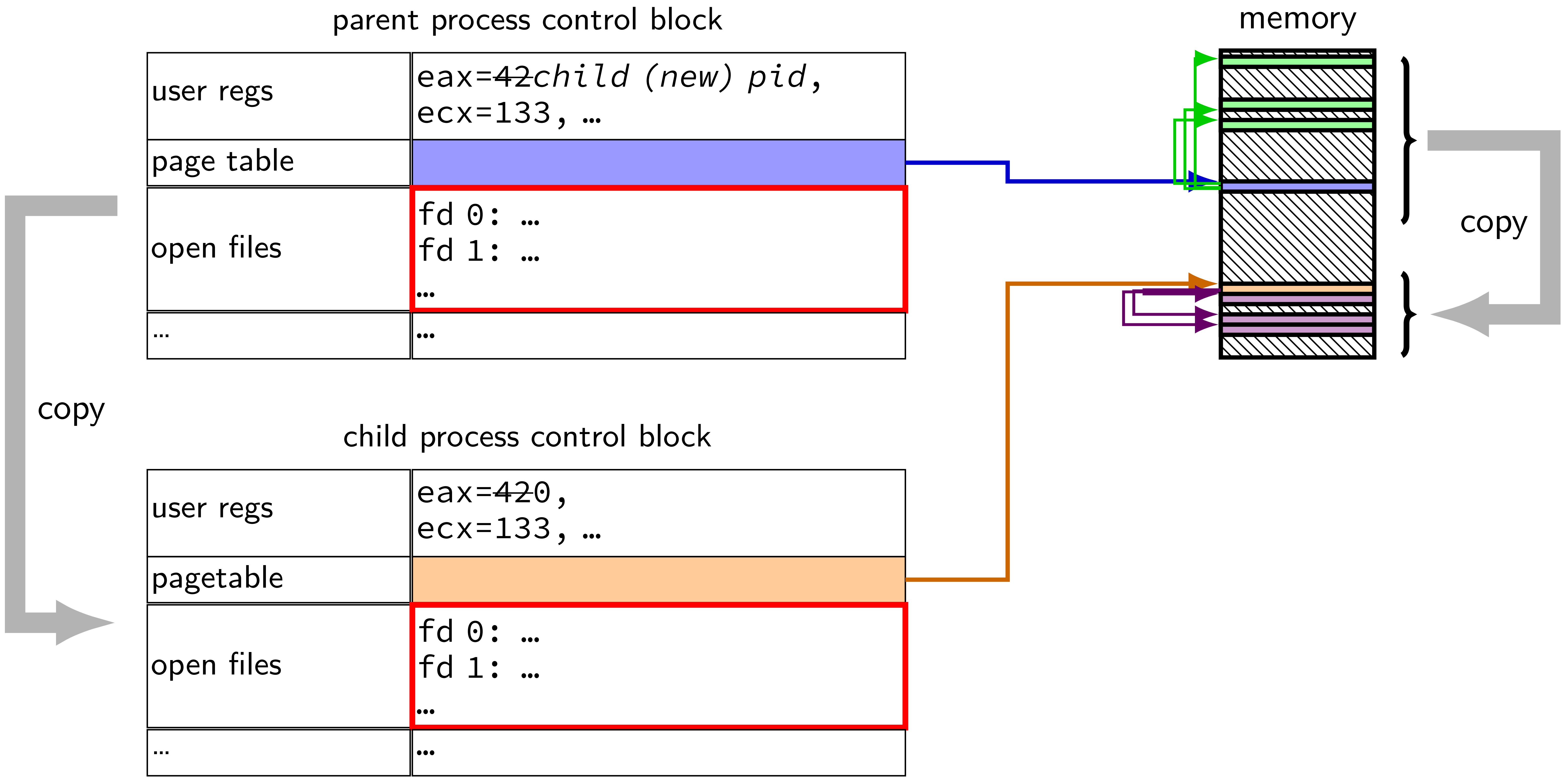
run `./my_program ...` but use `input.txt` as input
like we copied and pasted the file into the terminal

`echo foo > output.txt`:

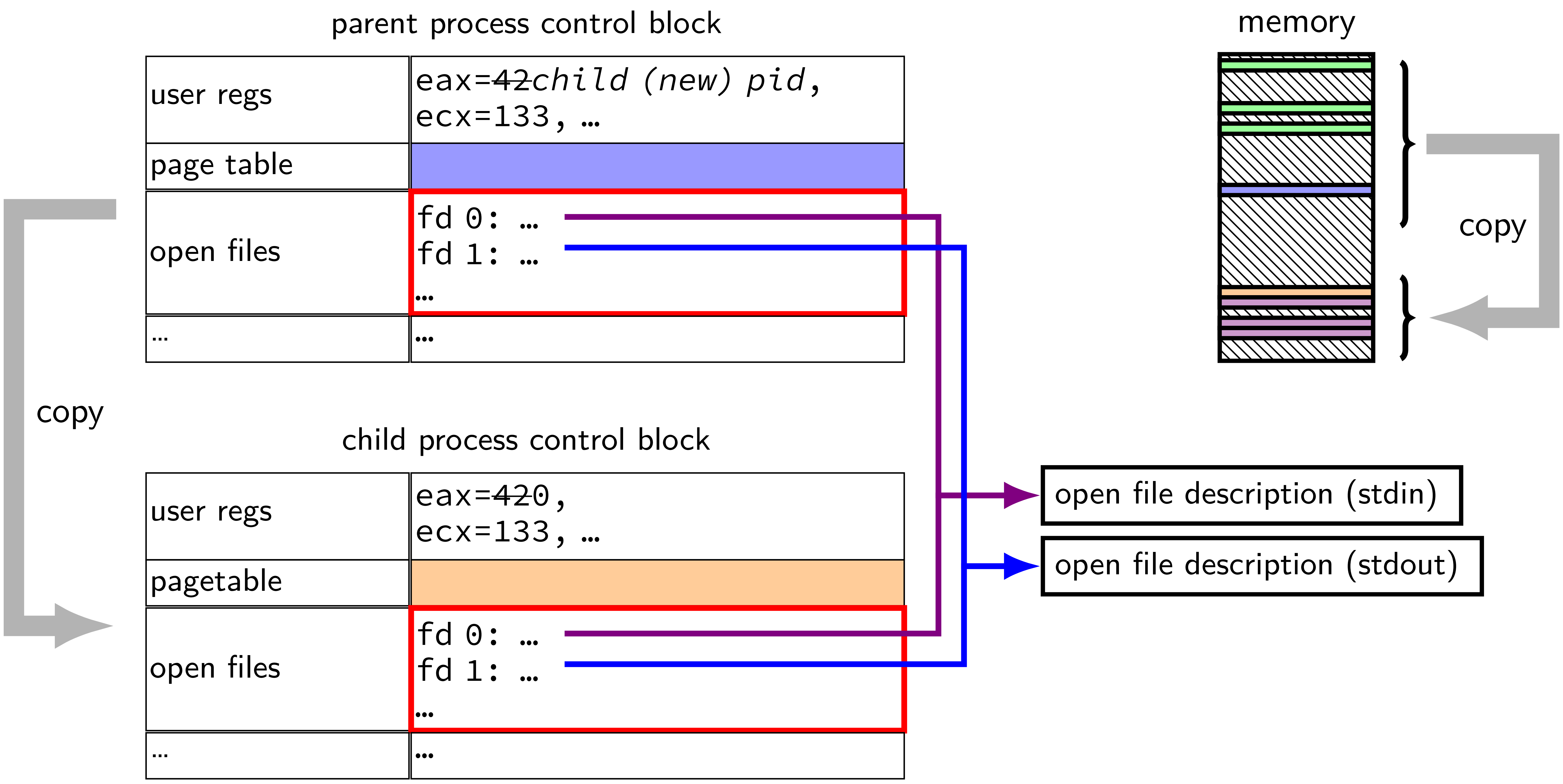
runs `echo foo`, sends output to `output.txt`
like we copied and pasted the output into that file
(as it was written)

fork copies file pointer list

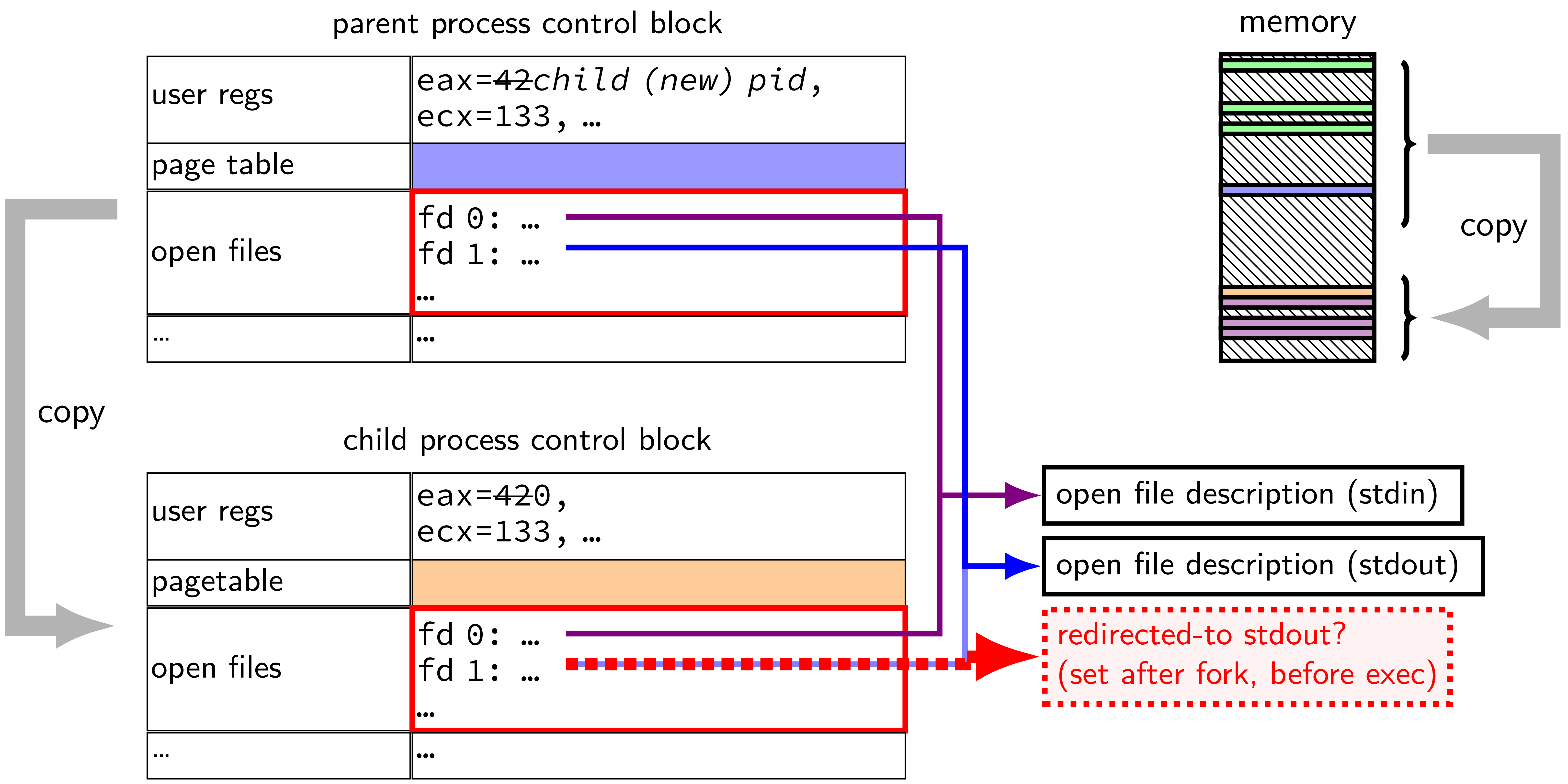
fork copies file pointer list



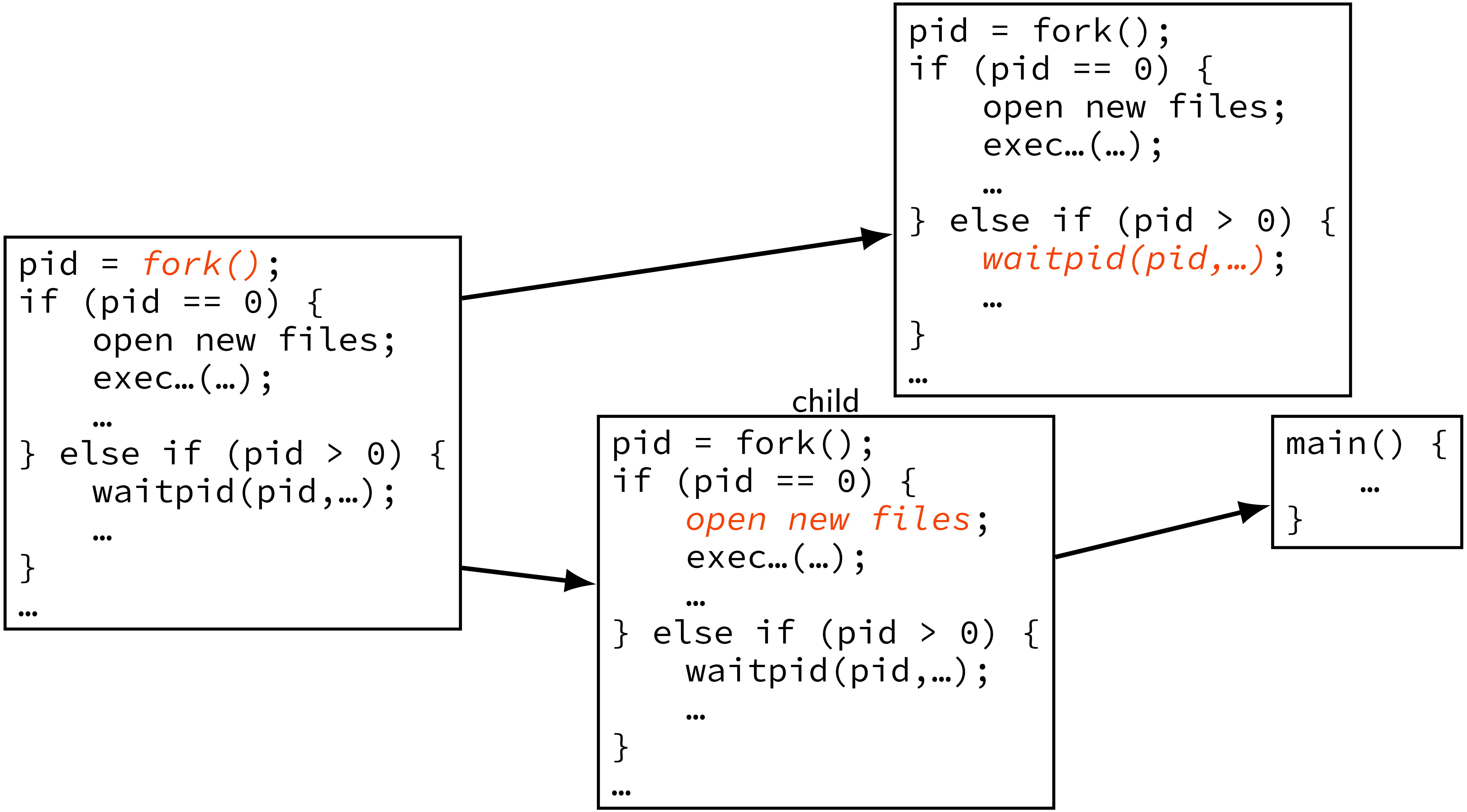
fork copies file pointer list



fork copies file pointer list



typical pattern with redirection



redirecting with exec

standard output/error/input are files

(C stdout/stderr/stdin; C++ cout/cerr/cin)

1. open files to redirect (probably after forking)
2. make them be standard output/error/input

   using `dup2()` library call
3. then exec, preserving new standard output/etc.

exec preserves open files

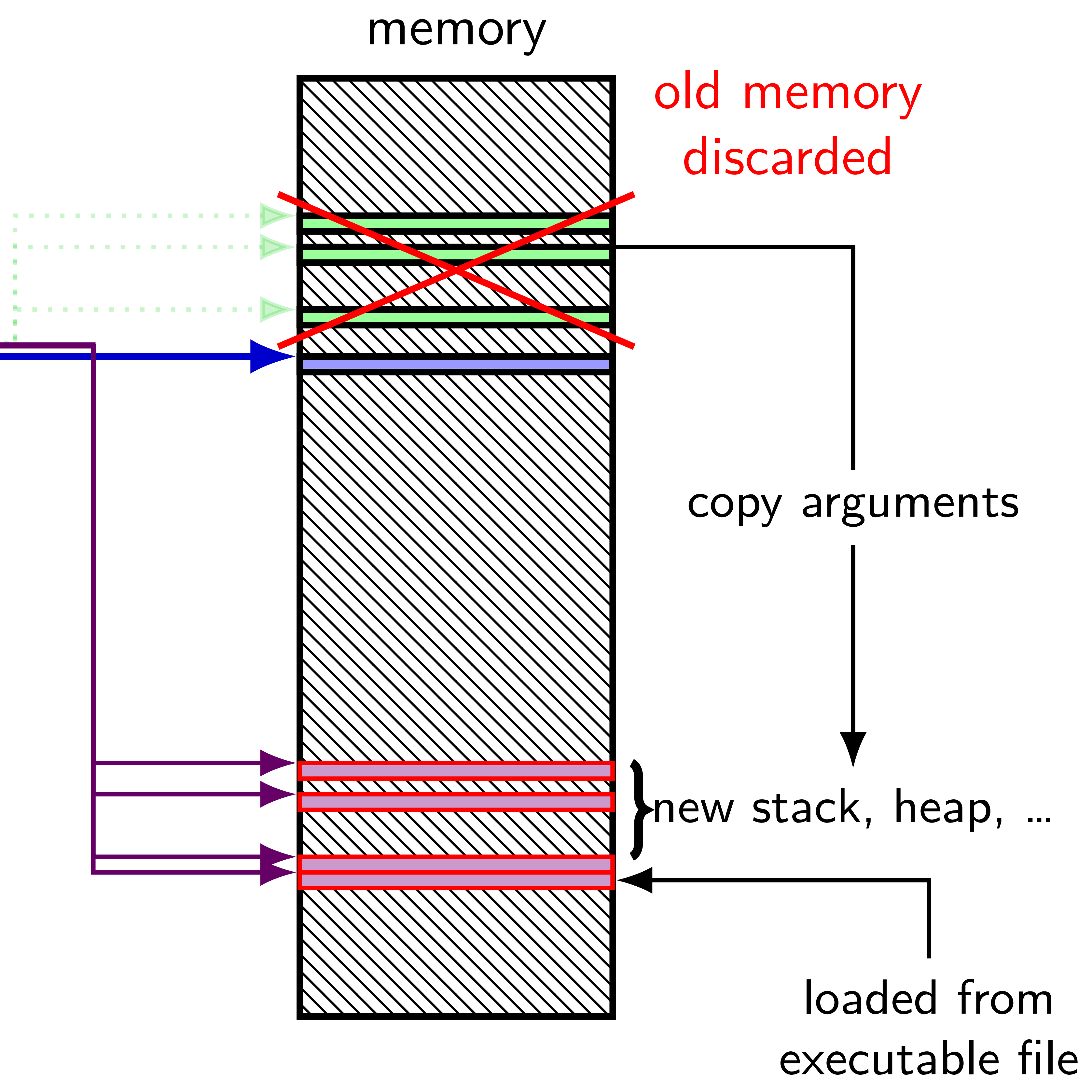
the process control block

user regs	eax=~~42~~*init. val.*, ecx=~~133~~*init. val.*, …
pagetable	
open files	`fd 0: (terminal …)` `fd 1: …`
…	…

not changed!

redirection/etc.:

setup stdin/stdout before exec



reassigning file descriptors

redirection: `./program >output.txt`

step 1: open output.txt for writing, get new file descriptor

step 2: make that new file descriptor stdout (number 1)

tool: `int dup2(int oldfd, int newfd)`

make `newfd` refer to same open file as `oldfd`

- same *open file description*
- shares the current location in the file
- (even after more reads/writes)

what if `newfd` already allocated — closed, then reused

reassigning and file table

```
// something like this in OS code
struct process_info {
    ...
    struct open_file_description *files[SIZE];
    ....
};
...
process->files[STDOUT_FILENO] = process->files[opened_fd];
```

syscall: `dup2(opened_fd, STDOUT_FILENO);`

dup2 example

redirects stdout to output to `output.txt`:

```
fflush(stdout);  /* clear printf's buffer */
int fd = open("output.txt",
              O_WRONLY | O_CREAT | O_TRUNC);
if (fd < 0)
    do_something_about_error();

dup2(fd, STDOUT_FILENO);
/* now both write(fd, ...) and write(STDOUT_FILENO, ...)
   write to output.txt
   */

close(fd); /* only close original, copy still works! */

printf("This will be sent to output.txt.\n");
```

shell output redirection example

Run `/bin/echo foo > output.txt`:

```c
char filename[] = "output.txt";
pid_t pid = fork();
if (pid == 0) {
    int fd = open(filename, O_WRONLY | O_CREAT | O_TRUNC);
    char* args[] = {"echo", "foo"};
    dup2(fd, STDOUT_FILENO);
    close(fd);
    execv("/bin/echo", args);
} else {
    waitpid(pid, NULL, 0);
}
```

shell input redirection example

Run `/bin/cat < input.txt`:

```
char filename[] = "input.txt";
pid_t pid = fork();
if (pid == 0) {
    int fd = open(filename, O_RDONLY);
    char* args[] = {"cat"};
    dup2(fd, STDIN_FILENO);
    close(fd);
    execv("/bin/cat", args);
} else {
    waitpid(pid, NULL, 0);
}
```

open/dup/close/etc. and fd array

```
// something like this in OS code
struct process_info {
  ...
  struct open_file_description *files[NUM];
};
```

open: `files[new_fd] = ...;`

dup2(from, to): `files[to] = files[from];`

close: `files[fd] = NULL;`

fork:

```
  for (int i = ...)
      child->files[i] = parent->files[i];
```

(plus extra work to avoid leaking memory)

dup2 exercise

recall: `dup2(old_fd, new_fd)`

```
int fd = open("output.txt", O_WRONLY | O_CREAT, 0666);
write(STDOUT_FILENO, "A", 1);
dup2(fd, STDOUT_FILENO);
pid_t pid = fork();
if (pid == 0) { /* child: */
    dup2(STDOUT_FILENO, fd);
    write(fd, "B", 1);
} else {
    write(STDOUT_FILENO, "C", 1);
}
```

Which outputs are possible?

A. stdout: ABC ; output.txt: empty

B. stdout: AC ; output.txt: B

C. stdout: A ; output.txt: CB

D. stdout: A ; output.txt: BC

E. more?

unshared seek pointers

if `foo.txt` contains “AB”:

```
int fd1 = open("foo.txt", O_RDONLY);
int fd2 = open("foo.txt", O_RDONLY);
char c;
read(fd1, &c, 1);
char d;
read(fd2, &d, 1);
```

expected result: c = ‘A’, d = ‘A’

shared seek pointers (1)

if `foo.txt` contains “AB”:

```
int fd = open("foo.txt", O_RDONLY);
dup2(fd, 100);
char c;
read(fd, &c, 1);
char d;
read(100, &d, 1);
```

expected result: c = ‘A’, d = ‘B’

shared seek pointers (2)

if `foo.txt` contains “AB”:

```
int fd = open("foo.txt", O_RDONLY);
pid_t p = fork();
if (p == 0) {
    char c;
    read(fd, &c, 1);
    ...
} else {
    char d;
    sleep(1);
    read(fd, &d, 1);
    ...
}
```

expected result: c = ‘A’, d = ‘B’

exercise

```
int fd = open("output.txt", O_WRONLY|O_CREAT|O_TRUNC, 0666);
write(fd, "A", 1);
dup2(STDOUT_FILENO, 100);
dup2(fd, STDOUT_FILENO);
write(STDOUT_FILENO, "B", 1);
write(fd, "C", 1);
close(fd);
write(STDOUT_FILENO, "D", 1);
write(100, "E", 1);
```

Assume `open()` and `dup2()` *do not fail*, `write()` does not fail as long as the fd it writes to is open, fd `100` was closed and is not what open returns, and `STDOUT_FILENO` is initially open. What is written to `output.txt`?

A. ABCDE
B. ABCD
C. ABC
D. ACD
E. something else

Unix API summary

spawn and wait for program: `fork` (copy), then

- in child: setup, then `execv`, etc. (replace copy)
- in parent: `waitpid`

files: open, read and/or write, close

- one interface for regular files, pipes, network, devices, …

file descriptors are indices into per-process array

- index 0, 1, 2 = stdin, stdout, stderr
- `dup2` — assign one index to another
- `close` — deallocate index

redirection

- open() to create new file descriptors
- dup2 in child to assign file descriptor to index 0, 1

fork exercise

What are two possible outputs of running this program?

```c
int main() {
    int x = 0;
    printf("begin\n"); fflush(stdout);
    pid_t p[2];
    for (int i = 0; i < 2; i += 1) {
        p[i] = fork();
        if (p[i] == 0) {
            x += 1;
            printf("loop x:%d\n", x); fflush(stdout);
            exit(0);
        } else {
            x += 5;
        }
    }
    for (int i = 0; i < 2; i += 1)
        waitpid(p[i], NULL, 0);
    printf("end x:%d\n", x); fflush(stdout);
}
```

Backup slides

shell

allow user (= person at keyboard) to run applications

user’s wrapper around process-management functions

aside: shell forms

POSIX: command line you have used before

also: graphical shells

  e.g. OS X Finder, Windows explorer

other types of command lines?

completely different interfaces?

searching for programs

POSIX convention: PATH *environment variable*

- example: `/home/cr4bd/bin:/usr/bin:/bin`
- list of directories to check in order

environment variables = key/value pairs stored with process

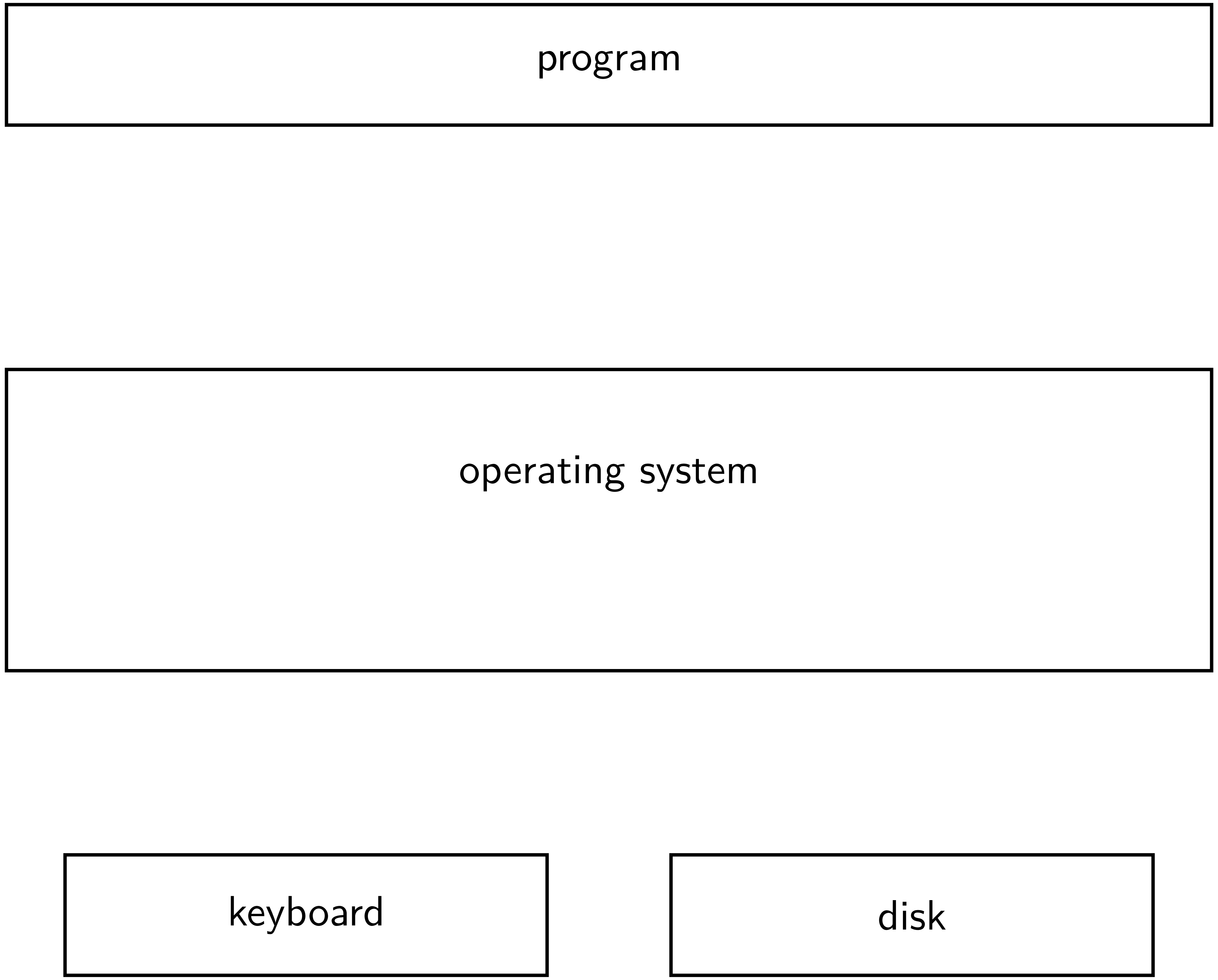
- by default, left unchanged on execve, fork, etc.

one way to implement: [pseudocode]

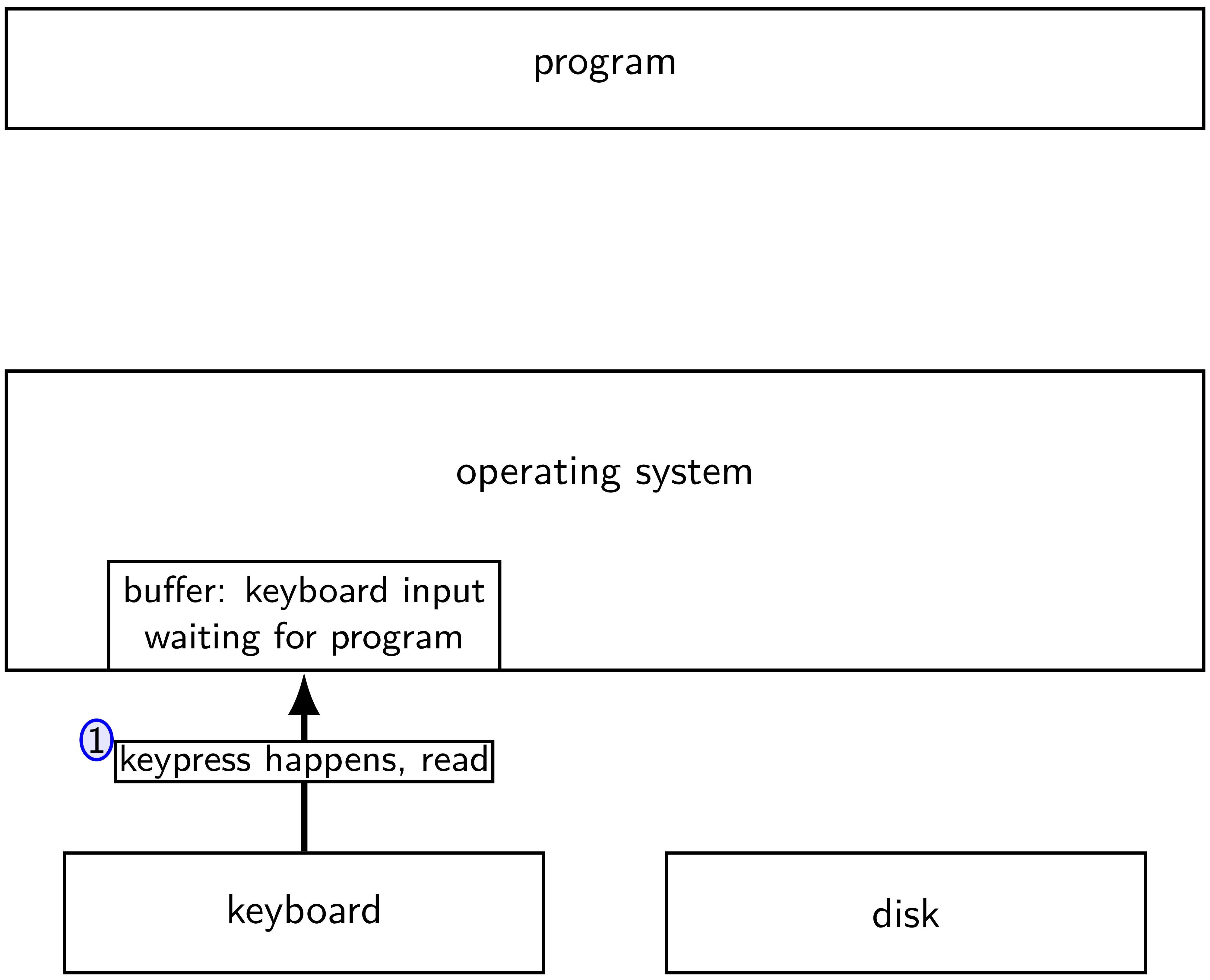
```
for (directory in path) {
    execv(directory + "/" + program_name, argv);
}
```

kernel buffering (reads)

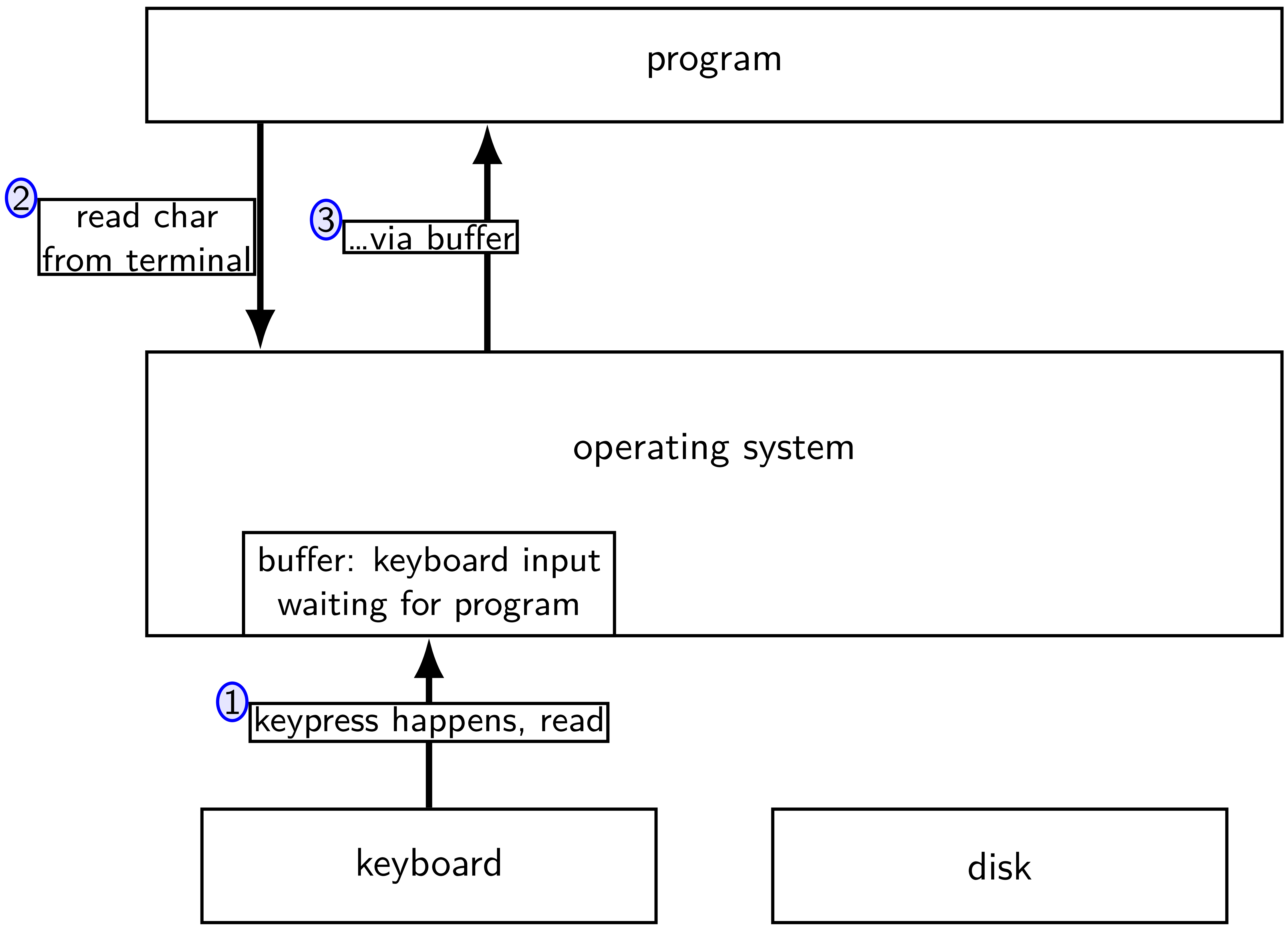
kernel buffering (reads)



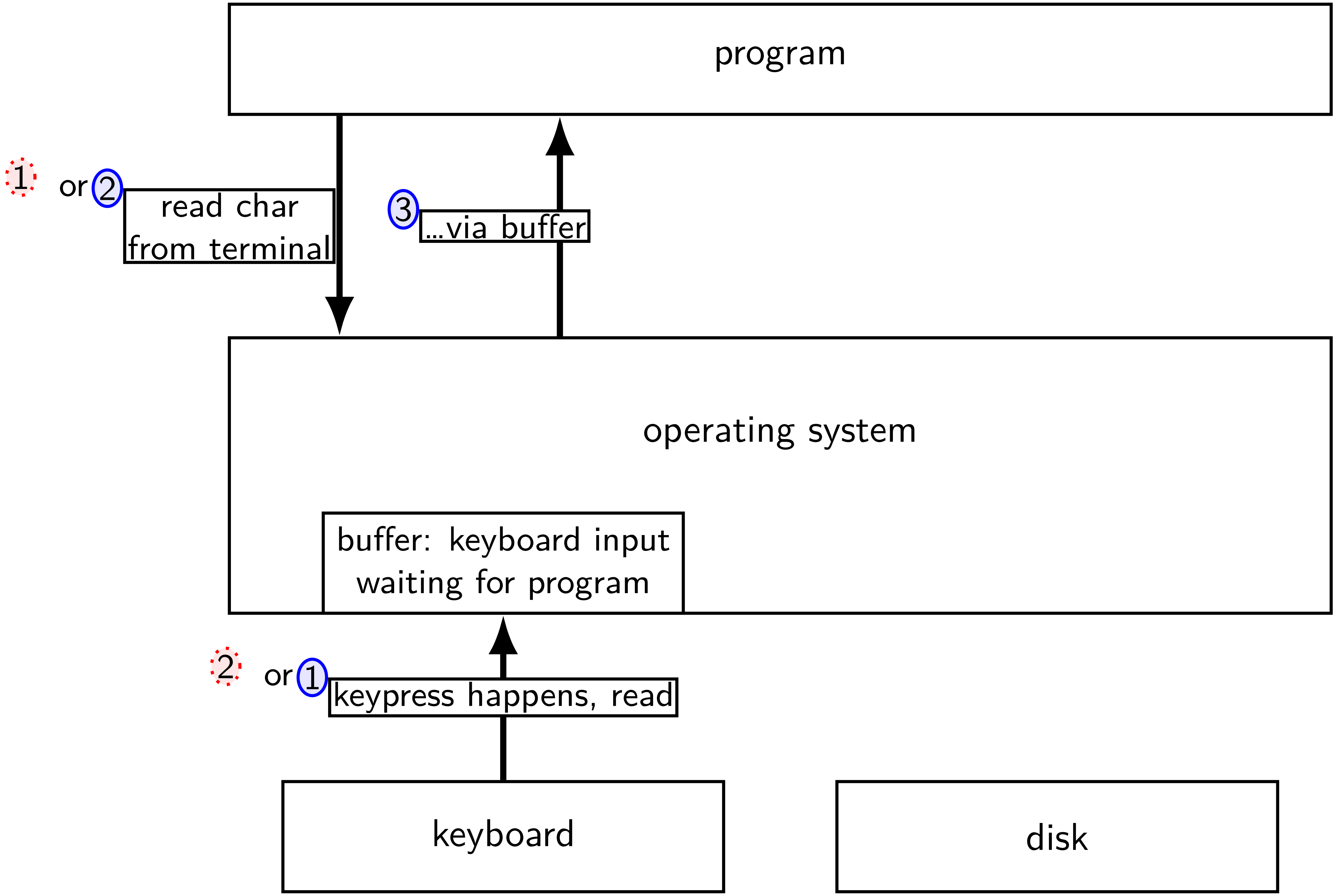
kernel buffering (reads)



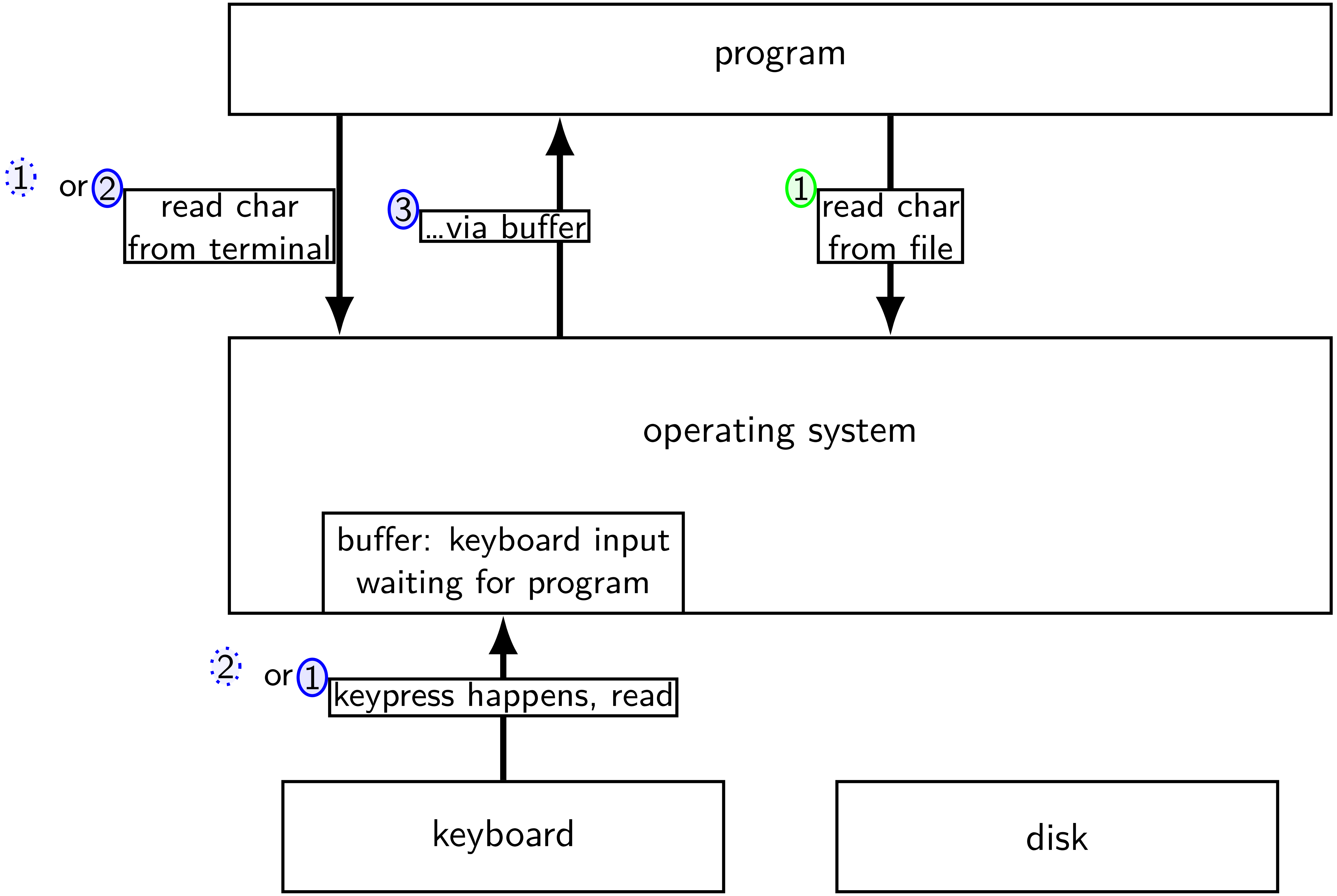
kernel buffering (reads)



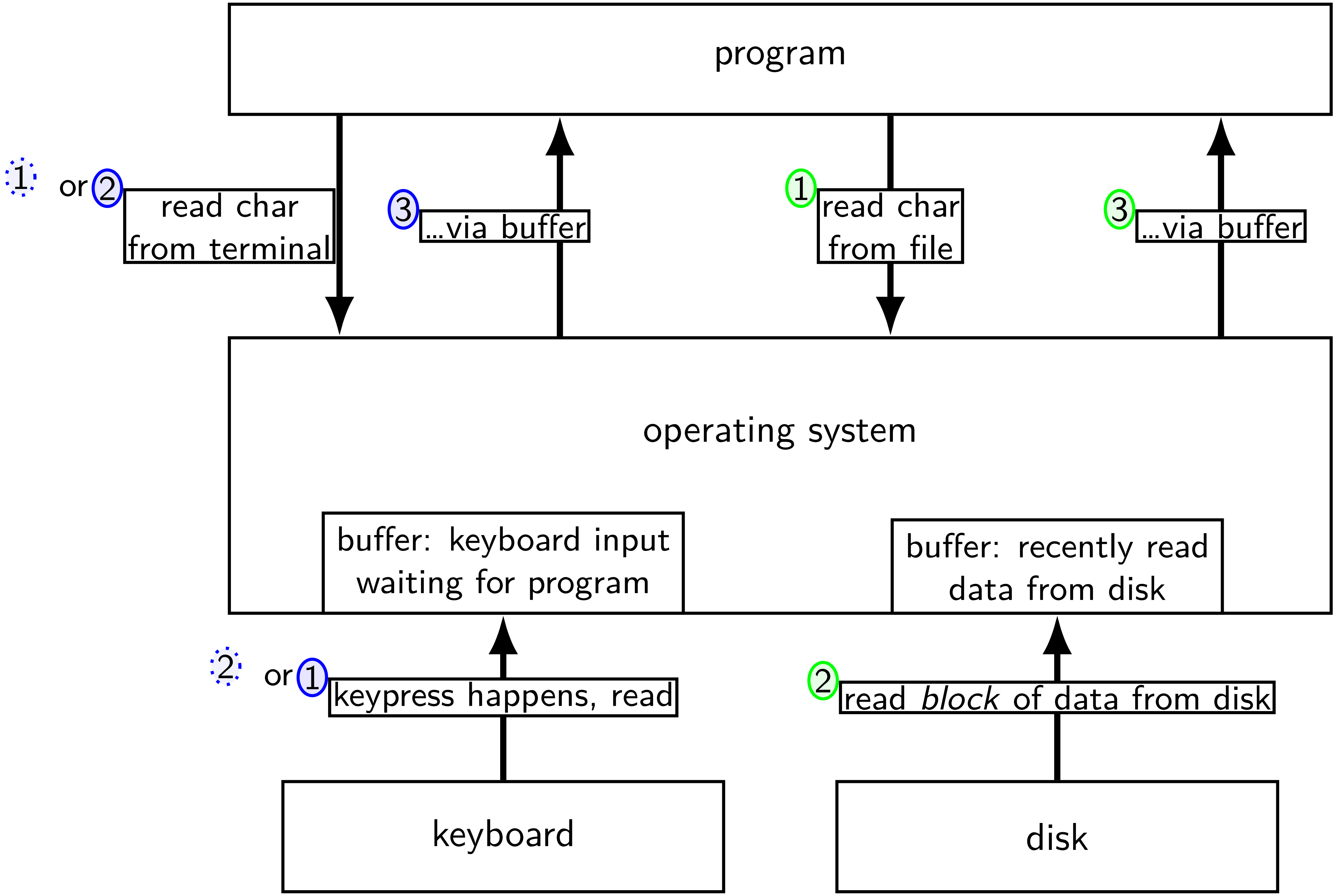
kernel buffering (reads)



kernel buffering (reads)

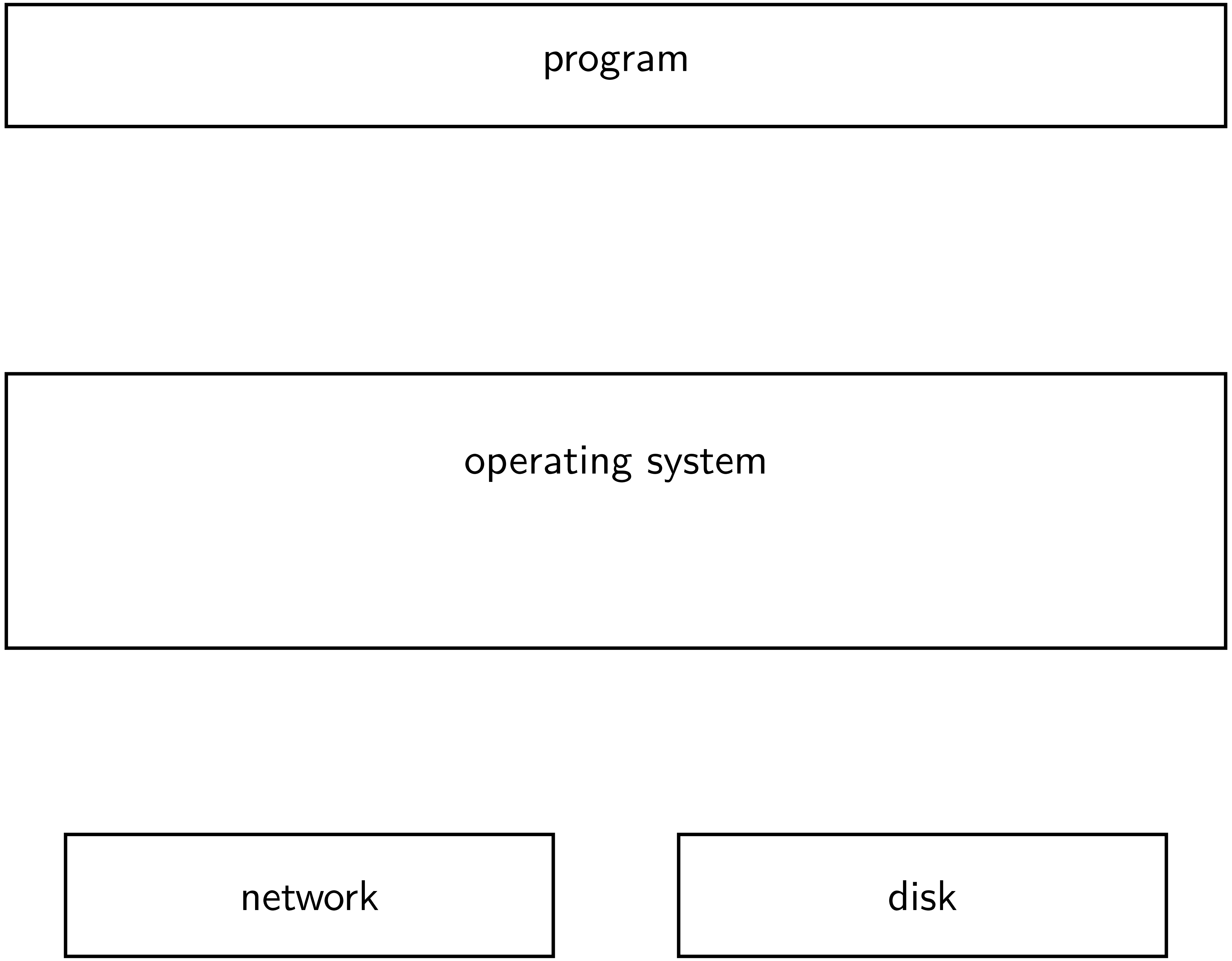


kernel buffering (reads)

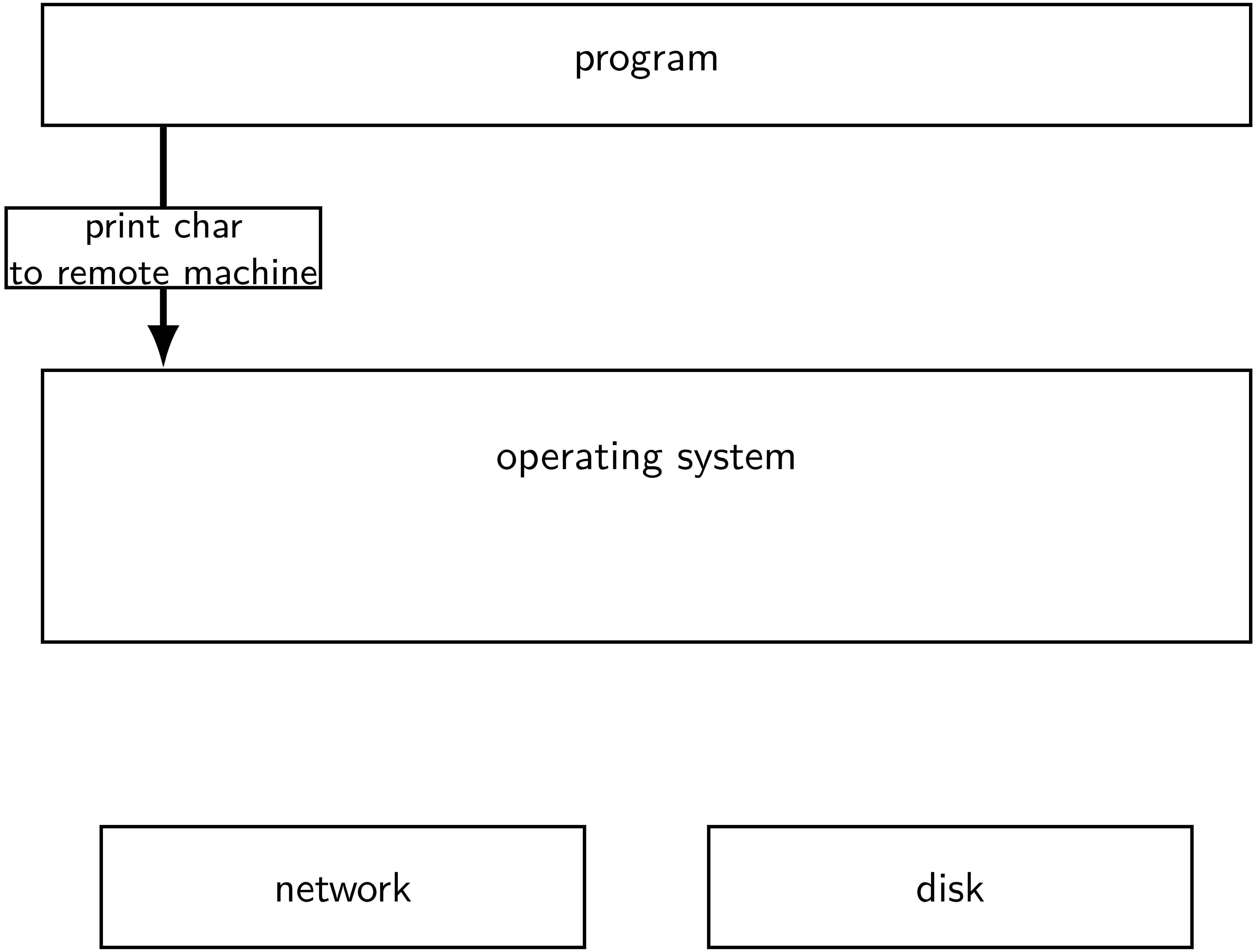


kernel buffering (writes)

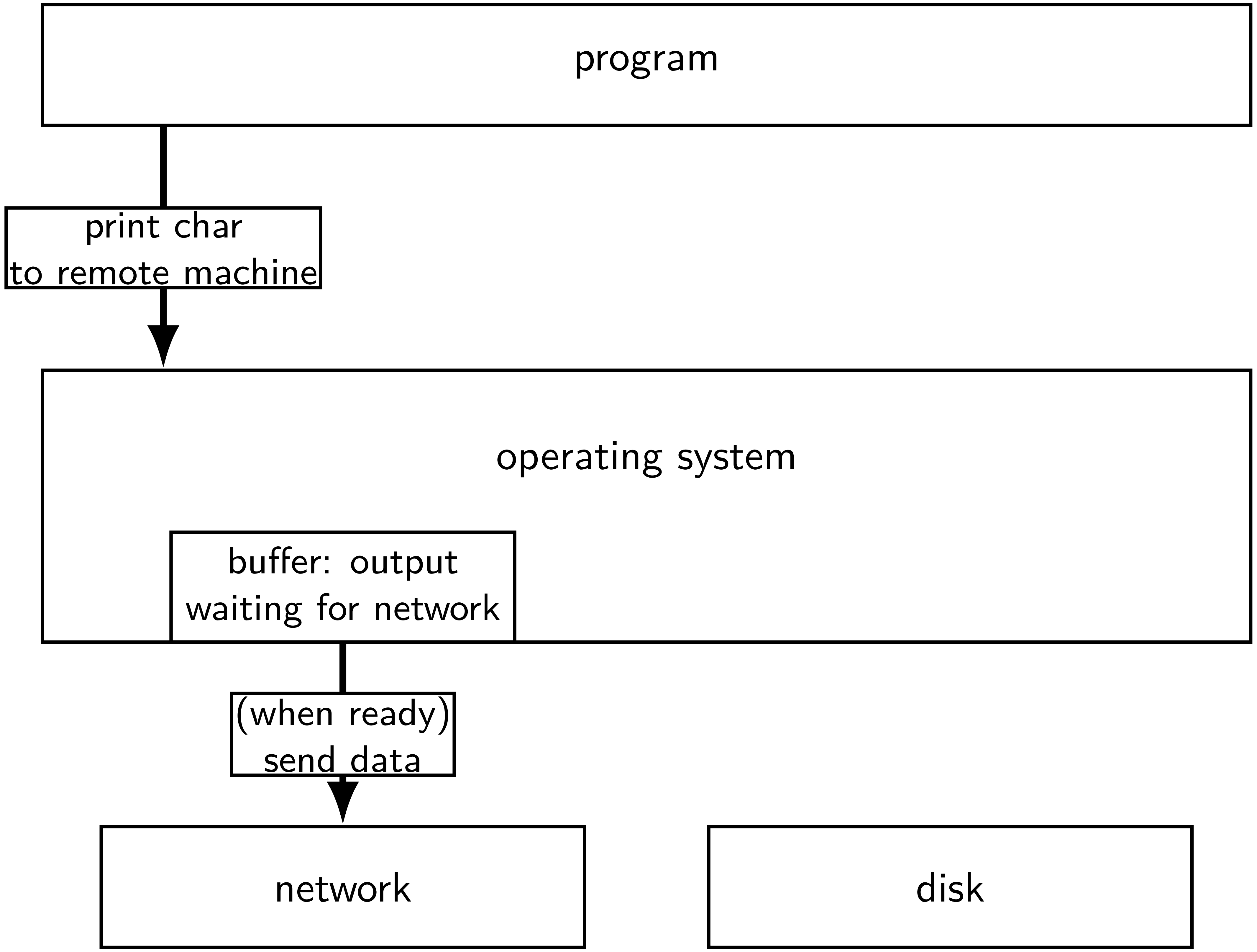
kernel buffering (writes)



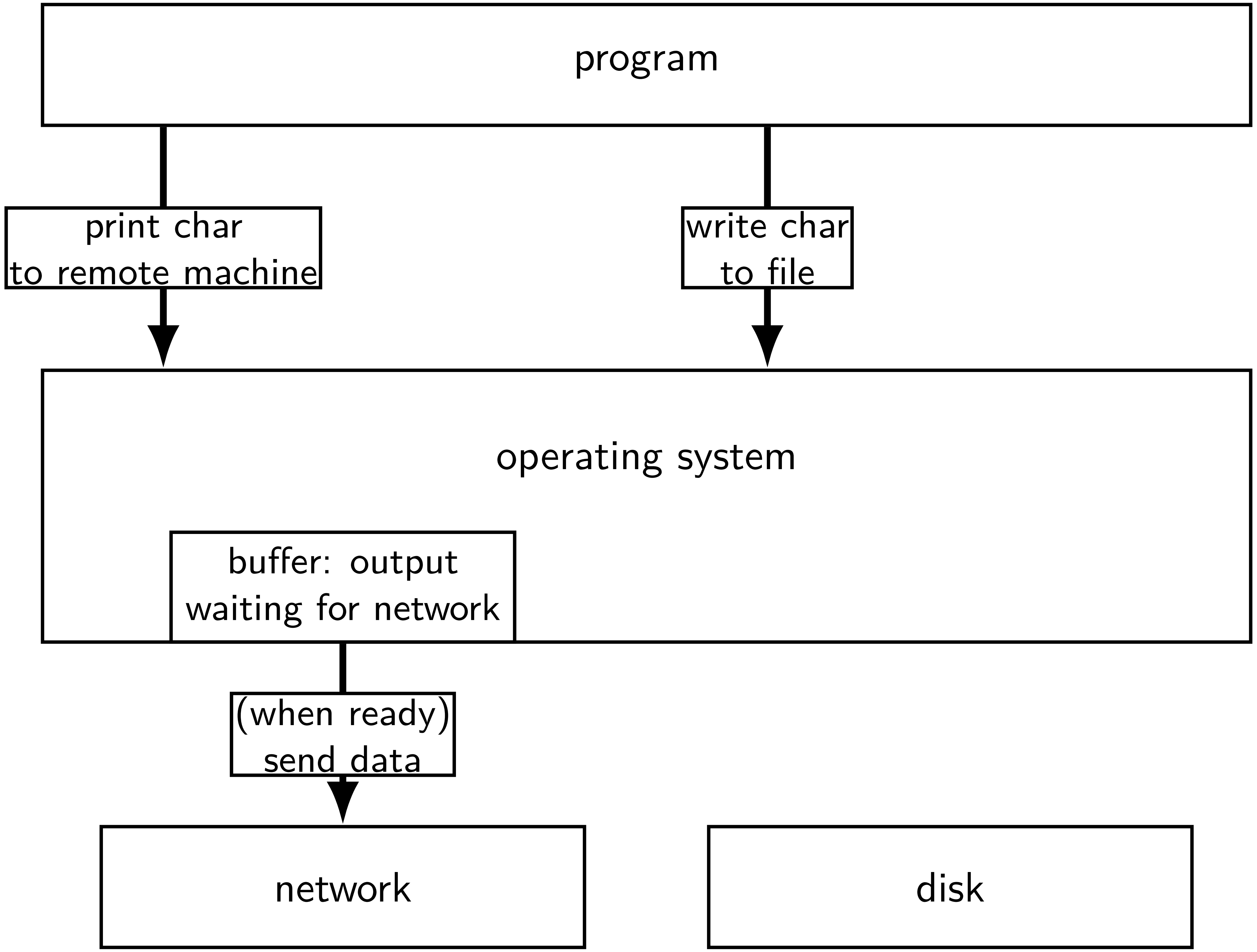
kernel buffering (writes)



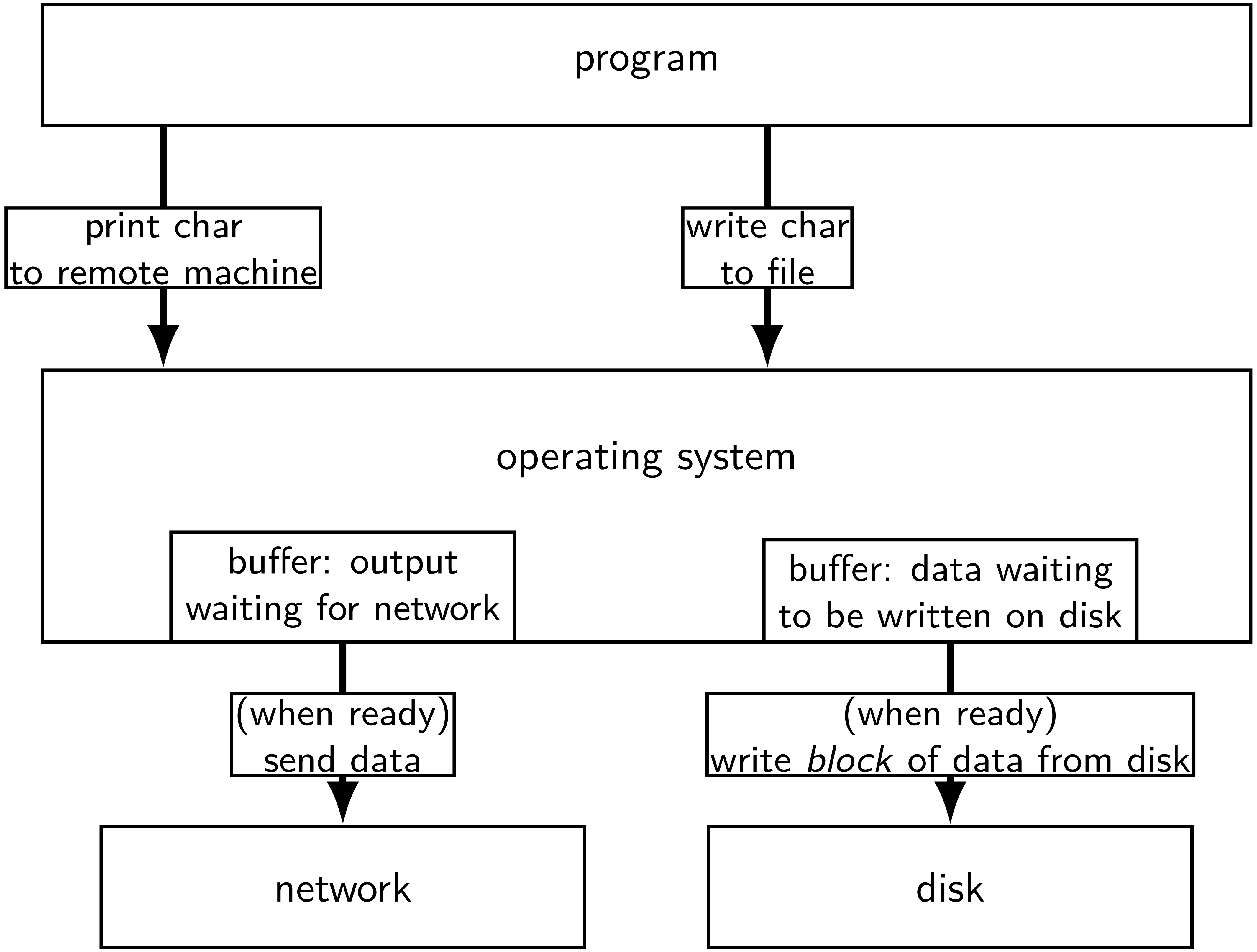
kernel buffering (writes)



kernel buffering (writes)



kernel buffering (writes)



read/write operations

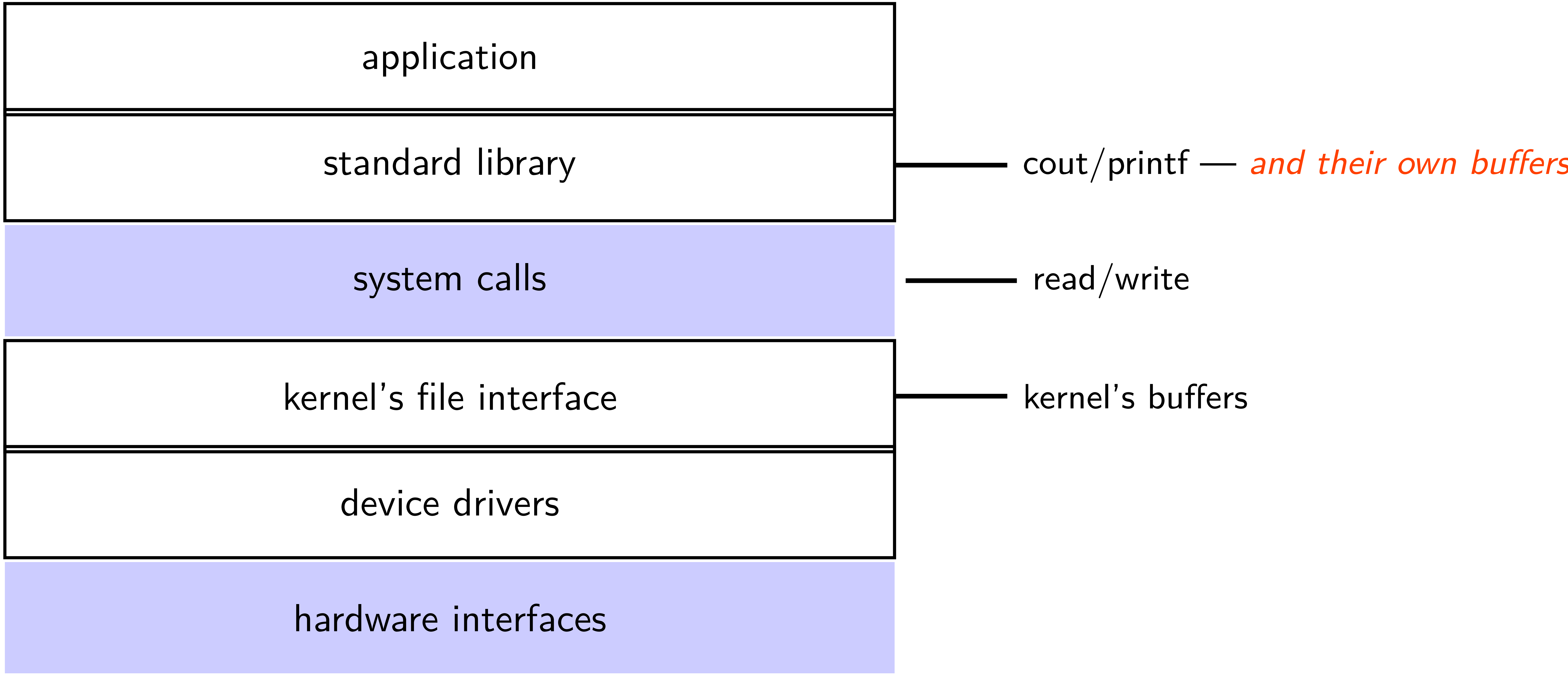
read()/write(): move data into/out of buffer

possibly wait if buffer is empty (read)/full (write)

actual I/O operations — wait for device to be ready

- trigger process to stop waiting if needed

layering


application
standard library
system calls
kernel's file interface
device drivers
hardware interfaces
cout/printf — *and their own buffers*
read/write
kernel's buffers

why the extra layer

better (but more complex to implement) interface:

- read line
- formatted input (scanf, cin into integer, etc.)
- formatted output

less system calls (bigger reads/writes) sometimes faster

- buffering can combine multiple in/out library calls into one system call

more portable interface

- cin, printf, etc. defined by C and C++ standards

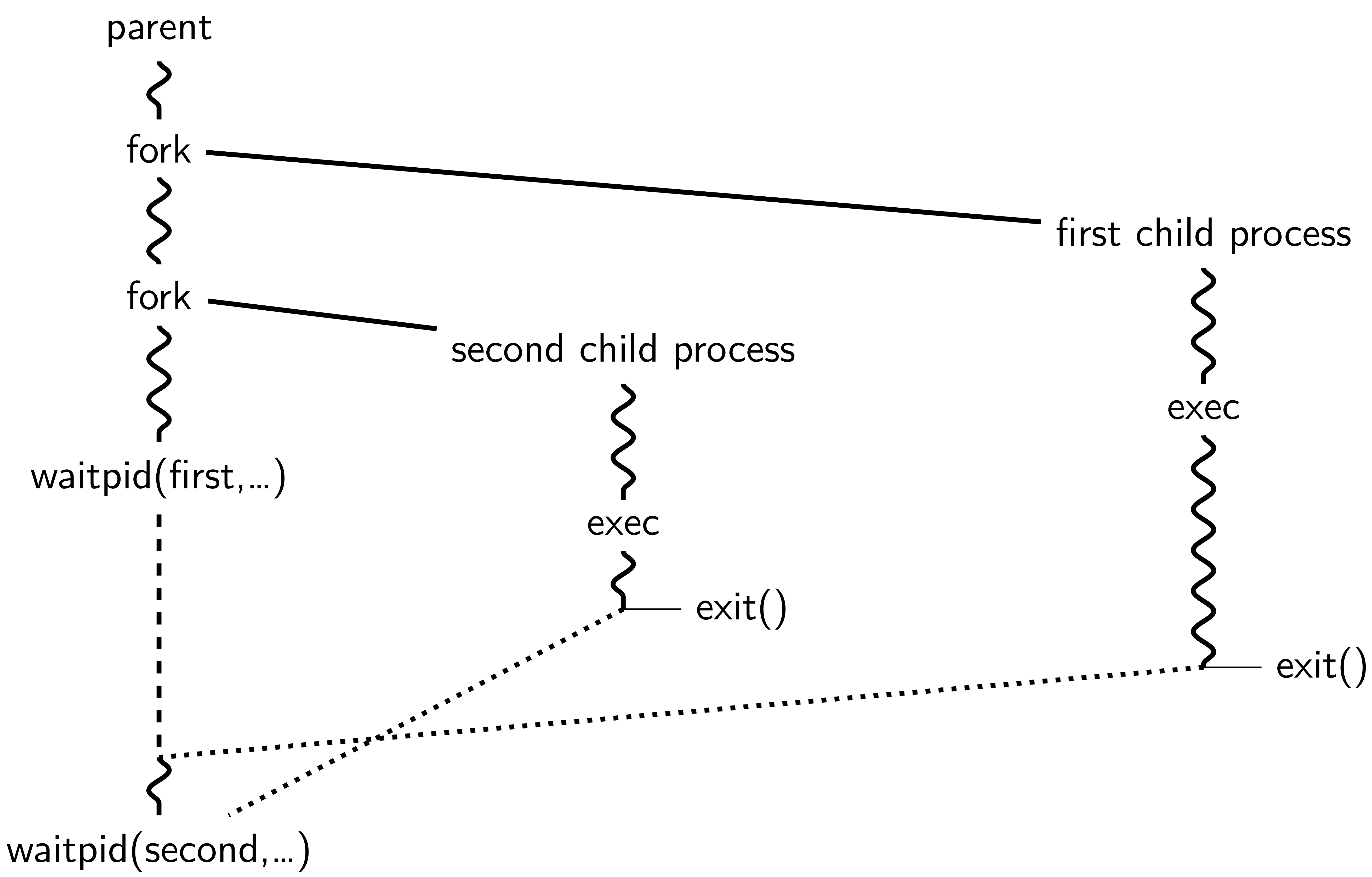
pipe() and blocking

BROKEN example:

```
int pipe_fd[2];
if (pipe(pipe_fd) < 0)
    handle_error();
int read_fd = pipe_fd[0];
int write_fd = pipe_fd[1];
write(write_fd, some_buffer, some_big_size);
read(read_fd, some_buffer, some_big_size);
```

This is likely to *not terminate*. What's the problem?

pattern with multiple?



this class: focus on Unix

Unix-like OSes will be our focus

we have source code

used to from 2150, etc.?

have been around for a while

xv6 imitates Unix

Unix history

image: Wikpedia/Eraserhead1+Infinity0+Sav_va

POSIX: standardized Unix

Portable Operating System Interface (POSIX)

“standard for Unix”

current version online: https://pubs.opengroup.org/onlinepubs/9699919799/

(almost) followed by most current Unix-like OSes

… but OSes add extra features

… and POSIX doesn’t specify everything

what POSIX defines

POSIX specifies the *library and shell interface*

- source code compatibility

doesn’t care what is/is not a system call…

doesn’t specify binary formats…

idea: write applications for POSIX, recompile and run on all implementations

- this was a very important goal in the 80s/90s
- at the time, no dominant Unix-like OS (Linux was very immature)

getpid

```
pid_t my_pid = getpid();
printf("my pid is %ld\n", (long) my_pid);
```

process ids in ps

```
cr4bd@machine:~$ ps
  PID TTY          TIME CMD
14777 pts/3    00:00:00 bash
14798 pts/3    00:00:00 ps
```

read/write

```
ssize_t read(int fd, void *buffer, size_t count);
ssize_t write(int fd, void *buffer, size_t count);
```

read/write up to *count* bytes to/from *buffer*

returns number of bytes read/written or -1 on error

- ssize_t is a signed integer type
- error code in `errno`

read returning 0 means end-of-file (*not an error*)

- can read/write less than requested (end of file, broken I/O device, …)

read/write

```
ssize_t read(int fd, void *buffer, size_t count);
ssize_t write(int fd, void *buffer, size_t count);
```

read/write *up to count* bytes to/from *buffer*

returns number of bytes read/written or -1 on error

- ssize_t is a signed integer type
- error code in `errno`

read returning 0 means end-of-file (*not an error*)

- can read/write less than requested (end of file, broken I/O device, ...)

againframe(readWrite)

read'ing one byte at a time

```
string s;
ssize_t amount_read;
char c;
/* cast to void * not needed in C */
while ((amount_read = read(STDIN_FILENO, (void*) &c, 1)) > 0) {
    /* amount_read must be exactly 1 */
    s += c;
}
if (amount_read == -1) {
    /* some error happened */
    perror("read"); /* print out a message about it */
} else if (amount_read == 0) {
    /* reached end of file */
}
```

write example

```
/* cast to void * optional in C */
write(STDOUT_FILENO, (void *) "Hello, World!\n", 14);
```

aside: environment variables (1)

key=value pairs associated with every process:

```
$ printenv
MODULE_VERSION_STACK=3.2.10
MANPATH=:/opt/puppetlabs/puppet/share/man
XDG_SESSION_ID=754
HOSTNAME=labsrv01
SELINUX_ROLE_REQUESTED=
TERM=screen
SHELL=/bin/bash
HISTSIZE=1000
SSH_CLIENT=128.143.67.91 58432 22
SELINUX_USE_CURRENT_RANGE=
QTDIR=/usr/lib64/qt-3.3
OLDPWD=/zf14/cr4bd
QTINC=/usr/lib64/qt-3.3/include
SSH_TTY=/dev/pts/0
QT_GRAPHICSSYSTEM_CHECKED=1
USER=cr4bd
LS_COLORS=rs=0:di=01;34:ln=01;36:mh=00:pi=40;33:so=01;35:do=01;35:bd=40;33;01:cd=40;33;01:or=40;31;01:mi=01;05;37
MODULE_VERSION=3.2.10
MAIL=/var/spool/mail/cr4bd
```

aside: environment variables (2)

environment variable library functions:

- `getenv(“KEY”)` $rightarrow$ *value*
- `putenv(“KEY=`*`value`*`”)` (sets KEY to *value*)
- `setenv(“KEY”, “value”)` (sets KEY to *value*)

```
char *envp[] = { "KEY1=value1", "KEY2=value2", NULL };
char *argv[] = { "somecommand", "some arg", NULL };
execve("/path/to/somecommand", argv, envp);
```

normal exec versions — keep same environment variables

aside: environment variables (3)

interpretation up to programs, but common ones…

`PATH=/bin:/usr/bin`

    to run a program 'foo', look for an executable in `/bin/foo`, then `/usr/bin/foo`

`HOME=/zf14/cr4bd`

    current user's home directory is '/zf14/cr4bd'

`TERM=screen-256color`

    your output goes to a 'screen-256color'-style terminal

…

multiple processes?

```
while (...) {
    pid = fork();
    if (pid == 0) {
        exec ...
    } else if (pid > 0) {
        pids.push_back(pid);
    }
}

/* retrieve exit statuses in order */
for (pid_t pid : pids) {
    waitpid(pid, ...);
    ...
}
```

waiting for all children

```
#include <sys/wait.h>
...
  while (true) {
    pid_t child_pid = waitpid(-1, &status, 0);
    if (child_pid == (pid_t) -1) {
      if (errno == ECHILD) {
        /* no child process to wait for */
        break;
      } else {
        /* some other error */
      }
    }
    /* handle child_pid exiting */
  }
```

multiple processes?

```
while (...) {
    pid = fork();
    if (pid == 0) {
        exec ...
    } else if (pid > 0) {
        pids.push_back(pid);
    }
}

/* retrieve exit statuses as processes finish */
while ((pid = waitpid(-1, ...)) != -1) {
    handleProcessFinishing(pid);
}
```

‘waiting’ without waiting

```
#include <sys/wait.h>
...
  pid_t return_value = waitpid(child_pid, &status, WNOHANG);
  if (return_value == (pid_t) 0) {
    /* child process not done yet */
  } else if (child_pid == (pid_t) -1) {
    /* error */
  } else {
    /* handle child_pid exiting */
  }
```

parent and child processes

every process (but process id 1) has a *parent process* (`getppid()`)

this is the process that can wait for it

creates tree of processes (Linux `pstree` command):

```
init(1)-+-ModemManager(919)-+-{ModemManager}(972)
        |                   `-{ModemManager}(1064)
        |-NetworkManager(1160)-+-dhclient(1755)
        |                      |-dnsmasq(1985)
        |                      |-{NetworkManager}(1180)
        |                      |-{NetworkManager}(1194)
        |                      `-{NetworkManager}(1195)
        |-accounts-daemon(1649)-+-{accounts-daemon}(1757)
        |                       `-{accounts-daemon}(1758)
        |-acpid(1338)
        |-apache2(3165)-+-apache2(4125)-+-{apache2}(4126)
        |               |               `-{apache2}(4127)
        |               |-apache2(28920)-+-{apache2}(28926)
        |               |                `-{apache2}(28960)
        |               |-apache2(28921)-+-{apache2}(28927)
        |               |                `-{apache2}(28963)
        |               |-apache2(28922)-+-{apache2}(28928)
        |               |                `-{apache2}(28961)
        |               |-apache2(28923)-+-{apache2}(28930)
```

parent and child questions

what if parent process exits before child?

- child's parent process becomes process id 1 (typically called *init*)

what if parent process never `waitpid()`s (or equivalent) for child?

- child process stays around as a "zombie"
- can't reuse pid in case parent wants to use `waitpid()`

what if non-parent tries to `waitpid()` for child?

- waitpid fails

read'ing a fixed amount

```
ssize_t offset = 0;
const ssize_t amount_to_read = 1024;
char result[amount_to_read];
do {
    /* cast to void * optional in C */
    ssize_t amount_read =
        read(STDIN_FILENO,
             (void *) (result + offset),
             amount_to_read - offset);
    if (amount_read < 0) {
        perror("read"); /* print error message */
        ... /* abort??? */
    } else {
        offset += amount_read;
    }
} while (offset != amount_to_read && amount_read != 0);
```

partial reads

on regular file: read reads what you request

but otherwise: usually gives you what's known to be available

after waiting for something to be available

reading from network — what's been received

reading from keyboard — what's been typed

write example (with error checking)

```
const char *ptr = "Hello, World!\n";
ssize_t remaining = 14;
while (remaining > 0) {
    /* cast to void * optional in C */
    ssize_t amount_written = write(STDOUT_FILENO,
                                   ptr,
                                   remaining);
    if (amount_written < 0) {
        perror("write"); /* print error message */
        ... /* abort??? */
    } else {
        remaining -= amount_written;
        ptr += amount_written;
    }
}
```

partial writes

usually only happen on error or interruption

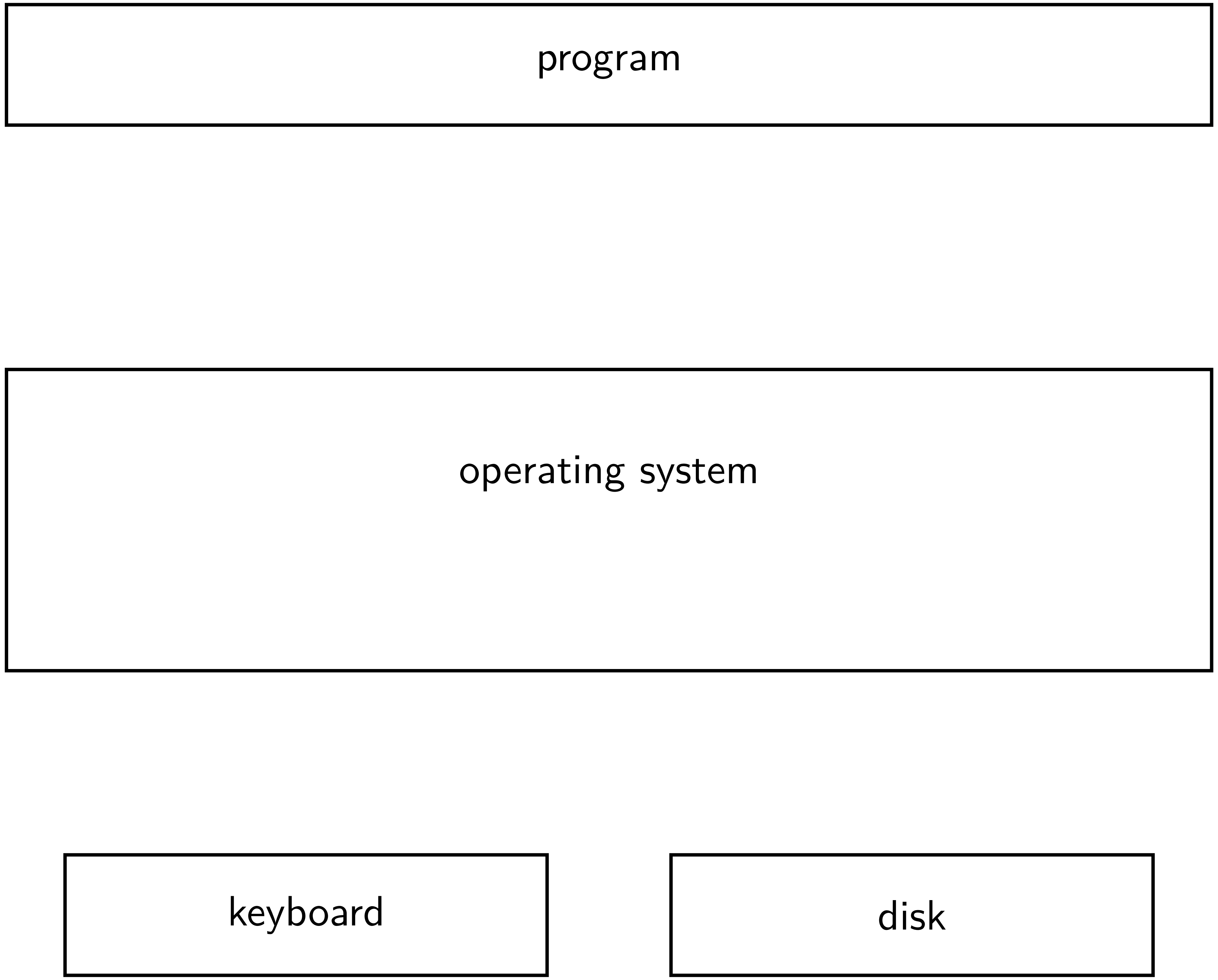
- but can request "non-blocking"
- (interruption: via *signal*)

usually: write *waits until it completes*

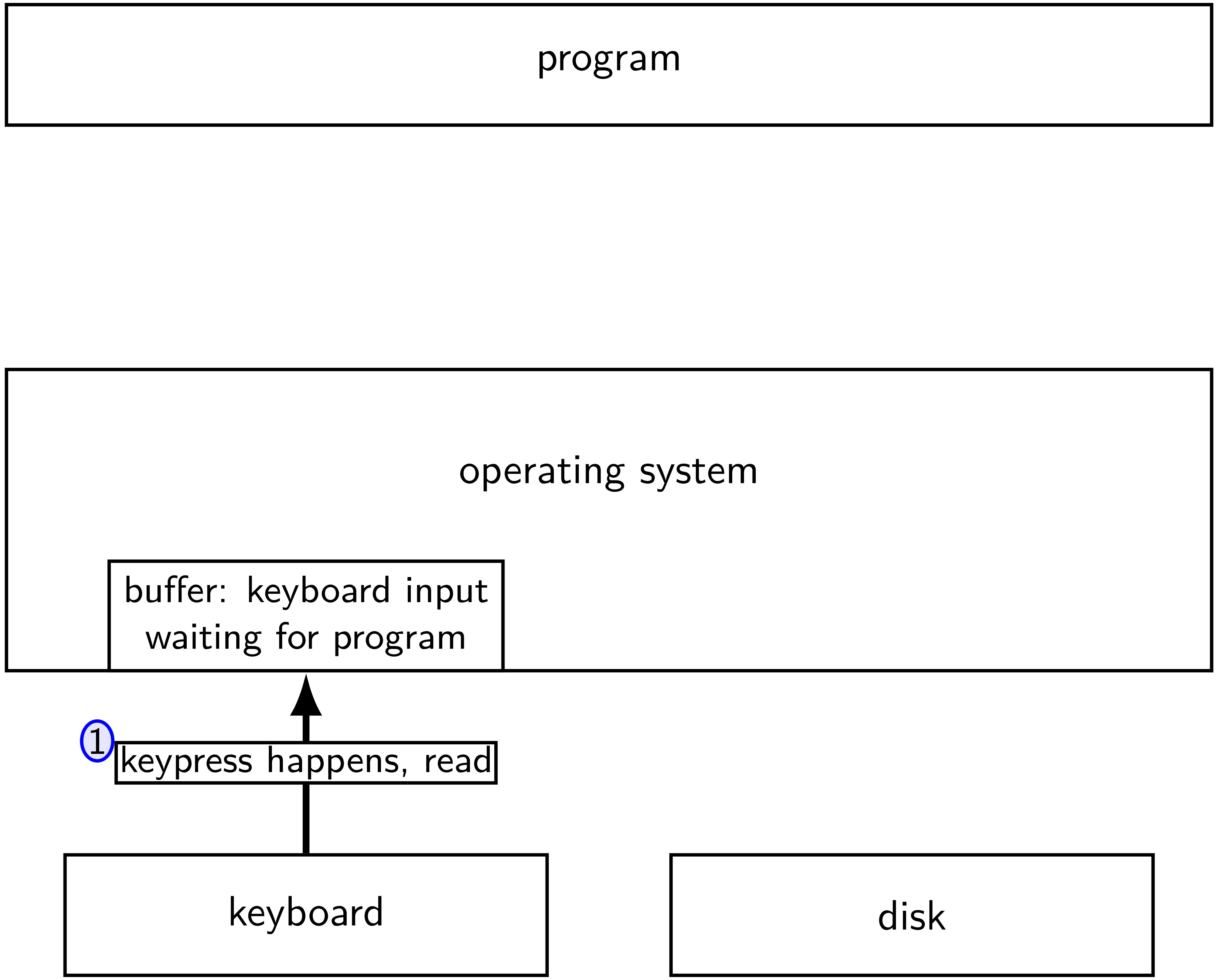
- = until remaining part fits in buffer in kernel
- does not mean data was sent on network, shown to user yet, etc.

kernel buffering (reads)

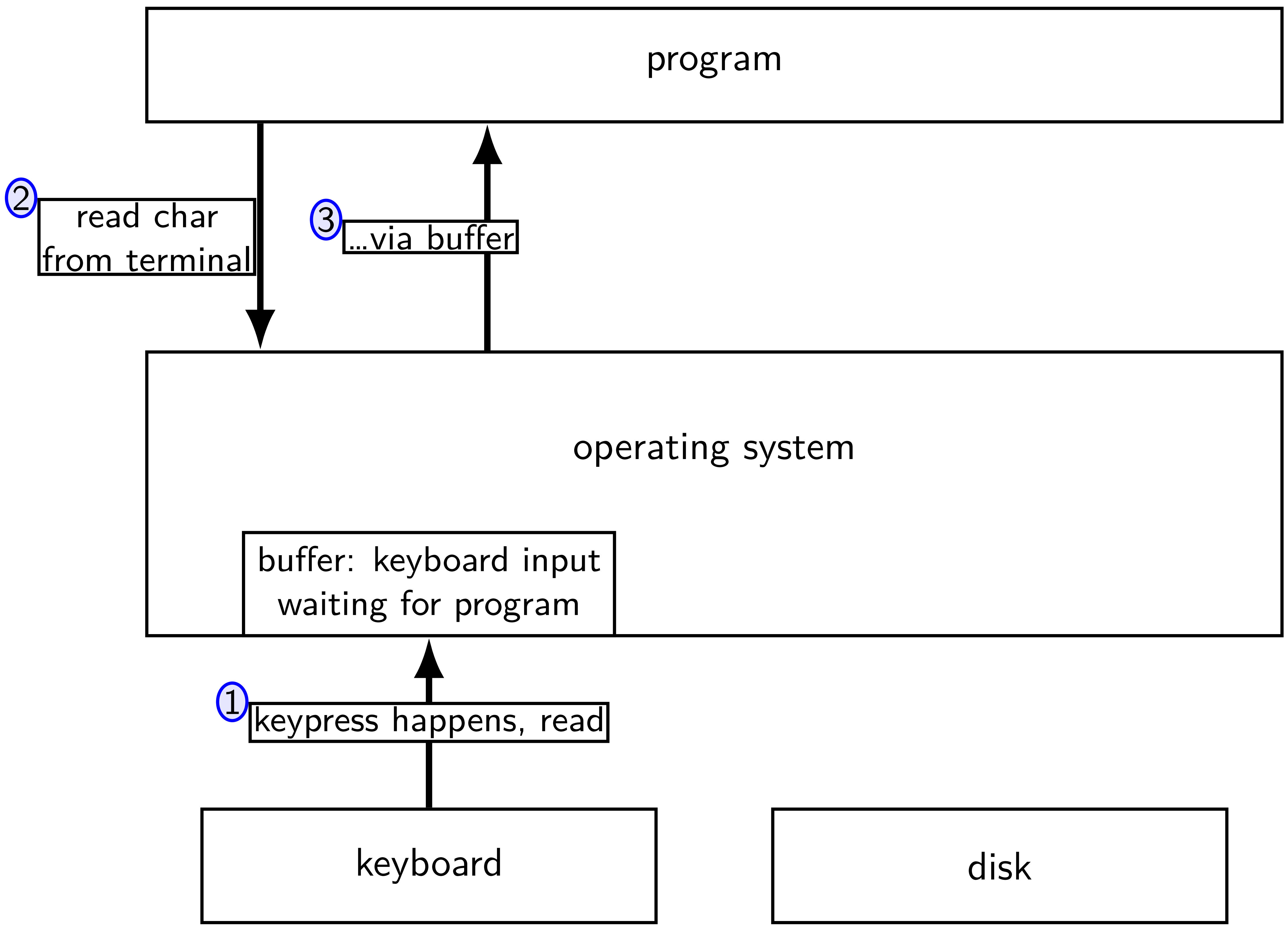
kernel buffering (reads)



kernel buffering (reads)



kernel buffering (reads)



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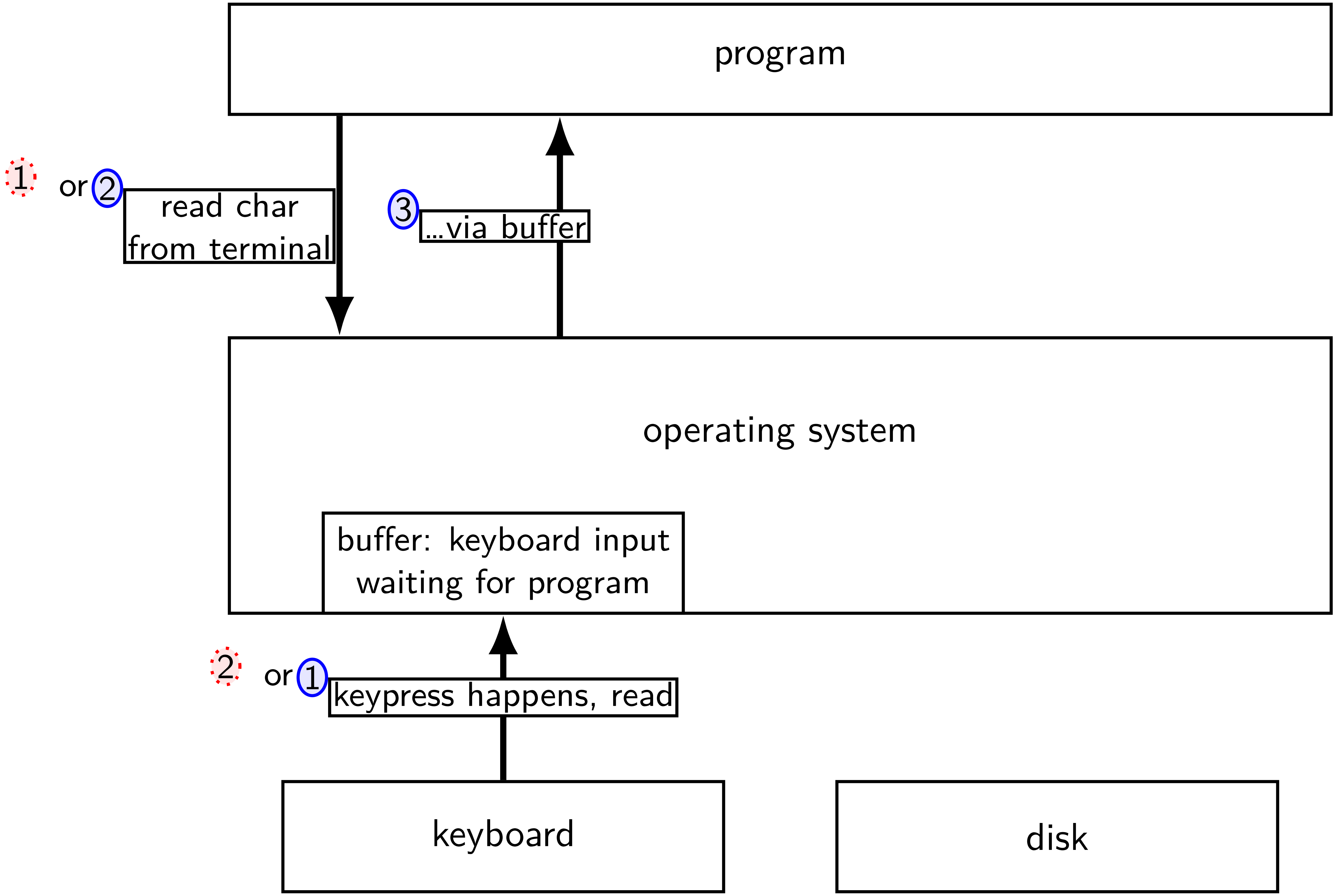
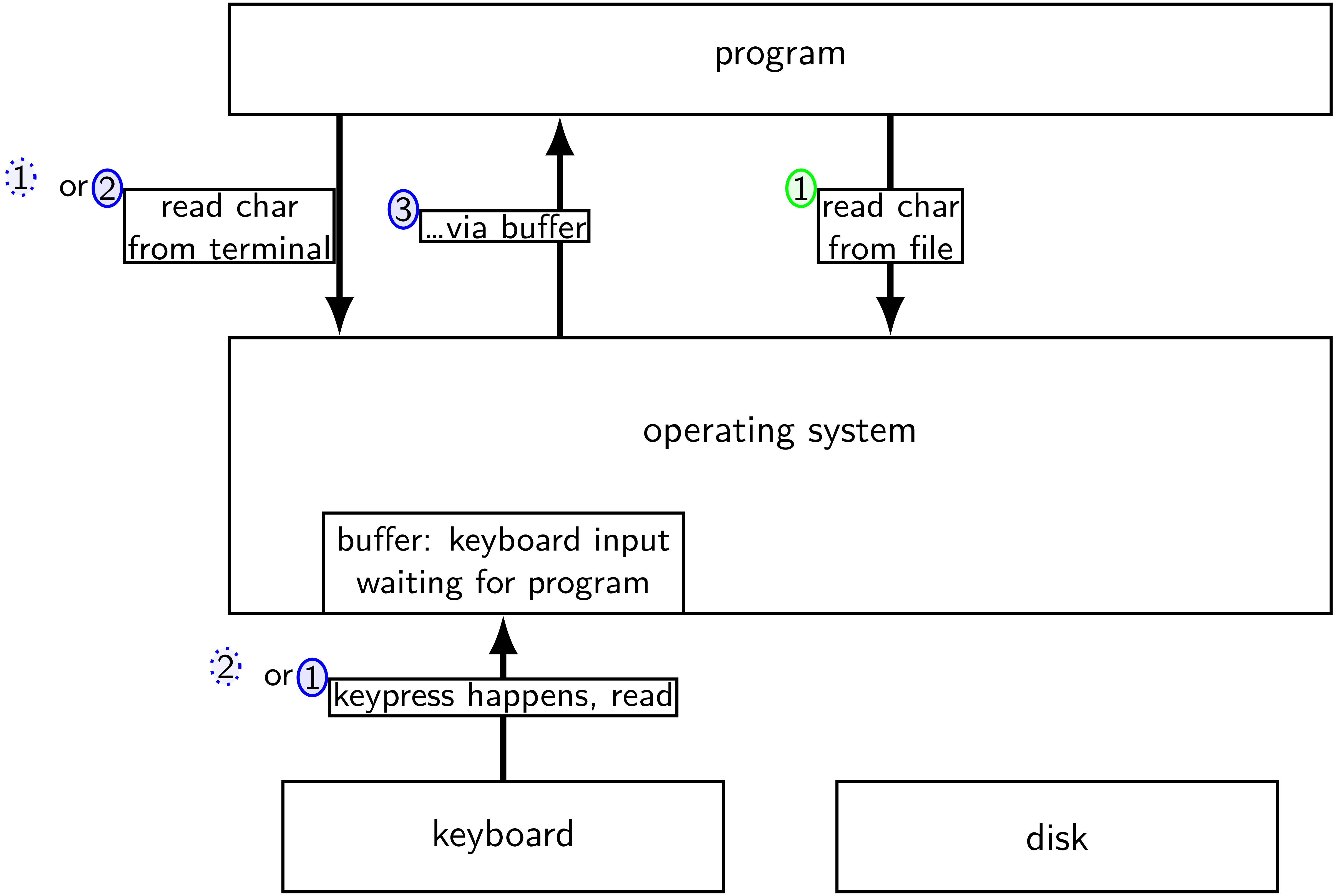
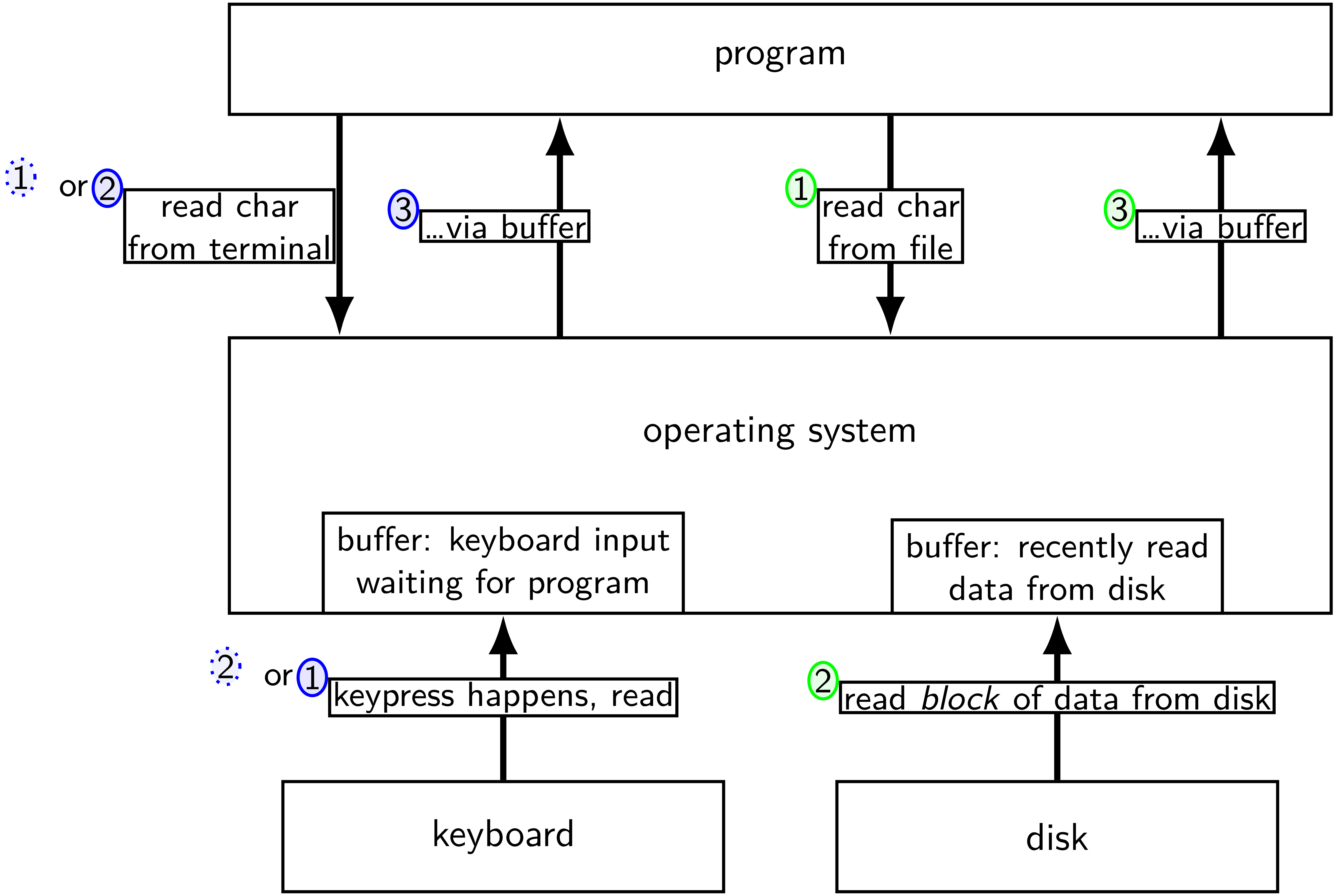


kernel buffering (reads)

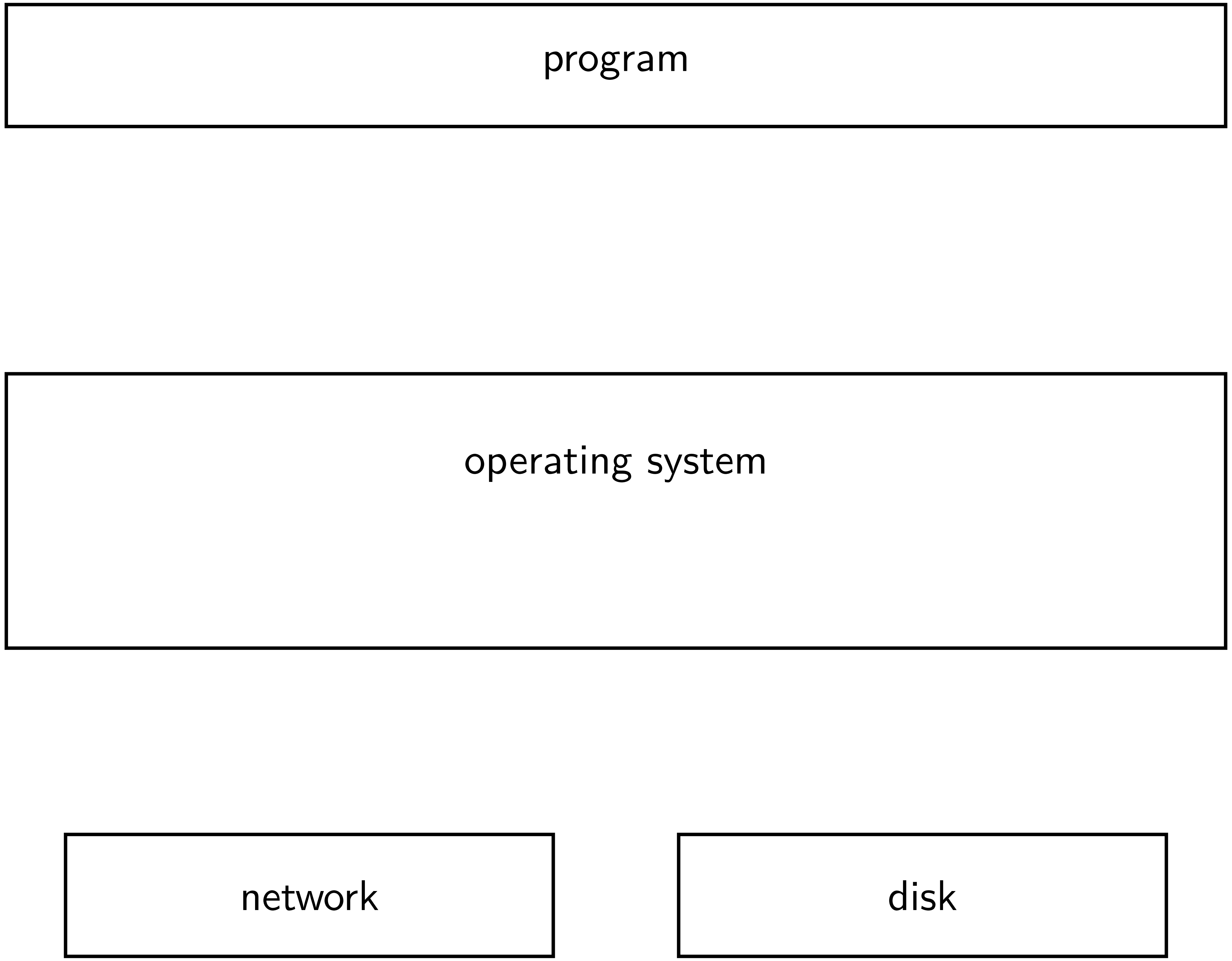


kernel buffering (reads)

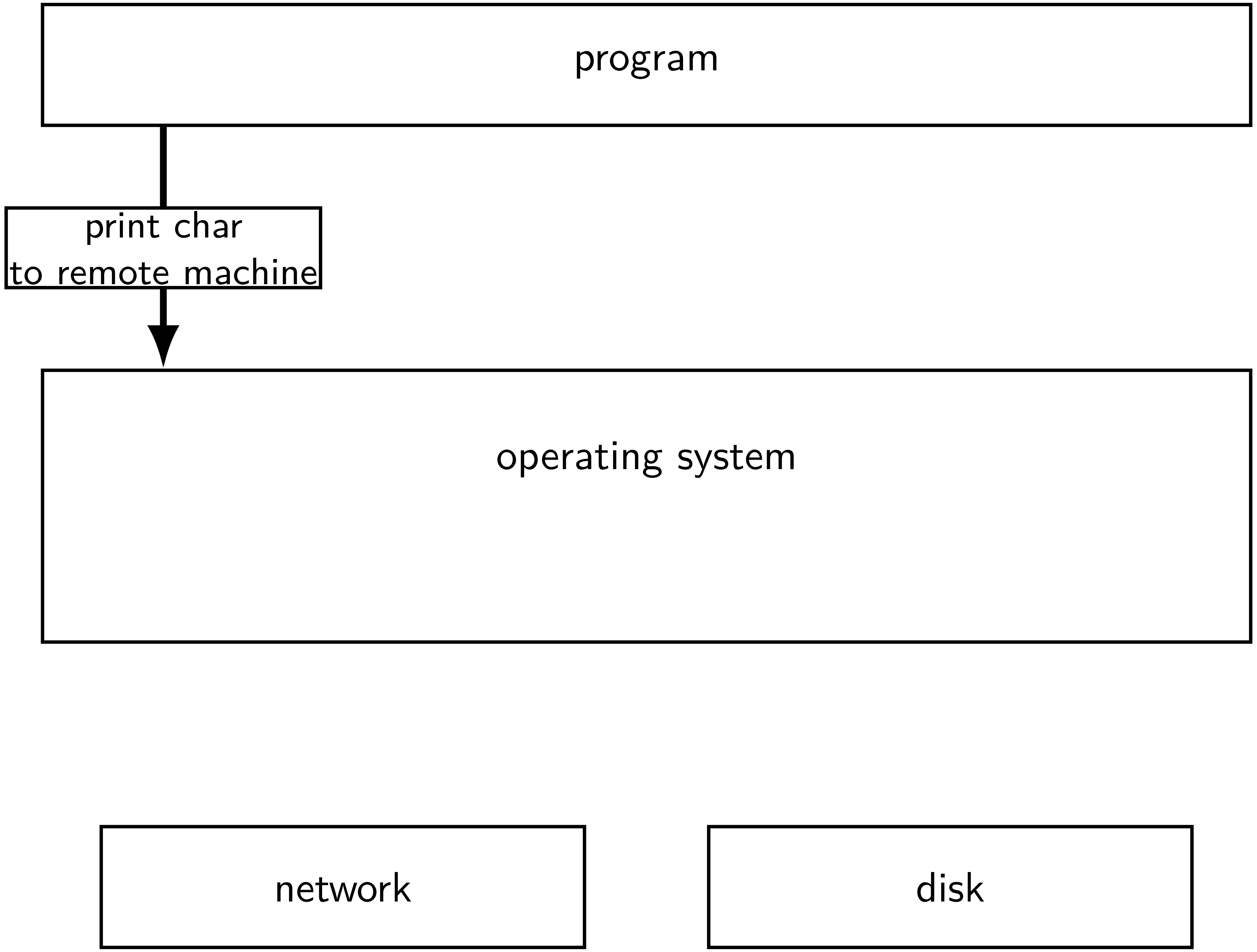


kernel buffering (writes)

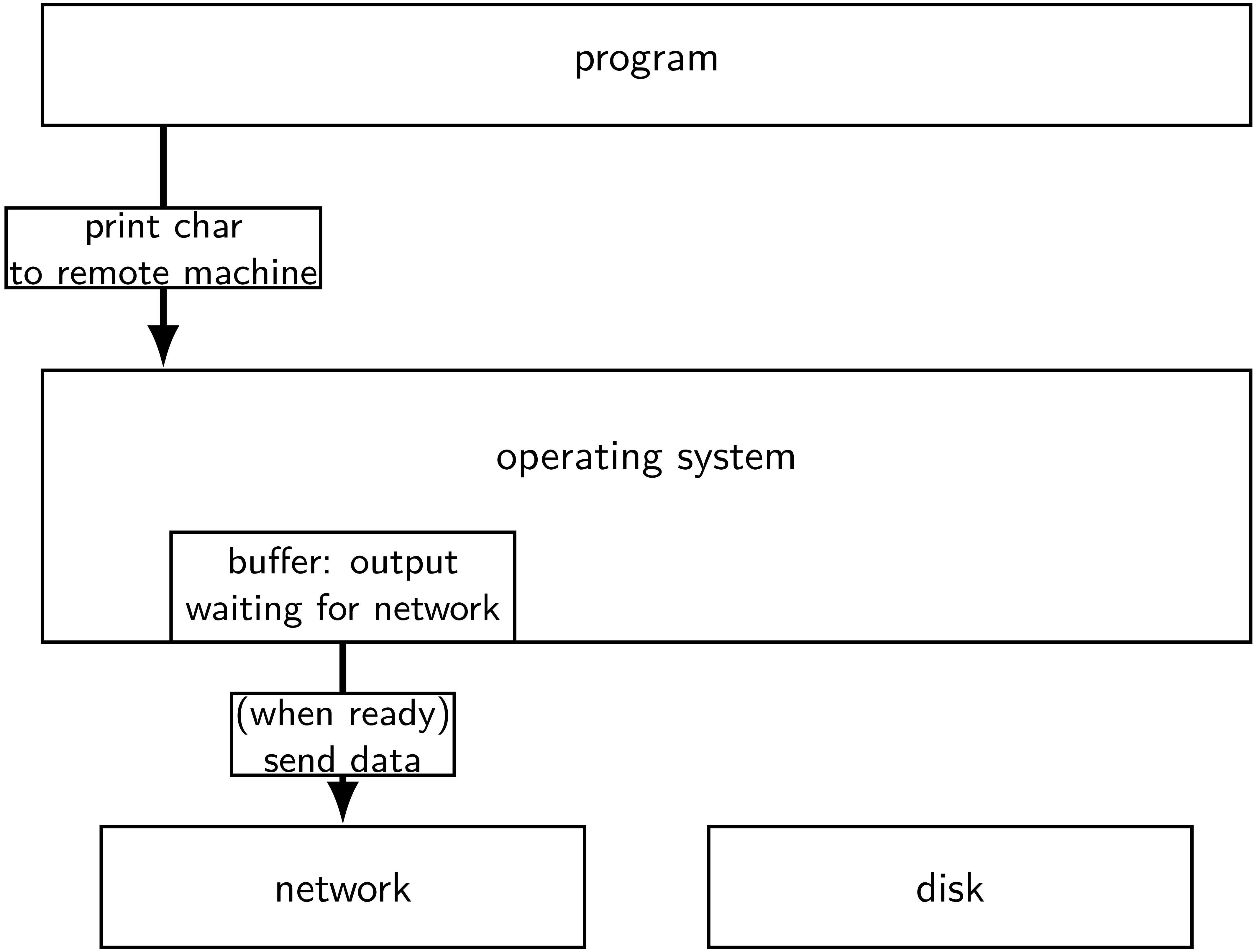
kernel buffering (writes)



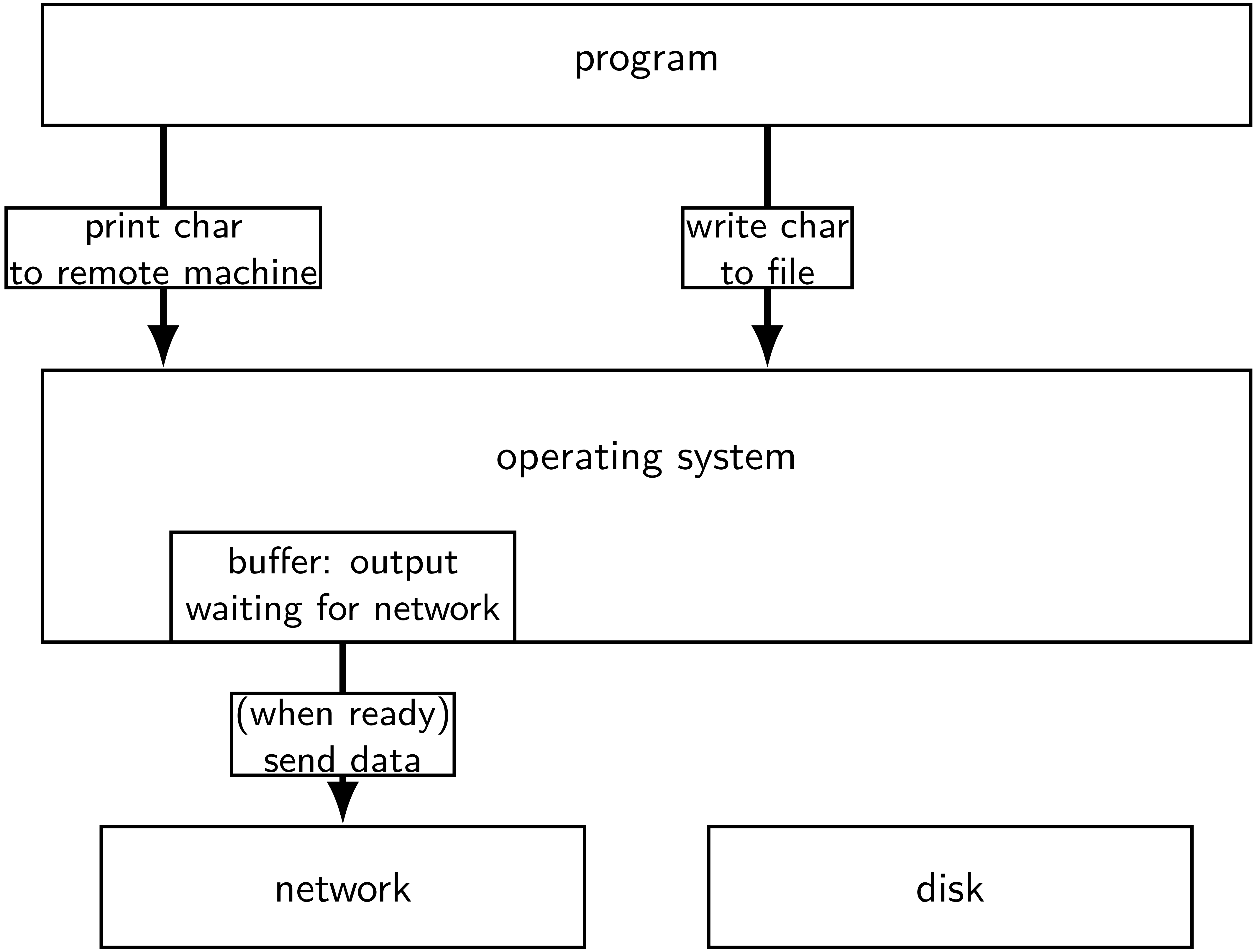
kernel buffering (writes)



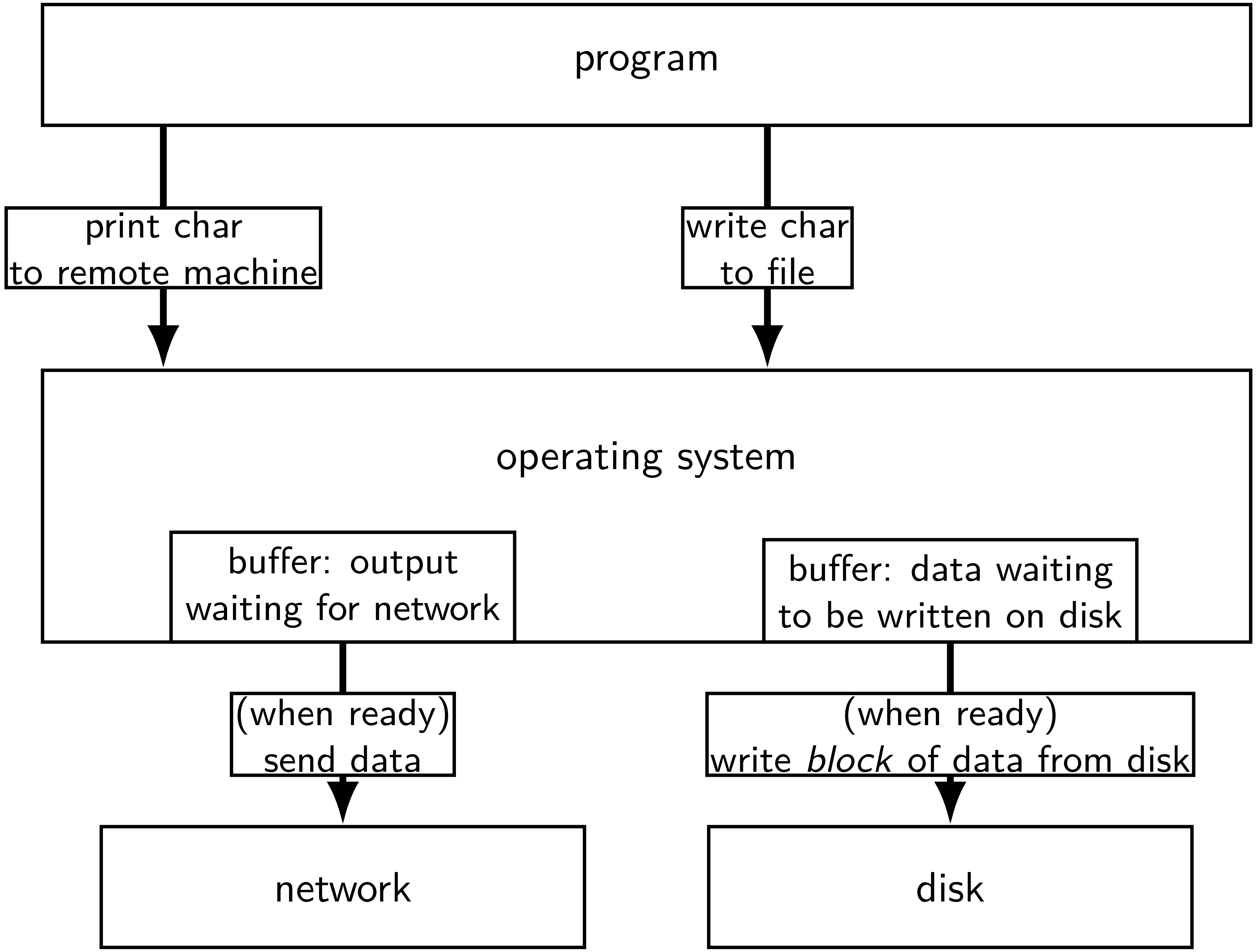
kernel buffering (writes)



kernel buffering (writes)



kernel buffering (writes)



read/write operations

read()/write(): move data into/out of buffer

possibly wait if buffer is empty (read)/full (write)

actual I/O operations — wait for device to be ready

- trigger process to stop waiting if needed

POSIX: everything is a file

the file: one interface for

- devices (terminals, printers, …)
- regular files on disk
- networking (sockets)
- local interprocess communication (pipes, sockets)

basic operations: open(), read(), write(), close()

againframe(commandLineFeatures)

exercise

```
int pipe_fds[2]; pipe(pipe_fds);
pid_t p = fork();
if (p == 0) {
  close(pipe_fds[0]);
  for (int i = 0; i < 10; ++i) {
    char c = '0' + i;
    write(pipe_fds[1], &c, 1);
  }
  exit(0);
}
close(pipe_fds[1]);
char buffer[10];
ssize_t count = read(pipe_fds[0], buffer, 10);
for (int i = 0; i < count; ++i) {
  printf("%c", buffer[i]);
}
```

Which of these are possible outputs (if pipe, read, write, fork don't fail)?

A. `0123456789` B. `0` C. (nothing)

D. A and B E. A and C F. A, B, and C

exercise

```
int pipe_fds[2]; pipe(pipe_fds);
pid_t p = fork();
if (p == 0) {
  close(pipe_fds[0]);
  for (int i = 0; i < 10; ++i) {
    char c = '0' + i;
    write(pipe_fds[1], &c, 1);
  }
  exit(0);
}
close(pipe_fds[1]);
char buffer[10];
ssize_t count = read(pipe_fds[0], buffer, 10);
for (int i = 0; i < count; ++i) {
  printf("%c", buffer[i]);
}
```

Which of these are possible outputs (if pipe, read, write, fork don't fail)?

A. `0123456789` B. `0` C. (nothing)

D. A and B E. A and C F. A, B, and C

againframe(pipeExtraEx2)

empirical evidence

```
  8 0
374 01
210 012
 30 0123
 12 01234
  3 012345
  1 0123456
  2 01234567
  1 012345678
359 0123456789
```

againframe(pipeExtraEx2More)

partial reads

read returning 0 always means end-of-file

- by default, read always waits *if no input available yet*
- but can set read to return *error* instead of waiting

read can return less than requested if not available

- e.g. child hasn't gotten far enough

pipe: closing?

if all write ends of pipe are closed

- can get end-of-file (read() returning 0) on read end
- exit()ing closes them

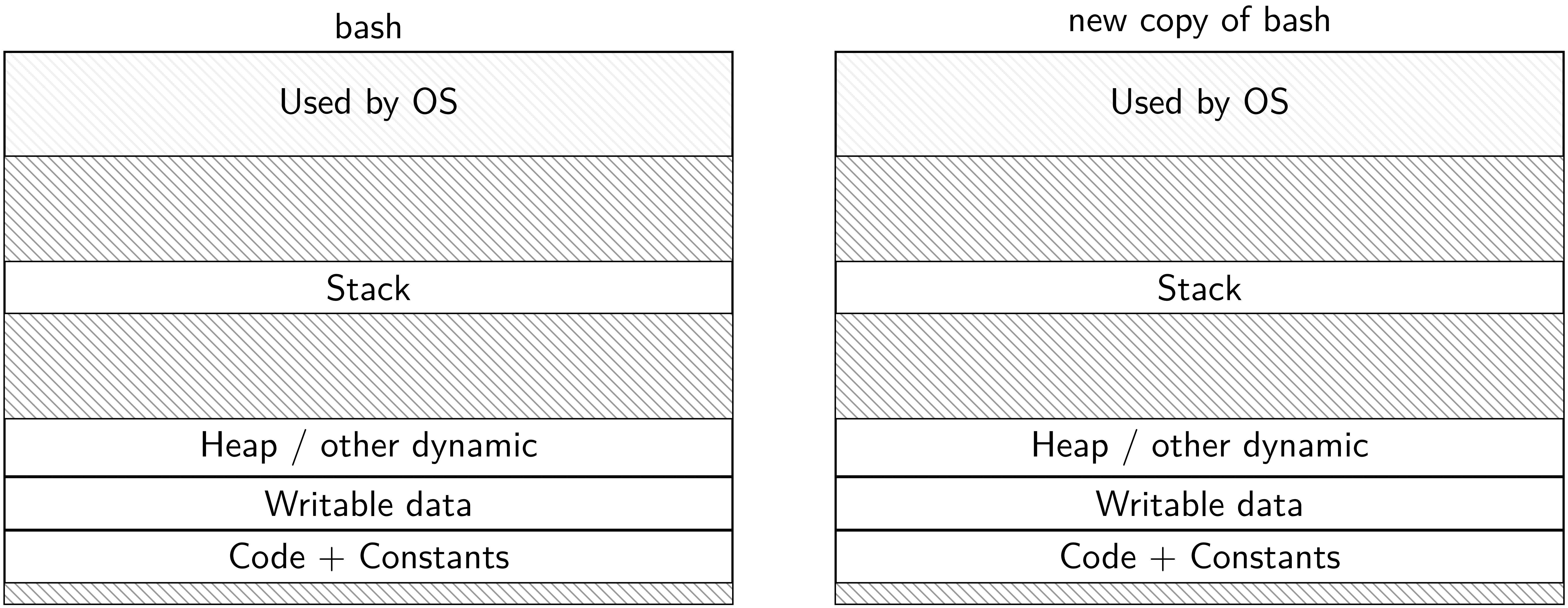
$rightarrow$
close write end when not using

generally: limited number of file descriptors per process

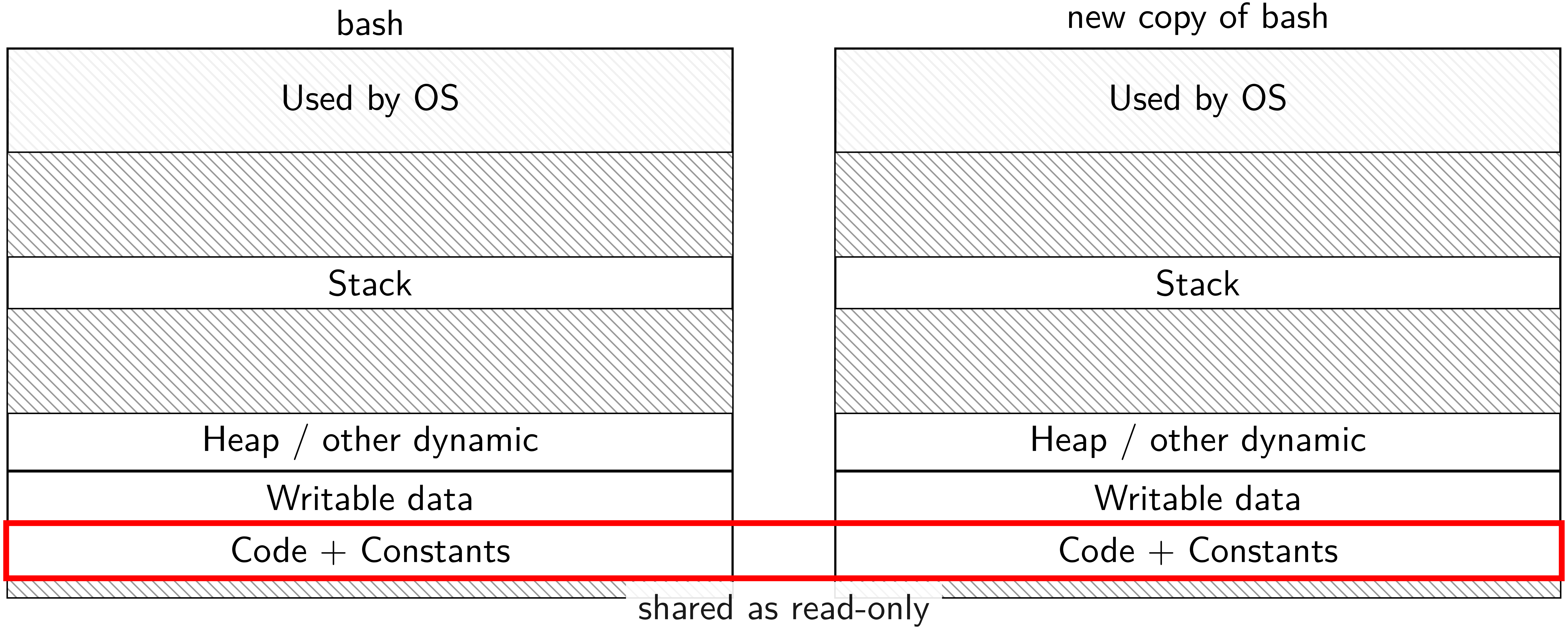
$rightarrow$
good habit to close file descriptors not being used

(but probably didn't matter for read end of pipes in example)

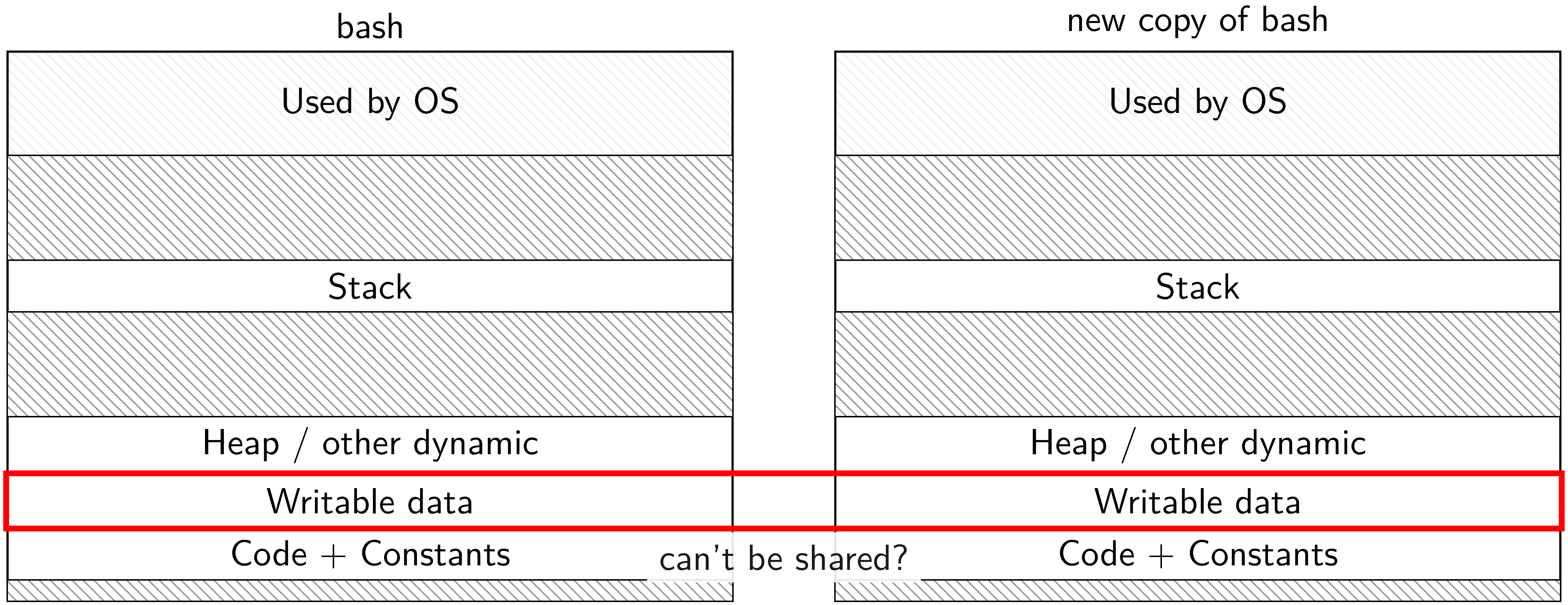
do we really need a complete copy?



do we really need a complete copy?



do we really need a complete copy?



trick for extra sharing

sharing writeable data is fine — until either process modifies it

- example: default value of global variables
- might typically not change
- (or OS might have preloaded executable's data anyways)

can we detect modifications?

trick: tell CPU (via page table) shared part is read-only

processor will trigger a fault when it's written

copy-on-write and page tables

VPN	valid?	write?	physical page
…	…	…	…
0x00601	1	1	0x12345
0x00602	1	1	0x12347
0x00603	1	1	0x12340
0x00604	1	1	0x200DF
0x00605	1	1	0x200AF
…	…	…	…

copy-on-write and page tables

VPN	valid?	write?	physical page
…	…	…	…
0x00601	1	0	0x12345
0x00602	1	0	0x12347
0x00603	1	0	0x12340
0x00604	1	0	0x200DF
0x00605	1		
…	…		

VPN	valid?	write?	physical page
…	…	…	…
0x00601	1	0	0x12345
0x00602	1	0	0x12347
0x00603	1	0	0x12340
0x00604	1	0	0x200DF
			0x200AF
…			…

copy operation actually duplicates page table

both processes *share all physical pages*

but marked *read-only in both tables*

copy-on-write and page tables

VPN	valid?	write?	physical page
…	…	…	…
0x00601	1	0	0x12345
0x00602	1	0	0x12347
0x00603	1	0	0x12340
0x00604	1	0	0x200DF
0x00605	1	0	0x200AF
…			

VPN	valid?	write?	physical page
…	…	…	…
0x00601	1	0	0x12345
0x00602	1	0	0x12347
0x00603	1	0	0x12340
0x00604	1	0	0x200DF
0x00605	1	0	0x200AF
…			

when either process tries to write a read-only page triggers a page fault — OS actually copies the page

copy-on-write and page tables

VPN	valid?	write?	physical page
...	...	...	...
0x00601	1	0	0x12345
0x00602	1	0	0x12347
0x00603	1	0	0x12340
0x00604	1	0	0x200DF
0x00605	1	0	0x200AF
...	...	...	...

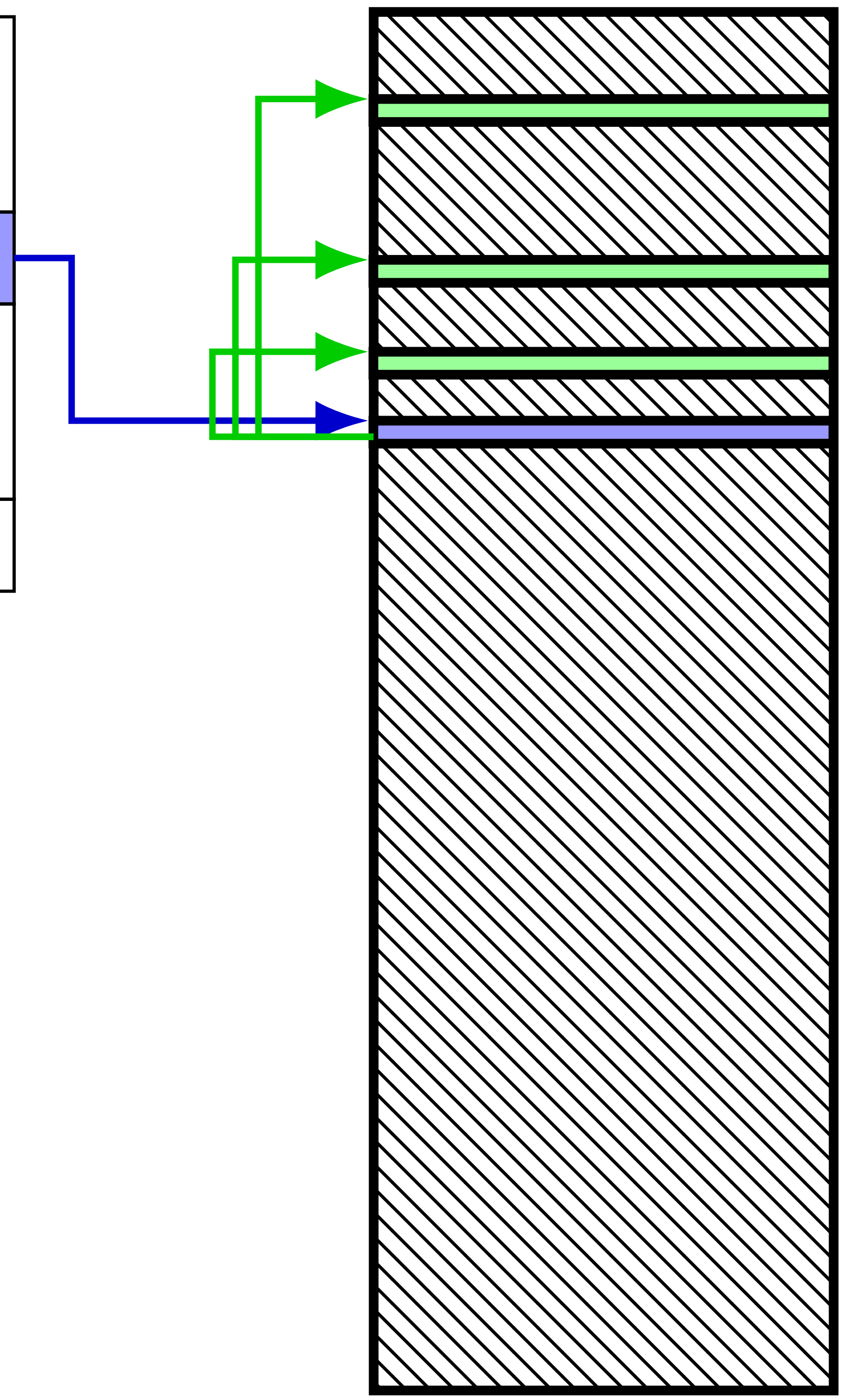
VPN	valid?	write?	physical page
...	...	...	...
0x00601	1	0	0x12345
0x00602	1	0	0x12347
0x00603	1	0	0x12340
0x00604	1	0	0x200DF
0x00605	1	1	0x300FD
...	...	...	...

after allocating a copy, OS reruns the write instruction

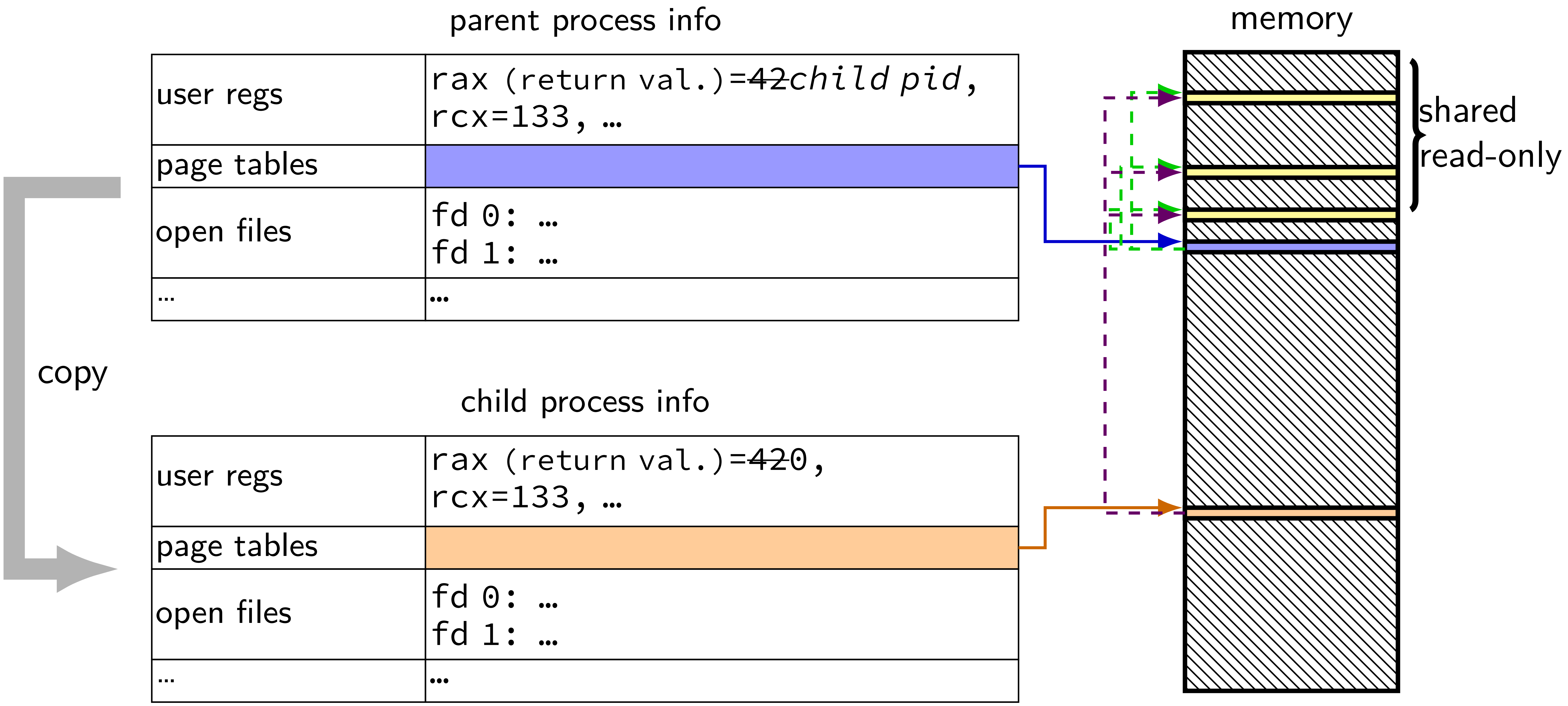
fork (w/ copy-on-write, if parent writes first)

parent process info

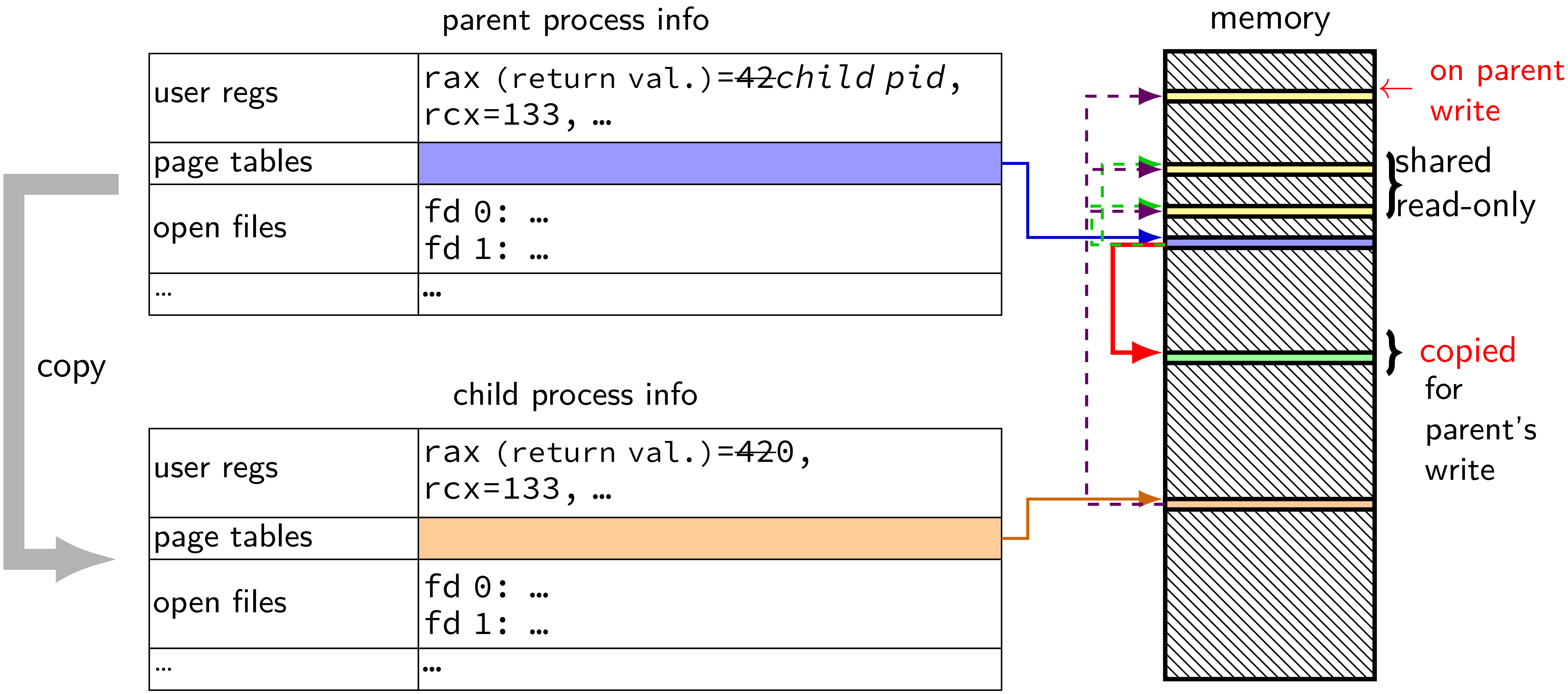
user regs	`rax (return val.)=`~~`42`~~*`child pid`*`,` `rcx=133, …`
page tables	
open files	`fd 0: …` `fd 1: …`
…	…



fork (w/ copy-on-write, if parent writes first)



fork (w/ copy-on-write, if parent writes first)



fork (w/ copy-on-write, if parent writes first)

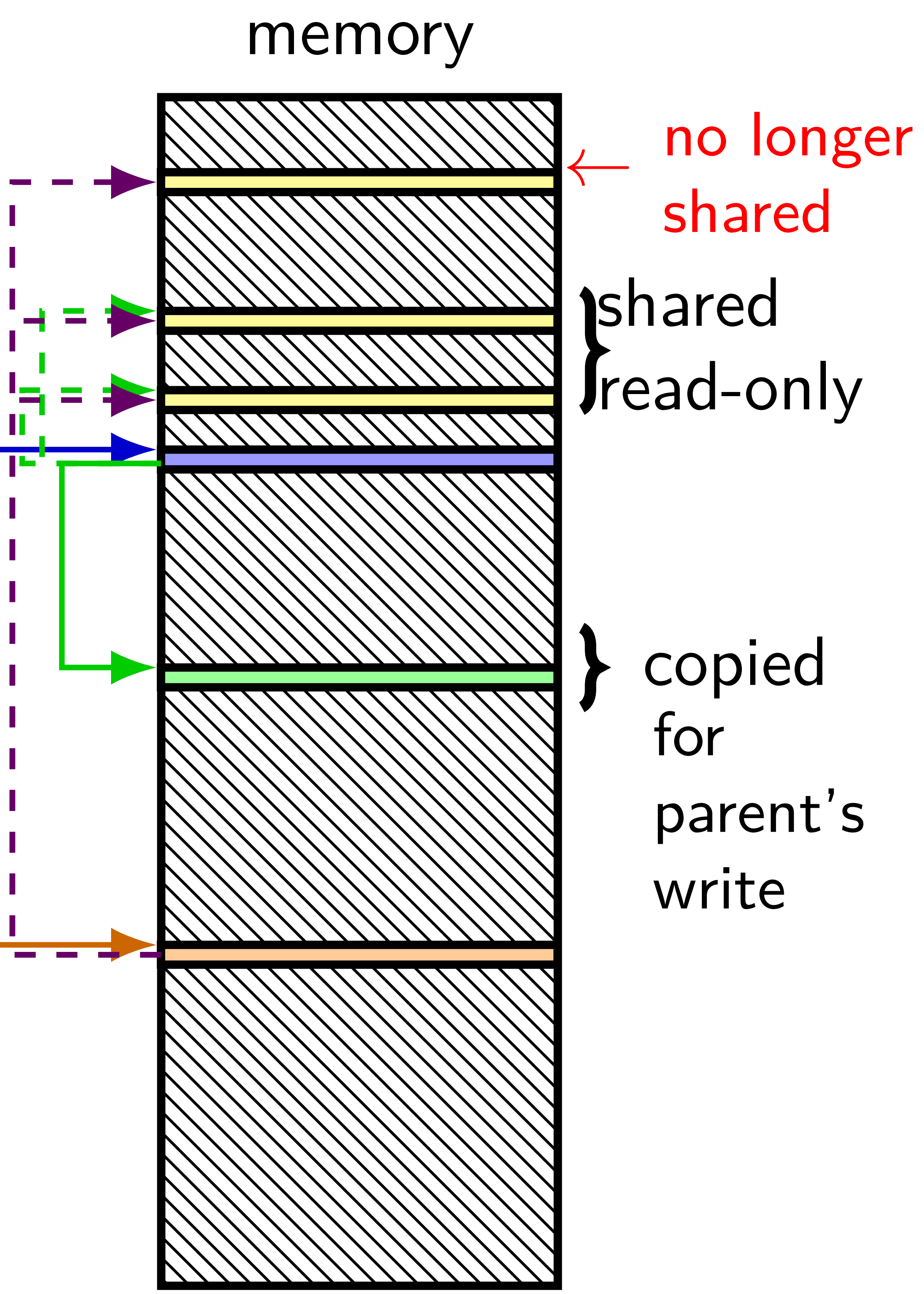
parent process info

user regs	`rax (return val.)=`~~`42`~~*`child pid`*`,` `rcx=133, …`
page tables	
open files	`fd 0: …` `fd 1: …`
…	…

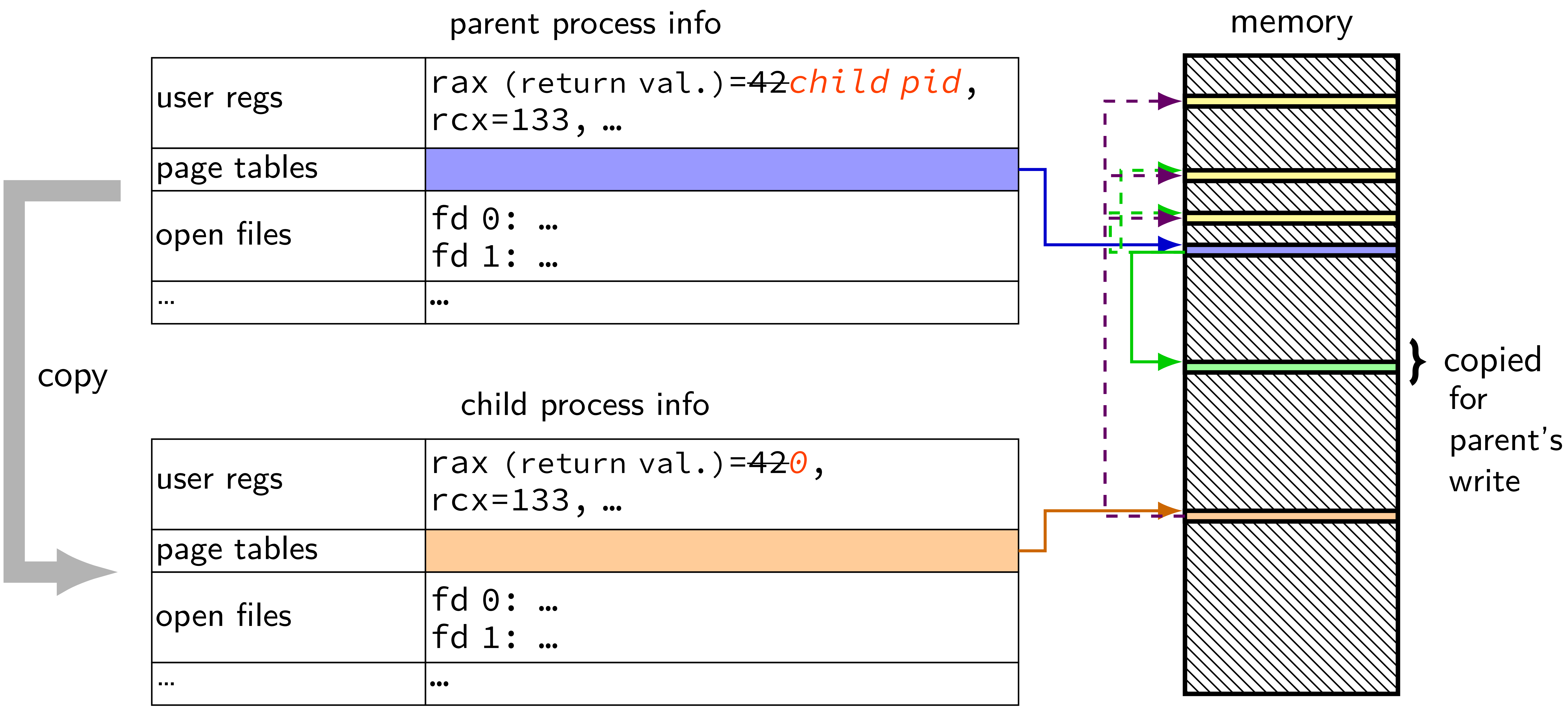
copy

child process info

user regs	`rax (return val.)=`~~`42`~~`0,` `rcx=133, …`
page tables	
open files	`fd 0: …` `fd 1: …`
…	…



fork (w/ copy-on-write, if parent writes first)



pipes

special kind of file: pipes

bytes go in one end, come out the other — once

created with `pipe()` library call

intended use: communicate between processes

- like implementing shell pipelines

pipe()

```
int pipe_fd[2];
if (pipe(pipe_fd) < 0)
    handle_error();
/* normal case: */
int read_fd = pipe_fd[0];
int write_fd = pipe_fd[1];
```

then from one process...

```
write(write_fd, ...);
```

and from another

```
read(read_fd, ...);
```

pipe example (1)

```
int pipe_fd\[2];
if (pipe(pipe_fd) < 0)
    handle_error(); /* e.g. out of file descriptors */
int read_fd = pipe_fd\[0];
int write_fd = pipe_fd\[1];
child_pid = fork();
if (child_pid  == 0) {
    /* in child process, write to pipe */
    close(read_fd);
    write_to_pipe(write_fd); /* function not shown */
    exit(EXIT_SUCCESS);
} else if (child_pid > 0) {
    /* in parent process, read from pipe */
    close(write_fd);
    read_from_pipe(read_fd); /* function not shown */
    waitpid(child_pid, NULL, 0);
    close(read_fd);
} else { /* fork error */ }
```

pipe example (1)

```
int pipe_fd\[2];
if (pipe(pipe_fd) < 0)
    handle_error(); /* e.g. out of file descriptors */
int read_fd = pipe_fd\[0];
int write_fd = pipe_fd\[1];
child_pid = fork();
if (child_pid  == 0) {
    /* in child process, write to pipe */
    close(read_fd);
    write_to_pipe(write_fd); /* function not shown */
    exit(EXIT_SUCCESS);
} else if (child_pid > 0) {
    /* in parent process, read from pipe */
    close(write_fd);
    read_from_pipe(read_fd); /* function not shown */
    waitpid(child_pid, NULL, 0);
    close(read_fd);
} else { /* fork error */ }
```

pipe example (1)

```
int pipe_fd\[2];
if (pipe(pipe_fd) < 0)
    handle_error(); /* e.g. out of file descriptors */
int read_fd = pipe_fd\[0];
int write_fd = pipe_fd\[1];
child_pid = fork();
if (child_pid  == 0) {
    /* in child process, write to pipe */
    close(read_fd);
    write_to_pipe(write_fd); /* function not shown */
    exit(EXIT_SUCCESS);
} else if (child_pid > 0) {
    /* in parent process, read from pipe */
    close(write_fd);
    read_from_pipe(read_fd); /* function not shown */
    waitpid(child_pid, NULL, 0);
    close(read_fd);
} else { /* fork error */ }
```

‘standard’ pattern with fork()

pipe example (1)

```
int pipe_fd\[2];
if (pipe(pipe_fd) < 0)
    handle_error(); /* e.g. out of file descriptors */
int read_fd = pipe_fd\[0];
int write_fd = pipe_fd\[1];
child_pid = fork();
if (child_pid  == 0) {
    /* in child process, write to pipe */
    close(read_fd);
    write_to_pipe(write_fd); /* function not shown */
    exit(EXIT_SUCCESS);
} else if (child_pid > 0) {
    /* in parent process, read from pipe */
    close(write_fd);
    read_from_pipe(read_fd); /* function not shown */
    waitpid(child_pid, NULL, 0);
    close(read_fd);
} else { /* fork error */ }
```

read() will not indicate end-of-file if write fd is open (any copy of it)

pipe example (1)

```
int pipe_fd\[2];
if (pipe(pipe_fd) < 0)
    handle_error(); /* e.g. out of file descriptors */
int read_fd = pipe_fd\[0];
int write_fd = pipe_fd\[1];
child_pid = fork();
if (child_pid  == 0) {
    /* in child process, write to pipe */
    close(read_fd);
    write_to_pipe(write_fd); /* function not shown */
    exit(EXIT_SUCCESS);
} else if (child_pid > 0) {
    /* in parent process, read from pipe */
    close(write_fd);
    read_from_pipe(read_fd); /* function not shown */
    waitpid(child_pid, NULL, 0);
    close(read_fd);
} else { /* fork error */ }
```

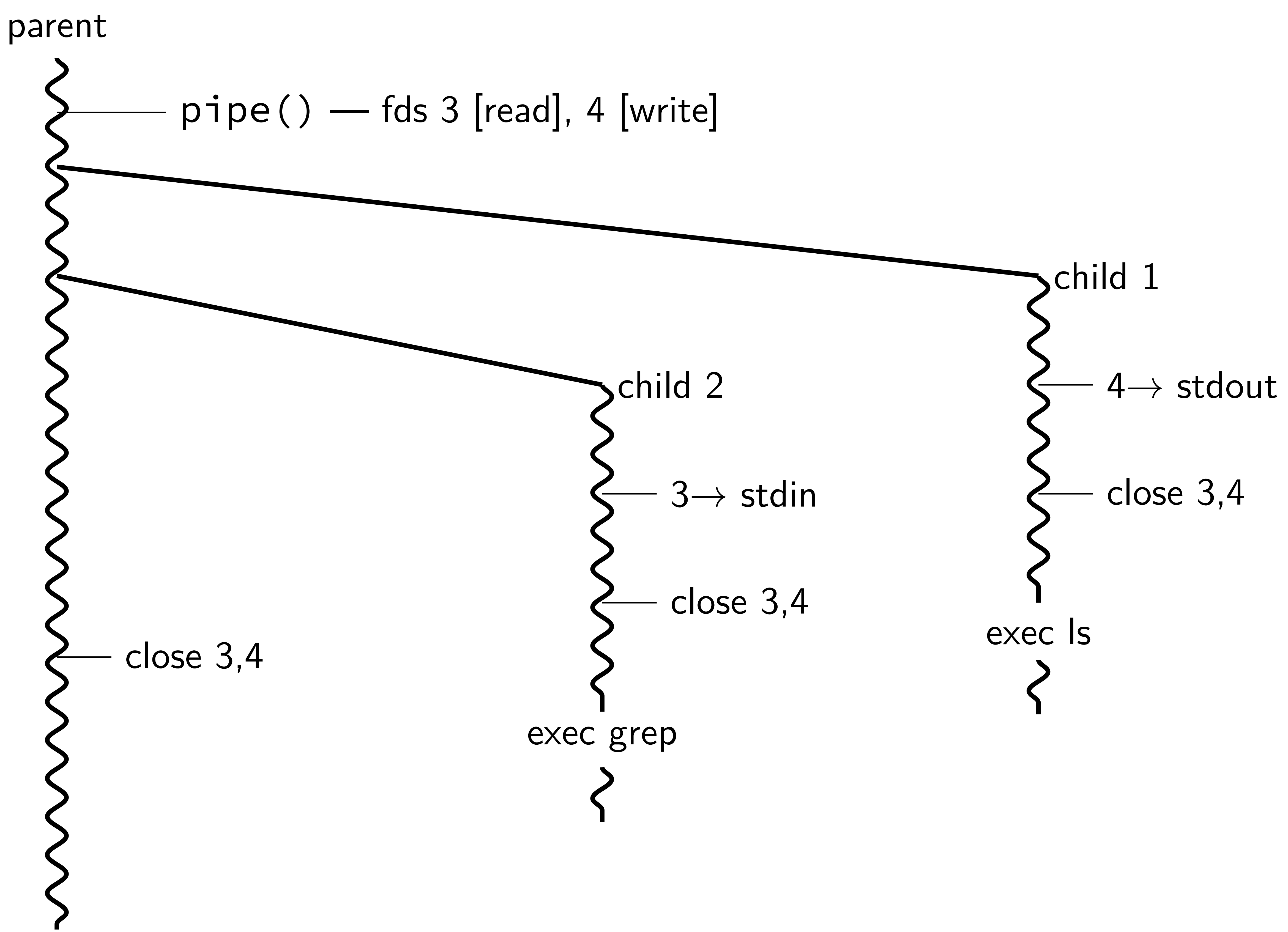
have habit of closing
to avoid ‘leaking’ file descriptors
you can run out

pipe and pipelines

```
ls -1 | grep foo
```

```
pipe(pipe_fd);
ls_pid = fork();
if (ls_pid == 0) {
    dup2(pipe_fd[1], STDOUT_FILENO);
    close(pipe_fd[0]); close(pipe_fd[1]);
    char *argv[] = {"ls", "-1", NULL};
    execv("/bin/ls", argv);
}
grep_pid = fork();
if (grep_pid == 0) {
    dup2(pipe_fd[0], STDIN_FILENO);
    close(pipe_fd[0]); close(pipe_fd[1]);
    char *argv[] = {"grep", "foo", NULL};
    execv("/bin/grep", argv);
}
close(pipe_fd[0]); close(pipe_fd[1]);
/* wait for processes, etc. */
```

example execution



exercise

```
pid_t p = fork();
int pipe_fds[2];
pipe(pipe_fds);
if (p == 0) { /* child */
  close(pipe_fds[0]);
  char c = 'A';
  write(pipe_fds[1], &c, 1);
  exit(0);
} else { /* parent */
  close(pipe_fds[1]);
  char c;
  int count = read(pipe_fds[0], &c, 1);
  printf("read %d bytes\n", count);
}
```

The child is trying to send the character `A` to the parent, but the above code outputs `read 0 bytes` instead of `read 1 bytes`. What happened?

exercise solution

\iftoggle{heldback pipe() is after fork — two pipes, one in child, one in parent}